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Triple Negative Breast Cancer Prevalence in Africa: a Systematic Review and Meta-analysis

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Triple Negative Breast Cancer Prevalence in Africa: a Systematic Review and Meta-analysis

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Abstract:

Introduction: The aggressive triple negative breast cancer (TNBC) subtype disproportionately affects women of African ancestry across the diaspora, but its prevalence across Africa has not been widely studied. This study seeks to estimate the prevalence of TNBC in African women.

Methods: For this systematic review and meta-analysis, PubMed, EMBASE, African Journals Online and Web of Science were searched for studies on TNBC frequency. All studies reporting TNBC frequencies among African women were included. Editorials, single case reports, and studies conducted in non-African nations without assessment of TNBC frequency were excluded. Screening and data abstraction were completed independently by two reviewers and a random-effects meta-analysis was performed to obtain the pooled frequency of TNBC in African populations.

Results: 1808 potentially eligible studies were identified of which 67 were included in the systematic review and 60 were included in the meta-analysis. Pooled frequency of TNBC was highest across West Africa, 46.0% [n= 15, 95% CI 38.8-53.3, I^2= 92%] and lowest in Central Africa, 14.9% [n= 1, 95% CI 8.1-23.3]. TNBC estimates were higher for studies that used Allred guidelines for ER/PR status compared to ASCO/CAP guidelines, and for studies that used older versions of ASCO/CAP guidelines for assessing HER2 status.

Conclusion: TNBC frequency was variable with the highest frequency reported in West Africa. Greater emphasis should be placed on establishing protocols for assessing receptor status due to the variability among studies.

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Article summary

Strengths and limitations of this study:

- Previous studies have reported higher ER-negativity among West African countries compared to East African countries, but no review or meta-analysis has been solely focused on the frequency of TNBC cases across the African continent.
- We found that estimates for TNBC frequency were variable across the African continent where highest pooled estimates were found in West African nations (~46% TNBC frequency). High frequency of TNBC and poor prognostic clinical factors highlight the need for early detection across the continent.
- Methods used to assess receptor status are not consistent across the African continent and there was a lack of cancer registry and population-based data. However, given the available data, this is the best estimate of TNBC frequency across Africa.
- More emphasis should be placed on establishing standard methods for receptor status testing to adequately diagnose and treat breast cancer. Implementation of cancer registries and/or population-based data for breast cancer across the continent should also be a priority.

Keywords

Triple negative breast cancer, TNBC, Africa, breast cancer, women of African ancestry

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Patient and Public Involvement

It was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

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Introduction

Breast cancer (BCa) mortality rates have markedly increased across Africa where estimated age-standardized rates in 2020 ranged from 20 deaths per 100,000 women across Northern Africa to 27 deaths per 100,000 women in Western Africa [1-3]. BCa has thus been dubbed an emerging epidemic in Africa [4]. Women of African ancestry (WAA) across the globe and indigenous African women are more likely to receive a poorer BCa prognosis compared to women of other ancestries [5]. Multiple studies often posit poorer prognosis as a result of healthcare systems across Africa, where there is limited capacity and health infrastructure, *e.g.* inadequate screening and diagnostic services [6]. However, BCa prevalence by subtype, as defined by receptor status, is strikingly different for indigenous African women compared to the BCa prevalence profile of Western countries and is not simply explained by limited access to healthcare [5].

Immunohistochemistry (IHC) is routinely used to classify BCa into molecular subtypes according to the presence or absence of the estrogen (ER) and progesterone (PR) hormone receptors and the human epidermal growth factor receptor 2 (HER2). Triple negative breast cancer (TNBC) is characterized by the lack of expression of all three biomarkers (ER, PR, HER2), which makes TNBC untreatable with targeted therapies such as Tamoxifen and Herceptin [7, 8]. TNBC is often associated with earlier disease-onset, advanced-stage tumors and aggressive disease progression when compared with other BCa subtypes [9, 10]. TNBC has also been shown to disproportionately affect African women, and younger WAA and Hispanic women in North America [10], where the prevalence of TNBC in WAA has been estimated to be more than twice the prevalence in non-Hispanic White women [11]. WAA also have a higher mortality rate from TNBC and more advanced stage at diagnosis [12]. Thus, it is important to investigate what seems to be an ancestral predisposition to TNBC since the reasons for this racial disparity in TNBC prevalence and outcomes are not fully understood. Studies to date have not compiled adequate information on TNBC frequency or considered reported frequency and routine practices associated with diagnosis and treatment across the African continent.

This systematic review and meta-analysis aims to increase understanding and knowledge regarding the frequency of TNBC across the African continent. The paucity of data and information on TNBC in Africa

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underscores the importance and urgency of such a review. A previous review reported higher ER-negativity among West African countries compared to East African countries [5] and a previous meta-analysis investigated ER-positivity across Africa [13] but no review or meta-analysis has been solely focused on the frequency of TNBC cases across the African continent. This review complements current biomedical research on TNBC and provides context for areas where TNBC research continues to expand. Improved understanding of TNBC frequency in continental Africa can further inform strategies for BCa detection and management for WAA globally. Due to shared ancestry between North American WAA and indigenous West African women [14], we hypothesize that there will be higher TNBC prevalence rates in countries across West Africa compared to other regions (North, East, Central, Southern) across Africa.

METHODS

Search Strategy and Selection Criteria

We used Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines as a framework for our systematic review and meta-analysis [15] as well as Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines [16]. On April 25th 2021, we searched PubMed, EMBASE, African Journals Online and Web of Science for relevant articles without date or language restrictions. Start date of the search was from inception of each database. A detailed version of our search strategy used in PubMed was modified for other databases. Search strategy and these modifications can be found in Table S1. Briefly, all search terms were Medical Subject Heading terms, including TNBC terms ("TNBC", "triple-negative*", "triple negative") and terms for African countries ("Africa", "African" and names of all 54 African countries) and outcome variables ("rate*", "prevalence", "epidemiology"). We included all studies that met the inclusion criteria. The inclusion criteria were as follows: studies that use breast cancer tissue samples from indigenous African women of any age, in any care setting (clinical, community, *etc.*) and at any geographic location; sample size of eligible participants ≥ 40 (as slightly more stringent criteria since normal distribution could be assumed at n= 30 [17]); studies that demonstrate at least one of the

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following: report on receptor status of breast tumours including ER, PR, and HER2; any primary study including but not limited to observational studies, cohort and case-control studies, cross-sectional studies, *etc.*

We excluded: editorials, single case reports, case series, and commentaries; studies that assessed diagnostic measures and treatment options for women with TNBC in the absence of assessment of its prevalence; studies conducted in non-African nations without assessment of indigenous African TNBC rates or that of first-generation African immigrants. Study selection began by screening titles and abstracts of articles collected after employing the search strategy. The full text of these articles was then reviewed to assess inclusion. Two data abstractors independently reviewed articles at both the title/abstract and full text review stages. When there were discrepancies, a consensus was made in consultation with a third reviewer. The protocol for this review was not registered. Non-English studies were included after translation through Google Translate followed by verification of translation by a French speaker.

Quality Assessment

Studies which passed full text review (Table S2) were evaluated for risk of bias using a tool developed by Hoy and colleagues specifically intended for prevalence-studies [18]. Each study was assessed according to ten items assessing internal validity (Table S3) and assigned to have either low (score of 1) or high (score of 0) risk of bias for each question by two independent reviewers. A third reviewer mediated discrepancy and a final score per question was agreed upon. Studies were then classified based on the total score for all questions in the quality assessment tool as having a high (<=5), moderate (6-8), or low (>= 8) risk of bias.

Data Analysis

After reviewing full-text articles, a heat map was constructed with the number of studies, TNBC frequency and number of participants per country across African populations for unique studies using Google Sheets. All meta-analyses, meta-regressions and sensitivity analyses were completed using R (version 4.0.2) [19]. Using the *metaprop* package in R, we conducted a meta-analysis of TNBC frequency among indigenous African women with breast cancer, stratified by country, region, risk of bias assessment, year of publication

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and the use of a validated tool for assessing receptor status. Freeman-Tukey double-arcsine transformation was used to stabilize the variances [20] and a random effects model for our meta-analyses. Pooled TNBC frequency was estimated separately per country and per region as two studies included data from more than one region. When studies investigated African and non-African participants, only data from African participants were included in meta-analyses. Heterogeneity between studies was assessed with Cochran's Q, I^2, and H statistics. Meta-regression was done to explore heterogeneity using *metareg* package. We used Egger's test to investigate publication bias and small study effects using the *metabias* package.

RESULTS

Of the 1808 records identified, 1032 remained after removing duplicates from the various databases. After screening titles and abstracts, 932 records were excluded due to irrelevance. The full text for the remaining 109 records were screened and an additional 42 studies were excluded because they did not meet eligibility criteria, leaving 67 relevant studies for inclusion (Figure 1). Our search strategy identified eligible studies from 20 countries across continental Africa. Nigeria and Tunisia had the highest number of eligible studies (8 studies each), followed by Morocco (7 studies), Algeria, Egypt, and South Africa (6 studies each), Ghana and Kenya (4 studies each) and South Africa (3 studies); there were only two studies from Uganda and only one study each from Botswana, Democratic Republic of Congo, Ethiopia, Guinea, Malawi, Mali, Mozambique, Rwanda, Senegal, and Sudan (Table S2). Five studies included data from multiple countries and regions across the continent (East and West Africa, Algeria, Egypt, Ethiopia, Ghana, Morocco, Namibia, Senegal, South Africa, and Tunisia). Five studies were translated from French to English and were all based in North Africa. Summary of clinical data can be found in Table 1 and Table S2.

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Table 1: Summary of study variables for included studies across Africa and by Region

Study variables	All studies (*n*= 67)	North African studies (*n*= 29)	East African studies (*n*= 10)	Southern African studies (*n*= 8)	Central African study (*n*= 1)	West African studies (*n*= 17)	West and East African studies (*n*= 2)
Mean age	30.62- 60.80 (*n*= 43)	30.62- 58.30 (*n*= 18)	43.00- 54.00 (*n*= 7)	53.60- 60.80 (*n*= 5)	50.00 (*n*= 1)	43.07- 52.05 (*n*= 12)	not investigated
Median age	36.00- 60.00 (*n*= 19)	36.00- 52.00 (*n*= 10)	48.00- 50.00 (*n*= 4)	54.00- 60.00 (*n*=3)	49.00 (*n*=1)	not investigated	56.00 (*n*= 1)
Min-max age recorded	13.00- 105.00 (*n*= 43)	17.00-105.00 (*n*= 26)	13.00-87.00 (*n*= 7)	27.00- 96.00 (*n*= 2)	29.00- 82.00 (*n*= 1)	20.00- 96.00 (*n*= 8)	14.00- 95.00 (*n*= 1)
Grade 1 tumors	0.00%- 60.64% (*n*= 53)	0.00%-22.00% (*n*= 24)	3.33%-25.00% (*n*= 8)	6.03%- 60.64% (*n*= 6)	1.15% (*n*= 1)	1.50%- 27.87% (*n*= 13)	9.56% [W], 15.25% [E] (*n*= 1)
Grade 2 tumors	18.18%- 82.29% (*n*= 54)	26.66%- 82.29% (*n*= 24)	35.00%-51.11% (*n*= 8)	29.79%- 53.22% (*n*= 7)	51.72% (*n*= 1)	18.18%- 63.53% (*n*= 13)	33.52% [W], 30.81% [E] (*n*= 1)
Grade 3 tumors	9.57%- 78.18% (*n*= 59)	13.94%- 73.33% (*n*= 28)	31.25%- 53.51% (*n*= 8)	8.40%- 62.07% (*n*= 7)	47.13% (*n*= 1)	36.83%- 78.18% (*n*= 14)	56.92% [W], 43.95% [E] (*n*= 1)
Positive lymph node status	12.10%- 99.20% (*n*= 42)	21.40%- 95.00% (*n*= 23)	49.00%- 77.19% (*n*= 6)	12.10%- 68.10% (*n*= 3)	not investigated	41.90%- 99.20% (*n*= 10)	not investigated
Premenopausal	29.10%- 73.0% (*n*= 30)	30.00%- 67.50% (*n*= 13)	34.00%- 64.56% (*n*= 5)	37.10% (n= 1)	60.92% (*n*= 1)	29.10%- 73.00% (*n*= 7)	not investigated
TNBC frequency	5.60%- 82.0% (*n*= 67)	6.00%- 66.25% (*n*= 29)	17.00%- 37.70% (*n*=10)	5.60%- 22.50% (*n*= 8)	14.94% (*n*= 1)	25.90%- 82.20% (*n*= 17)	24.10%- 32.80% (*n*= 2)

* two studies investigated both West and East African populations in their respective manuscripts, [W]- West African participants, [E]- East African participants

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All 67 studies reported TNBC frequency from specific hospital/health facility settings, 34 of which were conducted via academic and academic/university teaching hospitals. Fourteen studies were prospective whereas the others (n= 53) were retrospective studies. Additionally, 20 studies included some form of biased sampling (*e.g*., all metastatic cases, tissue microarrays, age cut-offs) whereas the remaining studies included were either random sampling or population based. Of the included studies, eight (12%), thirty-seven (55%) and twenty-two (33%) were classified as low, moderate and high risk of bias, respectively, (Figure S1-S2, Table S3-S4) after using the risk of bias assessment tool for prevalence studies by Hoy *et al.* [18]. Most studies were scored as high risk of bias due to data acquisition (*e.g.,* study population). However, according to criteria set by the Hoy *et al.* risk of bias tool, data were interpreted appropriately for most studies (e.g., having a clear definition of TNBC, appropriate numerator/denominator for frequencies). The eight studies that were low risk of bias, were based in Algeria (*n*= 1), Botswana (*n*= 1), Malawi (*n*= 1), Rwanda (*n*= 1), South Africa (*n*= 2) or multiple countries (*n*= 2).

After identifying unique study populations per country (*n*= 60, Figure 1), TNBC estimates from the meta-analysis, number of studies and participants per country were highlighted (Figure 2). Pooled TNBC frequency estimates per country (Table S5) ranged from 13.5% in South Africa (95% CI: 8.8%-19.1%, I^2 = 78%) to 55.3% in Ghana (95% CI: 45.1%-65.2%, I^2 = 83%). When investigating estimates per region (Figure 3), TNBC frequency was lowest in Central Africa (n=1, 14.9%; 95% CI 8.1%-23.3%) and highest in West Africa (n=15, 46.0%; 95%CI: 38.8%-29.3%, I^2 = 92%). For these two analyses with pooled estimates per country and per region, heterogeneity (I^2) was estimated at 94%, indicative of high between study variability. Pooled TNBC estimates were also stratified by use of validated tool for receptor status testing. Pooled TNBC frequency was higher (n= 25, 30.5%; 95%CI: 26.3%-34.9%, I^2 = 95%) in studies that reported the use of a validated tool for assessing receptor expression, when compared to studies that did not (n= 35, 24.8%; 95%CI: 21.9%-27.9%, I^2 = 93%) (Figure S4). Between study heterogeneity was high (I^2= 94%) and meta-regression showed that these estimates were different (Figure S5A, P = 0.032, β coefficient= 0.061). When investigating TNBC estimates by the tool used for ER/PR and HER2 cutoffs,

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pooled TNBC frequency was higher in studies that used Allred 1998 for ER/PR cutoffs and in older versions of ASCO/CAP guidelines when compared to more recent versions for HER2 status (Figure 4). There was also an association between publication year and TNBC estimates where meta-regression showed a decrease in effect size with increasing publication year (Figure S5B, $P < 0.001$, β coefficient= -0.017). When investigating the effect of risk of bias on study estimates, none was observed. Influence analyses showed that heterogeneity was not largely impacted by single studies. However, after conducting Egger's test, publication bias was identified (Figure S6, $P < 0.001$).

Discussion

In this systematic review and meta-analysis of 67 studies on African women with BCa, we found that there was a high frequency of TNBC (27.1%) in cases reported across Africa although it varied depending on country and region. TNBC frequency was highest in West African populations (46.0%) compared to other regions across continental Africa (14.9%-22.9%). This is consistent with increased TNBC/ER-negative prevalence observed in populations with high West African ancestry [14, 21, 22] in the Caribbean [23, 24] and in North America [25]. In a USA population-based study (2010-2014), TNBC prevalence in non-Hispanic White women was estimated to be ~ 8% whereas it was ~15% in non-Hispanic Black women [22]. Additionally, TNBC prevalence was estimated to be ~8% and ~25% in White and Black women respectively in a UK cancer registry-based population in London [26]. These high frequencies of TNBC across Africa and the African diaspora are concerning as triple negative breast tumors have a greater propensity to metastasize to vital organs such as the brain [27] and are typically more aggressive due to lack of targeted therapies.

When investigating clinical factors, the reported mean and/or median age at diagnosis was under 50 in 35 out of the 47 studies reporting age. Young age at diagnosis (under age 40) has been previously reported to be associated with triple negative and HER2-positive cancers as well as more aggressive clinical outcome [28]. Indeed, this poor prognosis of TNBC patients was evident as most of the included studies reported a

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high percentage of grade 3 tumours, lymph node positivity and TNBC frequency. It must however be noted that a younger age at diagnosis is also routinely observed in lower- and middle-income countries as this is also reflective of the population structure [29]. Therefore, this observed lower age at diagnosis may be indicative of the population distribution rather than the intrinsic aggressive biology of the tumors. To consider this possibility, we investigated associations with mean and median age at diagnosis and TNBC estimates and found no association (Figure S5C p= 0.230, β coefficient= -0.004, Figure S5D p= 0.320, β coefficient= -0.008). More advanced stage tumors and lymph node involvement at presentation may also be attributable to poor infrastructure and lack of BCa awareness and screening. A recent study of BCa across sub-Saharan Africa found that the majority of cases diagnosed were late-stage, emphasizing the need for early diagnosis [30]. Two separate studies from Nigeria and Ghana both found that most of the information obtained about BCa was from mass media and there was a general poor knowledge of BCa-associated risk factors [31, 32]. The Ghanaian-based study also found that the rate of breast self-examination (BSE), and clinical breast examination (CBE) were higher than that of obtaining mammograms [32] which emphasizes the need to promote screening programs in a culturally-relevant setting. It should be noted however that mammography has been associated with a two times higher chance of detecting ER-positive BCa compared to ER-negative BCa [33] which might be contributing to the relatively higher TNBC frequency observed across West African countries when compared to Southern African countries where mammography is more accessible.

Many studies were excluded on the account of not assessing ER, PR and HER2 status. With respect to receptor status, 30 out of the 67 studies used validated guidelines (ASCO/CAP [34-39], Allred [40] or Reiner's Scale [41]) for receptor status cut-offs. Such variability in classifying receptor status (i.e., the use of other guidelines with different cut-offs) affects the resulting treatment for patients with a diagnosis and how TNBC frequencies are calculated in each study. After stratifying studies by use of ASCO-CAP

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guidelines which account for specimen fixation and cut-offs for ER/PR expression at 1%, there was a decrease in pooled TNBC frequency (24.4%) compared to those that used Allred or Reiner's Scale (cutoff at 10%) resulting in TNBC frequency of 43.5% and 38.0% respectively (Figure 4A). A similar trend was observed for ASCO/CAP guidelines with respect to assessing HER2 status where more recent guidelines correlated with lower TNBC frequency compared to older guidelines (Figure 4B). Thus, meta-regression with publication year was done and indeed there was an association ($P < 0.001$, β coefficient= -0.017, Figure S5B). A similar trend was recently reported for East Africa- based studies conducted before and after 2013; ER/PR positivity was lower before 2013 compared to after 2013 [42]. The variability in how receptor status is assessed highlights a need for increased capacity to conduct immunohistochemical receptor status testing to further enhance BCa diagnoses and classification. It must be noted however, that IHC might not be feasible for many of the hospitals/health centers across Africa and international collaborations should be encouraged to assist with building such capacity. One Nigerian study noted cost to be a barrier – IHC was performed on only 31% of the reported cases [43]. It also stated that ASCO/CAP guidelines could not be adhered to for HER2 due to the high cost of fluorescence *in situ* hybridization in the case of an equivocal HER2 score. In contrast, South Africa has an extensive healthcare system with a comprehensive standardized national public health system for routinely assessing breast cancer receptor status [44]. The disparity in access to diagnostic and therapeutic tools across the continent could also be contributing to the lack of receptor status data and higher BCa burden reported here.

To our knowledge, this is the first systematic review and meta-analysis with an in-depth analysis on TNBC frequency across continental Africa. However, there are some limitations to be considered. IHC and specimen collection and processing are not equally accessible, and neither are they uniformly done across the African continent. Specimen fixation, storage time of the samples and other pre-analytical IHC variables have been previously shown to impact accuracy of IHC results [45]. It was estimated that up to 20% of IHC

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results globally are inaccurate based off of these pre-analytical variables [46]. The true frequency of TNBC is therefore not ascertainable due to lack of population-based data. Additionally, there are no validated search strategies for observational studies thus, some studies may be missed. As expected, there was considerable heterogeneity. High heterogeneity could be due to the large number of studies, or the varying ethnicities across Africa. However, given the available data, this is the best estimate of TNBC frequency.

This study provides the closest estimate of TNBC frequency across the different regions of continental Africa. Considerations should be made at the country-level to address IHC protocols and adherence to ASCO-CAP guidelines wherever possible. There is a clear disparity across the continent with respect to diagnostic and therapeutic tools that needs to be effectively addressed to prevent BCa burden. Priority should also be given to implementing culturally relevant BCa awareness programs as these have been proven to increase cancer awareness knowledge and thus could decrease preventable deaths from BCa [47]. There is also a dearth of knowledge across the continent about BCa subtype prevalence in general. This should be addressed as soon as possible by the establishment of cancer registries before the burden of BCa and other chronic diseases drastically increase with the epidemiological transition that has already started to take place across Africa.

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Data sharing statement:
The data underlying this article are available in the article and in its online supplementary material.

Ethical approval statement:
Not necessary for this manuscript.

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Figure legends:

Figure 1: PRISMA flowchart accounting for all articles included in the narrative synthesis and meta-analysis.

Figure 2: Pooled TNBC frequencies out of all BCa subtypes from studies done across African countries. Data represent pooled TNBC frequencies among all BCa subtypes reported in unique studies done in populations from stated countries. Pooled frequencies were calculated if the country had more than one study as stated within the meta-analysis. Estimates do not account for heterogeneity, IHC cut-offs and size of the respective populations.

Figure 3: Pooled TNBC frequency in Africa by region. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

Figure 4: Pooled TNBC frequency in Africa by tool used for (A) ER/PR status (B) HER2 status. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

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Figure 1: PRISMA flowchart accounting for all articles included in the narrative synthesis and meta-analysis

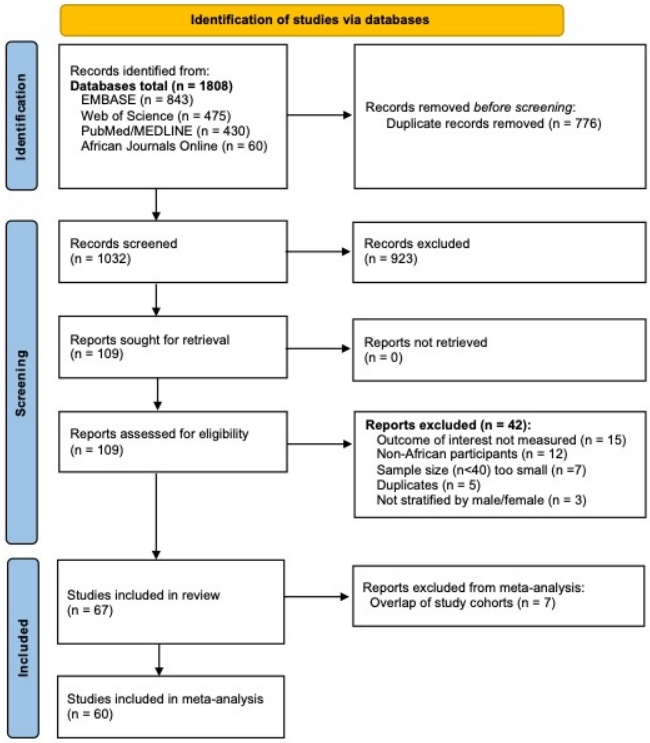


Figure 1: PRISMA flowchart accounting for all articles included in the narrative synthesis and meta-analysis.

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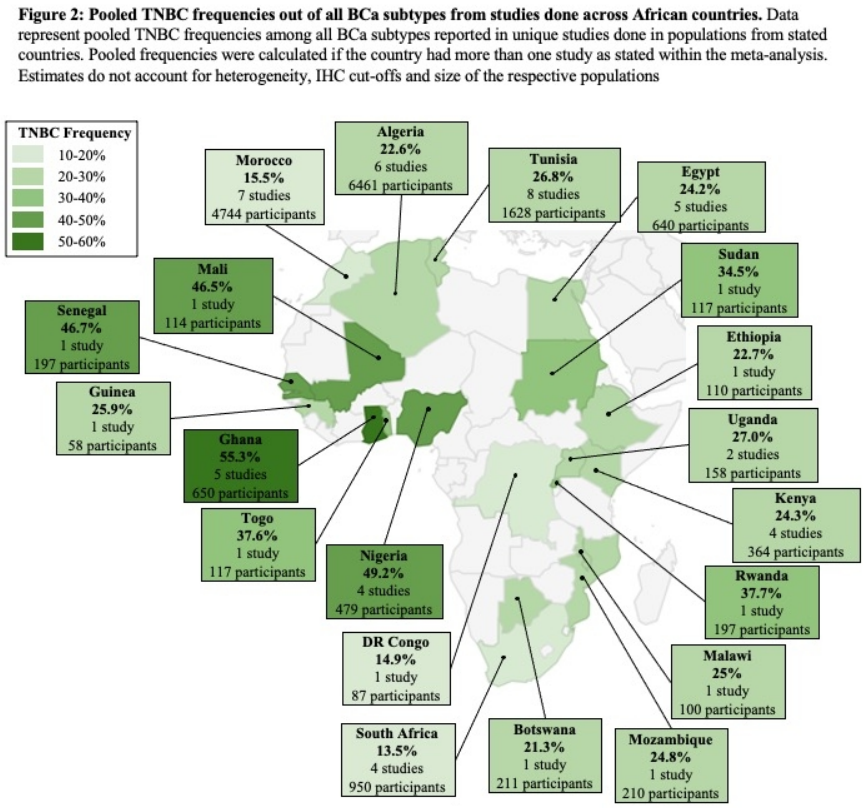


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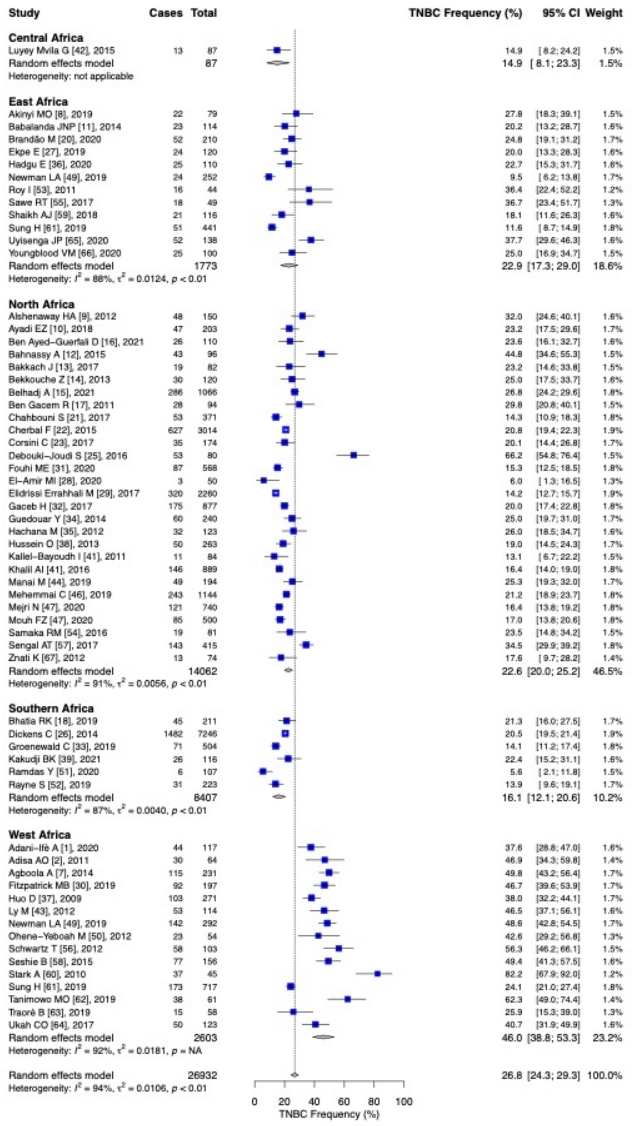


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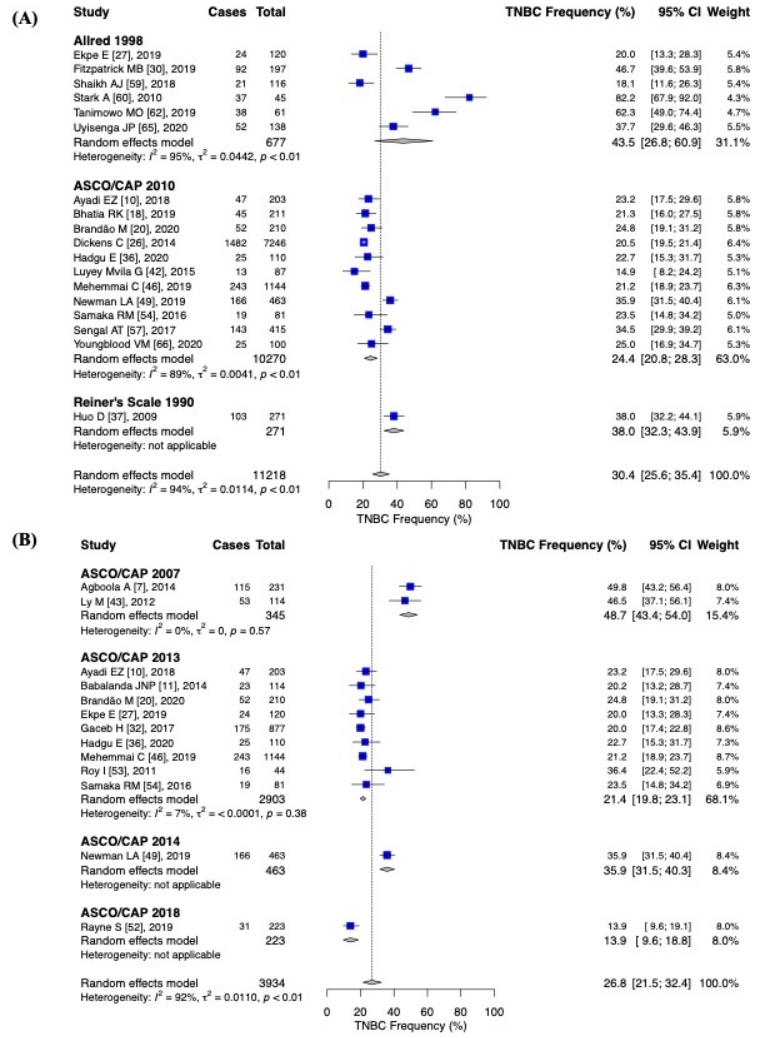


Figure 4: Pooled TNBC frequency in Africa by tool used for (A) ER/PR status (B) HER2 status. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

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SUPPORTING DOCUMENTS

Table S1: Search strategy used for Databases

Database	Search terms
Pubmed/MEDLINE	(Africa OR African OR "West* Africa" OR "East* Africa" OR "South* Africa" OR "North* Africa" OR "Central Africa" OR Algeria OR Angola OR Benin OR Botswana OR "Burkina Faso" OR Burundi OR Cabo Verde OR Cameroon OR Central African Republic OR "CAR" OR Chad OR Comoros OR "DRC" OR Congo OR "Republic of Congo" OR "Ivory Coast" OR Cote d'Ivoire OR Djibouti OR Egypt OR "Equatorial Guinea" OR Eritrea OR Eswatini OR Swaziland OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR "Guinea-Bissau" OR Kenya OR Lesotho OR Liberia OR Libya OR Madagascar [PL] OR Malawi OR Mali OR Mauritania OR Mauritanie OR Mauritius OR Morocco OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR "Sao Tome and Principe" OR Senegal OR Seychelles OR "Sierra Leone" OR Somalia OR "South Africa" OR "South Sudan" OR Sudan OR Tanzania OR Togo OR Tunisia OR Uganda OR Zambia OR Zimbabwe) AND (TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND (rate* OR prevalence OR incidence OR epidemiology)
Web of Science	TS = (Africa or African or "West* Africa" or "East* Africa" or "South* Africa" or "North* Africa" or "Central Africa" or Algeria or Angola OR Benin OR Botswana OR "Burkina Faso" OR Burundi OR Cabo Verde OR Cameroon OR Central African Republic OR "CAR" OR Chad OR Comoros OR "DRC" OR Congo OR "Republic of Congo" OR "Ivory Coast" OR Cote d'Ivoire OR Djibouti OR Egypt OR "Equatorial Guinea" OR Eritrea OR Eswatini OR Swaziland OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR "Guinea-Bissau" OR Kenya OR Lesotho OR Liberia OR Libya OR Madagascar [PL] OR Malawi OR Mali OR Mauritania OR Mauritanie OR Mauritius OR Morocco OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR "Sao Tome and Principe" OR Senegal OR Seychelles OR "Sierra Leone" OR Somalia OR "South Africa" OR "South Sudan" OR Sudan OR Tanzania OR Togo OR Tunisia OR Uganda OR Zambia OR Zimbabwe) AND TS= (TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND TS= (rate* OR prevalence OR incidence OR epidemiology)
Embase	(Africa OR African OR "West* Africa" OR "East* Africa" OR "South* Africa" OR "North* Africa" OR "Central Africa" OR Algeria OR Angola OR Benin OR Botswana OR "Burkina Faso" OR Burundi OR Cabo Verde OR Cameroon OR Central African Republic OR "CAR" OR Chad OR Comoros OR "DRC" OR Congo OR "Republic of Congo" OR "Ivory Coast" OR Cote d'Ivoire OR Djibouti OR Egypt OR "Equatorial Guinea" OR Eritrea OR Eswatini OR Swaziland OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR "Guinea-Bissau" OR Kenya OR Lesotho OR Liberia OR Libya OR Madagascar OR Malawi OR Mali OR Mauritania OR Mauritanie OR Mauritius OR Morocco OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR "Sao Tome and Principe" OR Senegal OR Seychelles OR "Sierra Leone" OR Somalia OR "South Africa" OR "South Sudan" OR Sudan OR Tanzania OR Togo OR Tunisia OR Uganda OR Zambia OR Zimbabwe) AND (TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND (rate* OR prevalence OR incidence OR epidemiology)
African Journals Online (AJOL)	(TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND (rate* OR prevalence OR incidence OR epidemiology)

Table S2: Summary of data from included studies

Author	Population	Region	Mean age ± SD	Median age (min-max)	Grade 3 tumors	Positive Nodes	Number of TNBC cases	Participants studied with known receptor status (all participants)	Frequency of TNBC	Risk of Bias Assessment
Adani-Ifè A, 2020 [1]	Togo	West Africa	52.05 ±12.38	(30-85)	42.70%	41.90%	44	117 (312)	37.60%	Moderate
Adisa AO, 2011 [2]	Nigeria	West Africa	45.90	(20-81)	/	/	30	64 (385)	46.88%	High
Agboola A, 2014 [3]	Nigeria	West Africa	/	/	35.70%	99.20%	99	155 (199)	54.36%	High
Agboola A, 2014 [4]	Nigeria	West Africa	/	/	/	94.20%	55	100 (308)	55.00%	Moderate
Agboola A, 2013 [5]	Nigeria	West Africa	/	/	/	92.70%	81	149 (308)	63.87%	High
Agboola A, 2016 [6]	Nigeria	West Africa	/	/	34.12%	91.76%	114	187 (255)	60.96%	Moderate
Agboola A, 2014 [7]	Nigeria	West Africa	/	/	38.50%	92.20%	115	231 (231)	49.78%	High
Akinyi MO, 2019 [8]	Kenya	East Africa	48.00 ±14.5	/	/	71%	22	79 (79)	27.85%	High
Alshenawy HA, 2012 [9]	Egypt	North Africa	58.30 ±10.1	/	20.00%	/	48	150 (150)	26.67%	Moderate
Ayadi EZ, 2018 [10]	Tunisia	North Africa	35.00 ±4.2	(17-40)	46.85%	64.10%	47	203 (238)	23.15%	Moderate
Babalanda JNP, 2014 [11]	Uganda	East Africa	47	(13-87)	53.51%	77.19%	23	114 (114)	20.18%	Moderate
Bahnassy A, 2015 [12]	Egypt	North Africa	TNBC: 51.91± 12.34 non-TNBC: 52.77 ± 12.13 Controls: 35 ± 13.94	TNBC: (30-78) non-TNBC: (27-81) Controls: (22-64)	15.63%	72.91%	43	96 (96)	44.79%	Moderate
Bakkach J, 2017 [13]	Morocco	North Africa	/	36 (/)	40.20%	58.50%	19	82 (331)	23.00%	High
Bekkouche Z, 2013 [14]	Algeria	North Africa	49.72	(30-84)	73.33%	35.83%	30	120 (120)	25.00%	High
Belhadj A, 2021 [15]	Algeria	North Africa	/	50 (21-88)	38.86%	64.70%	286	1066 (1140)	27.00%	Moderate
Ben Ayed-Guerfali D, 2021 [16]	Tunisia	North Africa	/	42.78 (27-65)	26.78%	59.70%	26	110 (134)	23.60%	High
Ben Gacem R, 2011 [17]	Tunisia	North Africa	48.70	47 (31-75)	43.62%	55.26%	28	94 (94)	29.79%	Moderate
Bhatia RK, 2019 [18]	Botswana	Southern Africa	/	54 (27-96)	25.76%	/	45	211 (384)	21.30%	Low

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Bogan D, 2017 [19]	Egypt	North Africa	52.38 ±12.17	(27-81)	13.94%	66.67%	43	96 (96)	44.79%	Moderate
Brandão M, 2020 [20]	Mozambique	East Africa	/	48 (/)	31.25%	61.43%	52	210 (262)	25.00%	Moderate
Chahbouni S, 2017 [21]	Morocco	North Africa	48.00	(22-99)	/	/	53	371 (390)	13.59%	High
Cherbal F, 2015 [22]	Algeria	North Africa	/	48.5 (22-84)	31.30%	/	627	3014 (3014)	20.80%	Moderate
Corsini C, 2017 [23]	Multiple (France, Algeria, Egypt, Morocco, Tunisia)	North Africa	/	/	44.74%	42%	35	174 (456)	20.10%	Moderate
Cubasch H, 2018 [24]	South Africa	Southern Africa	54.40 ±14.2	/	39.53%	/	126	581 (602)	21.70%	Low
Debouki-Joudi S, 2016 [25]	Tunisia	North Africa	49.30	(24-80)	26.22%	55.20%	53	80 (91)	66.25%	High
Dickens C, 2014 [26]	Multiple (South Africa & Namibia)	Southern Africa	/	/	33.87%	/	1482	7246 (12361)	20.45%	Low
Ekpe E, 2019 [27]	Kenya	East Africa	47.00 ±12	(21-79)	50.00%	/	24	120 (125)	20.00%	Moderate
El-Amir MI, 2020 [28]	Egypt	North Africa	54.20 ±10.6	(32-77)	18.00%	72%	3	50 (50)	6.00%	High
Elidrissi Errahhali M, 2017 [29]	Morocco	North Africa	48.70	(19-105)	18.00%	64.80%	320	2260 (2406)	14.20%	Moderate
Fitzpatrick MB, 2019 [30]	Senegal	West Africa	47.00 ±12.7	/	76.17%	/	92	197 (522)	46.70%	Moderate
Fouhi ME, 2020 [31]	Morocco	North Africa	51.60	(23-89)	32.60%	35.70%	87	568 (668)	15.30%	High
Gaceb H, 2017 [32]	Algeria	North Africa	48.44	(22-84)	28.50%	/	175	877 (877)	19.95%	Low
Groenewald C, 2019 [33]	South Africa	Southern Africa	53.60 ±13.4	/	/	68.10%	71	504 (505)	14.10%	Moderate
Guedouar Y, 2014 [34]	Algeria	North Africa	47.00	(23-78)	24.58%	38.33%	60	240 (240)	25.00%	High
Hachana M, 2012 [35]	Tunisia	North Africa	/	/	40.18%	21.40%	32	123 (123)	26.02%	Moderate
Hadgu E, 2020 [36]	Ethiopia	East Africa	43.00 ±14	50 (22-75)	50.00%	/	25	110 (112)	23.00%	Moderate
Huo D, 2009 [37]	Multiple (Nigeria & Senegal)	West Africa	44.80	/	39.15%	72%	103	271 (378)	27.00%	Low
Hussein O, 2013 [38]	Egypt	North Africa	53.00 ±11.6	52	17.39%	/	50	263 (263)	19.00%	High

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Kakudji BK, 2021 [39]	South Africa	Southern Africa	56.35 ±14.09	/	62.07%	64.66%	26	116 (116)	22.50%	High
Kallel-Bayoudh I, 2011 [40]	Tunisia	North Africa	50.00 ±11	(27-75)	35.44%	65%	11	84 (84)	13.00%	High
Khalil AI, 2016 [41]	Morocco	North Africa	50.20 ±11.34	(17-93)	35.48%	49.40%	146	889 (1277)	16.40%	Moderate
Luyey Mvila G, 2015 [42]	Democratic Republic of Congo	Central Africa	50.00	49 (29-82)	47.13%	/	13	87 (87)	14.94%	Moderate
Ly M, 2012 [43]	Mali	West Africa	46.00	(25-82)	78.18%	91%	53	114 (114)	46.00%	Moderate
Manai M, 2019 [44]	Tunisia	North Africa	/	42 (24-62)	42.20%	95%	49	194 (210)	25.00%	Moderate
McCormack VA, 2013 [45]	South Africa	Southern Africa	55.30	/	42.34%	/	209	1027 (1247)	20.40%	Low
Mehemmai C, 2019 [46]	Algeria	North Africa	/	48.5 (21-84)	26.99%	/	243	1144 (1162)	21.24%	Moderate
Mejri N, 2020 [47]	Tunisia	North Africa	/	48.5 IBCa 49 non-IBCa (/)	46.20% IBCa 36.10% non-IBCa	50%	121	740 (740)	16.40%	High
Mouh FZ, 2020 [48]	Morocco	North Africa	/	47 (/)	34.92%	60.10%	85	500 (500)	17.00%	High
Newman LA, 2019 [49]	Multiple (Ghana, Ethiopia, United States)	West and East Africa	/	56 (14-95)	/	/	Ghanaian: 142 Ethiopian: 24	Ghana: 292 (618) Ethiopia: 144 (252)	Ghana: 49.00% Ethiopia: 17.00%	High
Ohene-Yeboah M, 2012 [50]	Ghana	West Africa	49.10 ±15.3	(20-96)	53.70%	/	23	54 (330)	42.70%	Moderate
Ramdas Y, 2020 [51]	South Africa	Southern Africa	60.80 ±9.3	60 (43-86)	9.57%	12.10%	6	107 (107)	5.60%	High
Rayne S, 2019 [52]	South Africa	Southern Africa	/	56	51.10%	/	31	223 (365)	14.00%	Moderate
Roy I, 2011 [53]	Uganda	East Africa	/	/	40.00%	/	16	44 (64)	36.00%	Moderate
Samaka RM, 2016 [54]	Egypt	North Africa	/	(30-81)	44.44%	77.80%	19	81 (81)	23.50%	Moderate
Sawe RT, 2017 [55]	Kenya	East Africa	51.90	48.5 (16-84)	42.22%	/	18	49 (49)	36.70%	Moderate
Schwartz T, 2012 [56]	Ghana	West Africa	49.00 ±13.4	/	52.52%	/	58	103 (104)	56.30%	Moderate
Sengal AT, 2017 [57]	Sudan	North Africa	48.80 ±13.5	(20-90)	41.79%	52.70%	143	415 (560)	34.50%	Moderate
Seshie B, 2015 [58]	Ghana	West Africa	52.50 ±12.1	(24-77)	31.58%	66.00%	77	156 (165)	49.40%	Moderate
Shaikh AJ, 2018 [59]	Kenya	East Africa	54.00	(27-83)	33.33%	49%	21	116 (123)	17.00%	Moderate
Stark A, 2010 [60]	Ghana	West Africa	48.00 ±13.7	/	76.00%	/	37	45 (75)	82.20%	Moderate

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Sung H, 2019 [61]	Multiple (USA, Caribbean, West Africa, East Africa)	West and East Africa	/	/	West Africa: 56.92% East Africa: 43.95%	/	West Africa: 173 East Africa: 51	West Africa: 717 (811) East Africa: 441 (484)	West Africa: 24.10% East Africa: 11.60%	Moderate
Tanimowo MO, 2019 [62]	Nigeria	West Africa	43.07 ±11.19	(23-70)	29.51%	/	38	61 (61)	62.30%	High
Traoré B, 2019 [63]	Guinea	West Africa	48.20 ±10.9	(31-80)	36.21%	62.10%	15	58 (58)	25.90%	Moderate
Ukah CO, 2017 [64]	Nigeria	West Africa	46.30 ±12.3	(24-85)	26.83%	/	50	123 (397)	40.70%	Moderate
Uyisenga JP, 2020 [65]	Rwanda	East Africa	49.70 ±12.98	(17-86)	/	70.06%	52	138 (138)	37.70%	Low
Youngblood VM, 2020 [66]	Malawi	East Africa	/	49 (21-80)	37.00%	77%	25	100 (100)	25.00%	Low
Znati K, 2012 [67]	Morocco	North Africa	30.62	(18-35)	45.07%	51%	13	74 (74)	18.00%	High

/ data not stated

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Table S3: Questions for Risk of Bias Assessment, modified from Hoy *et al*., 2012

1. Was the study's target population a close representation of the national population in relation to relevant variables, e.g. age, sex, occupation?
2. Was the sampling frame a true or close representation of the target population?
3. Was some form of random selection used to select the sample, or was a census undertaken?
4. Was the likelihood of non-response bias minimal?
5. Were data collected directly from the subjects (as opposed to a proxy)?
6. Was an acceptable case definition used in the study? Was TNBC defined as ER, PR, HER2 negative?
7. Was the study instrument that measured the parameter of interest (e.g. appropriate guidelines for cut-offs ASCO-CAP) shown to have reliability and validity?
8. Was the same mode of data collection used for all subjects? Was the information from each patient collected in the same way (e.g clinical referral appointment)?
9. Was there a shortest prevalence or incidence period identified?
10.Were the numerator(s) and denominator(s) for the parameter of interest appropriate?

Table S4: Raw Values for Risk of Bias Assessment for Included Studies

Author	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	TOTAL	Overall ROB
Adani-Ifè A, 2020 [1]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
Adisa AO, 2011 [2]	0	0	1	0	1	0	0	0	1	0	3	**High**
Agboola A, 2014 [3]	0	0	0	0	1	1	1	1	1	0	5	**High**
Agboola A, 2014 [4]	0	0	1	0	1	1	1	1	1	1	7	**Moderate**
Agboola A, 2013 [5]	0	0	0	0	1	1	1	1	0	1	5	**High**
Agboola A, 2016 [6]	0	0	0	1	1	0	1	1	1	1	6	**Moderate**
Agboola A, 2014 [7]	0	0	0	0	1	0	1	1	1	1	5	**High**
Akinyi MO, 2019 [8]	0	0	0	1	1	1	0	1	0	0	4	**High**
Alshenawy HA, 2012 [9]	0	0	1	1	1	1	0	1	1	1	7	**Moderate**
Ayadi EZ, 2018 [10]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Babalanda JNP, 2014 [11]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Bahnassy A, 2015 [12]	0	0	1	1	1	1	0	1	1	1	7	**Moderate**
Bakkach J, 2017 [13]	0	0	0	1	1	0	0	1	1	0	4	**High**
Bekkouche Z, 2013 [14]	0	0	1	0	1	1	0	1	1	0	5	**High**

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Belhadj A, 2021 [15]	1	0	1	1	1	1	0	1	1	1	8	**Moderate**
Ben Ayed-Guerfali D, 2021 [16]	0	0	0	0	1	0	0	1	1	0	3	**High**
Ben Gacem R, 2011 [17]	0	0	1	1	1	1	0	1	0	1	6	**Moderate**
Bhatia RK, 2019 [18]	1	1	1	1	1	1	1	1	1	1	10	**Low**
Bogan D, 2017 [19]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Brandão M, 2020 [20]	0	0	1	1	1	1	1	0	1	1	7	**Moderate**
Chahbouni S, 2017 [21]	0	0	0	0	1	1	0	1	1	0	4	**High**
Cherbal F, 2015 [22]	1	1	1	0	1	1	0	1	1	1	8	**Moderate**
Corsini C, 2017 [23]	0	0	1	1	1	0	0	1	1	1	6	**Moderate**
Cubasch H, 2018 [24]	1	1	1	1	1	1	0	1	1	1	9	**Low**
Debouki-Joudi S, 2016 [25]	0	0	0	0	1	1	0	1	0	1	4	**High**
Dickens C, 2014 [26]	1	1	1	1	1	1	1	1	1	1	10	**Low**
Ekpe E, 2019 [27]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
El-Amir MI, 2020 [28]	0	0	0	0	1	0	0	1	1	1	4	**High**
Elidrissi Errahhali M, 2017 [29]	0	1	1	0	1	1	0	1	1	1	7	**Moderate**
Fitzpatrick MB, 2019 [30]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
Fouhi ME, 2020 [31]	0	1	0	0	1	0	0	1	1	0	4	**High**
Gaceb H, 2017 [32]	1	1	1	0	1	1	1	1	1	1	9	**Low**
Groenewald C, 2019 [33]	0	0	0	1	1	1	0	1	1	1	6	**Moderate**
Guedouar Y, 2014 [34]	0	1	0	0	1	1	0	1	1	0	5	**High**
Hachana M, 2012 [35]	0	0	1	1	1	1	0	1	1	1	7	**Moderate**
Hadgu E, 2020 [36]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Huo D, 2009 [37]	0	1	1	1	1	1	1	1	1	1	9	**Low**
Hussein O, 2013 [38]	0	0	1	0	1	0	0	1	1	1	5	**High**
Kakudji BK, 2021 [39]	0	0	0	1	1	1	0	0	1	1	5	**High**
Kallel-Bayoudh I, 2011 [40]	0	0	0	0	1	1	0	1	1	1	5	**High**
Khalil AI, 2016 [41]	0	1	1	0	1	1	0	1	1	1	7	**Moderate**
Luyey Mvila G, 2015 [42]	0	0	1	0	1	1	1	1	1	1	7	**Moderate**
Ly M, 2012 [43]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Manai M, 2019 [44]	0	1	1	0	1	1	0	1	1	1	7	**Moderate**
McCormack VA, 2013 [45]	1	1	1	1	1	1	0	1	1	1	9	**Low**
Mehemmai C, 2019 [46]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Mejri N, 2020 [47]	0	0	0	0	1	0	0	1	1	0	3	**High**
Mouh FZ, 2020 [48]	0	0	0	0	1	1	0	1	1	1	5	**High**
Newman LA, 2019 [49]	0	0	0	0	1	0	1	1	1	1	5	**High**
Ohene-Yeboah M, 2012 [50]	0	0	1	0	1	1	0	1	1	1	6	**Moderate**

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Ramdas Y, 2020 [51]	0	0	0	1	1	0	0	1	1	1	5	**High**
Rayne S, 2019 [52]	0	0	1	0	1	1	1	0	1	1	6	**Moderate**
Roy I, 2011 [53]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
Samaka RM, 2016 [54]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Sawe RT, 2017 [55]	0	1	1	0	1	1	0	1	1	0	6	**Moderate**
Schwartz T, 2012 [56]	0	0	0	1	1	1	0	1	1	1	6	**Moderate**
Sengal AT, 2017 [57]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Seshie B, 2015 [58]	0	1	1	1	1	1	0	1	1	0	7	**Moderate**
Shaikh AJ, 2018 [59]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Stark A, 2010 [60]	0	0	1	0	1	1	1	1	1	0	6	**Moderate**
Sung H, 2019 [61]	0	1	1	1	1	1	0	1	1	1	8	**Moderate**
Tanimowo MO, 2019 [62]	0	0	0	1	1	0	1	0	1	1	5	**High**
Traoré B, 2019 [63]	0	1	1	0	1	1	0	1	1	0	6	**Moderate**
Ukah CO, 2017 [64]	0	1	0	1	1	1	0	1	1	1	7	**Moderate**
Uyisenga JP, 2020 [65]	0	1	1	1	1	1	1	1	1	1	9	**Low**
Youngblood VM, 2020 [66]	0	1	1	1	1	1	1	1	1	1	9	**Low**
Znati K, 2012 [67]	0	0	0	0	1	0	0	1	1	0	3	**High**

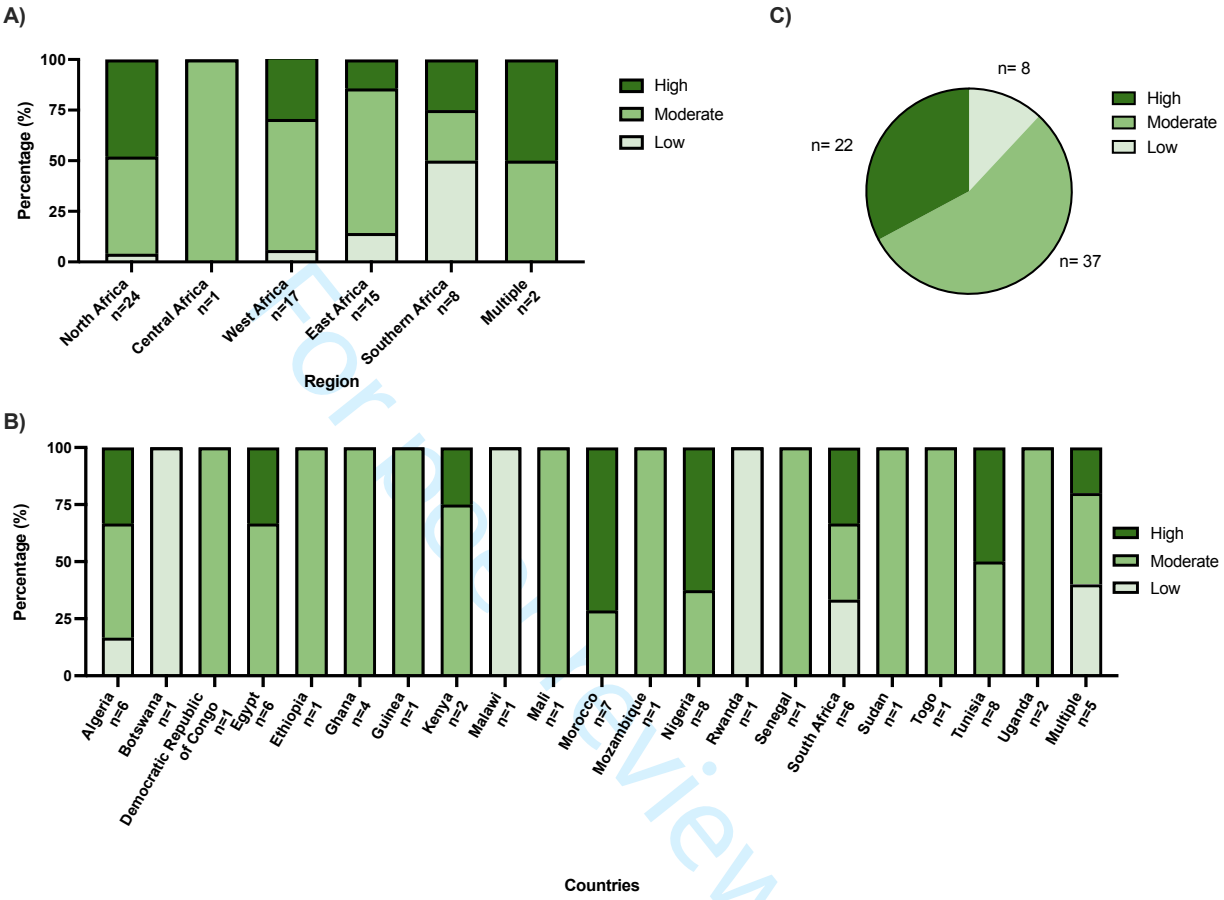
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Table S5: Pooled TNBC Frequcny in Africa by Country

Country (no. of studies)	Total participants	Pooled TNBC Frequency, % [95% CI]	Heterogeneity (I^2), %	Weight, %
Algeria (*n*= 6)	6461	22.6 [20.3- 25.1]	76	12.0
Botswana (*n*= 1)	211	21.3 [16.0- 27.1]	not applicable	1.9
Democratic Republic of Congo (*n*= 1)	87	14.9 [8.1- 23.3]	not applicable	1.7
Egypt (*n*= 5)	640	24.2 [14.0- 36.1]	90	8.8
Ethiopia (*n*= 1)	110	22.7 [15.3- 31.1]	not applicable	1.8
Ghana (*n*= 4)	358	57.6 [43.1- 71.4]	86	6.6
Guinea (*n*= 1)	58	25.9 [15.3- 38.0]	not applicable	1.6
Kenya (*n*= 4)	364	24.3 [17.3- 32.0]	62	6.8
Malawi (*n*= 1)	100	25.0 [17.0- 34.0]	not applicable	1.8
Mali (*n*= 1)	114	46.5 [37.4- 55.7]	not applicable	1.8
Morocco (*n*= 7)	4744	15.5 [14.1- 16.9]	30	13.5
Mozambique (*n*= 1)	210	24.8 [19.1- 30.8]	not applicable	1.9
Nigeria (*n* = 4)	479	49.2 [41.4- 57.1]	62	6.9
Rwanda (*n*= 1)	138	37.7 [29.8- 45.9]	not applicable	1.8
Senegal (*n*= 1)	197	46.7 [39.8- 53.7]	not applicable	1.9
South Africa (*n*= 4)	950	13.5 [8.8- 19.1]	78	7.5
Sudan (*n*= 1)	415	34.5 [30.0- 39.1]	not applicable	2.0
Togo (*n*= 1)	117	37.6 [29.0- 46.6]	not applicable	1.8
Tunisia (*n*= 8)	1628	26.8 [18.6- 35.9]	93	14.6
Uganda (*n*= 2)	158	27.0 [12.7- 44.1]	76	3.2
Total	17539	27.4 [24.5- 30.3]	94	100

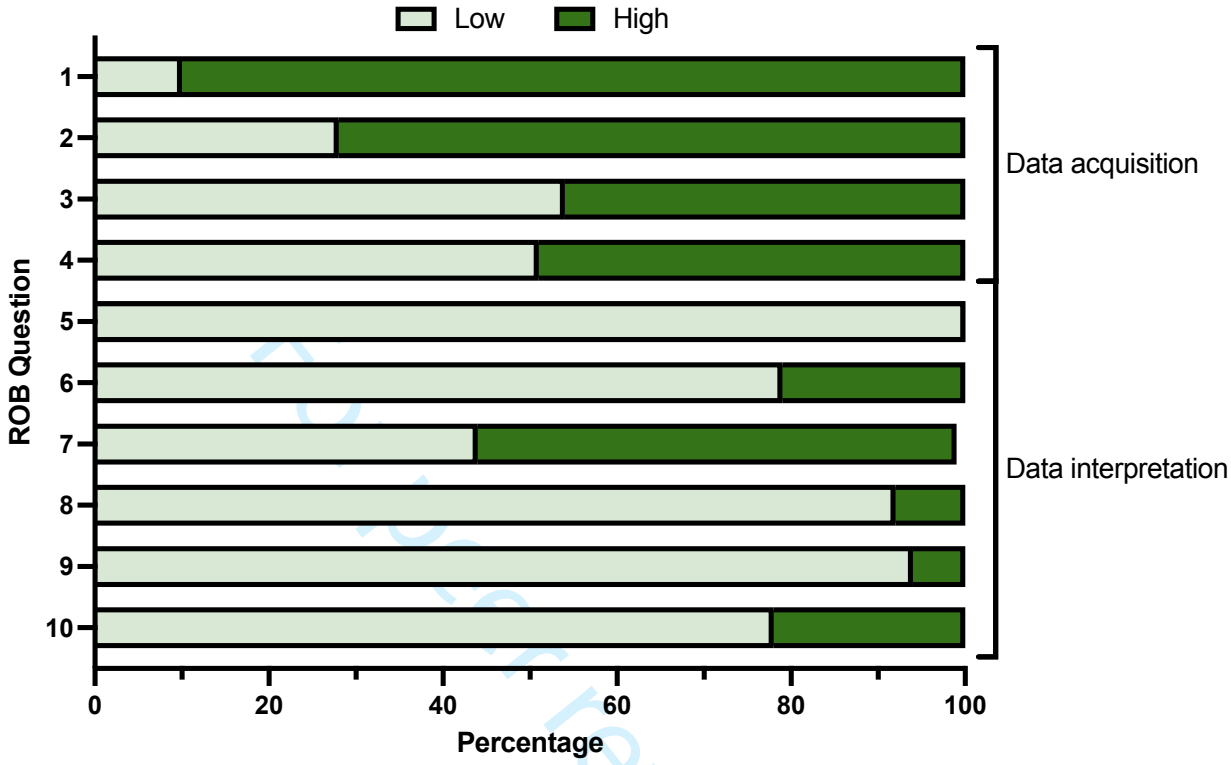
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Figure S1: Risk of bias assessment of all articles by (A) region, (B) country and (C) overall risk of bias assessment for all articles included. Southern Africa had the highest percentage of low risk of bias. Multiple denotes more than one country and/or region or was investigated within that study.



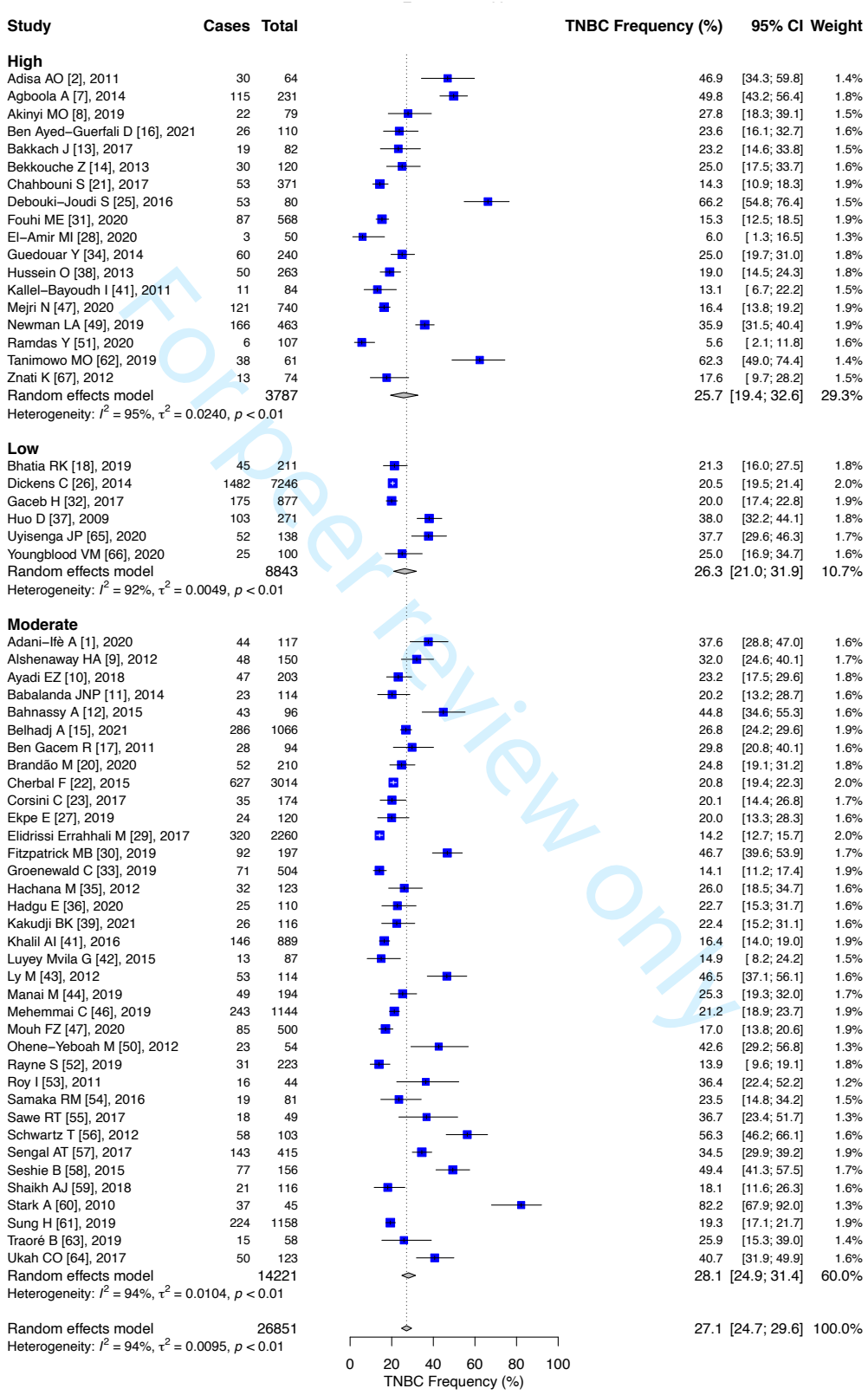
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Figure S2: Risk of bias (ROB) assessment per question for all articles included in systematic review. High risk of bias was noted specifically for questions on data acquisition



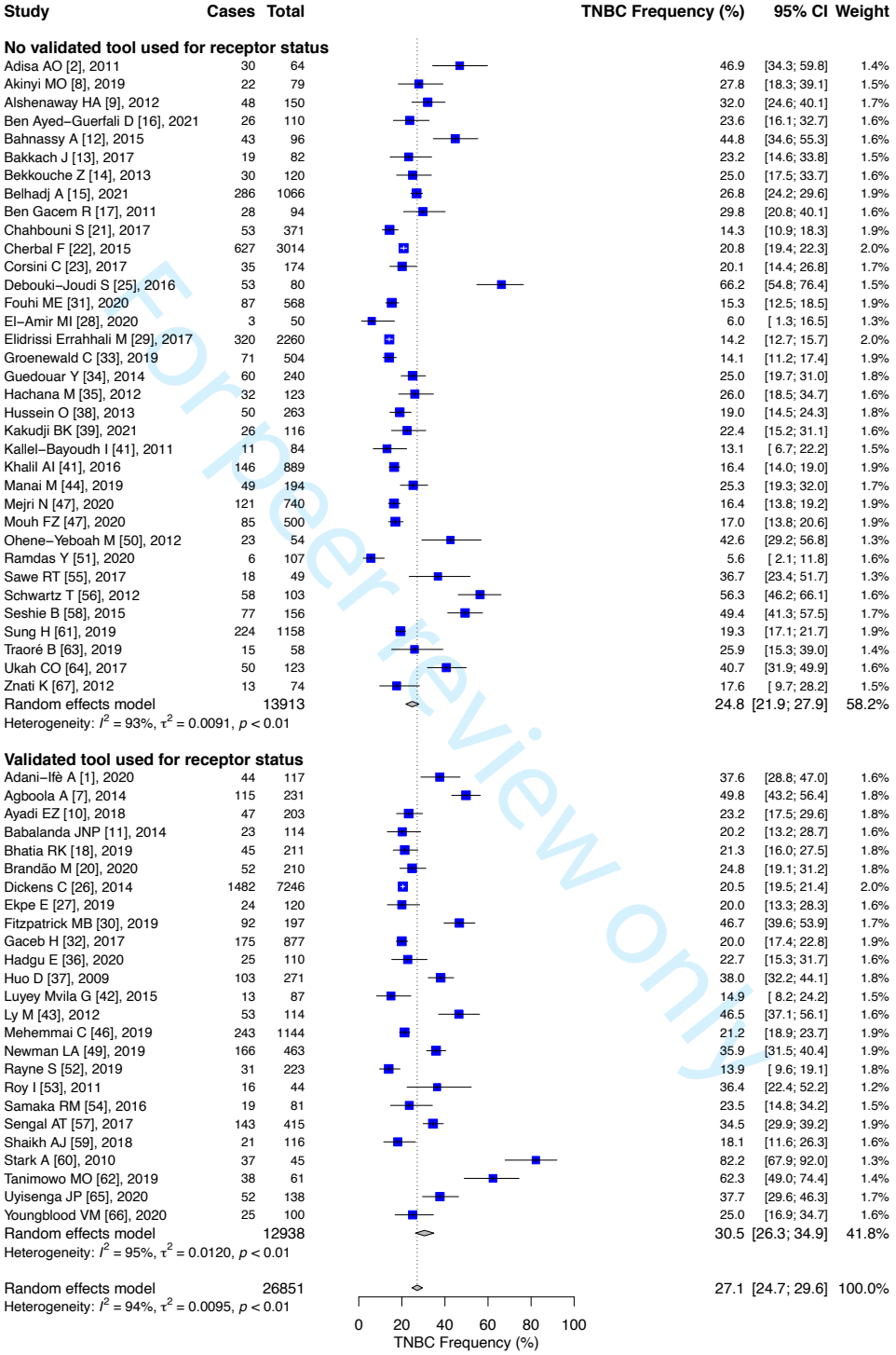
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Figure S3: Pooled TNBC frequency in Africa by risk of bias assessment. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.



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Figure S4: Pooled TNBC frequency in Africa by use of validated tool for receptor status. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.



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Figure S5: Meta-regression shows association with year of publication but not with age or use of validated tool for receptor status. The association of use of validated tool to assess receptor status (A), year of publication (B), mean age at diagnosis (C) and median age at diagnosis (D) on heterogeneity of treatment effects between studies was investigated by a meta-regression. P <0.05 indicates significant association.

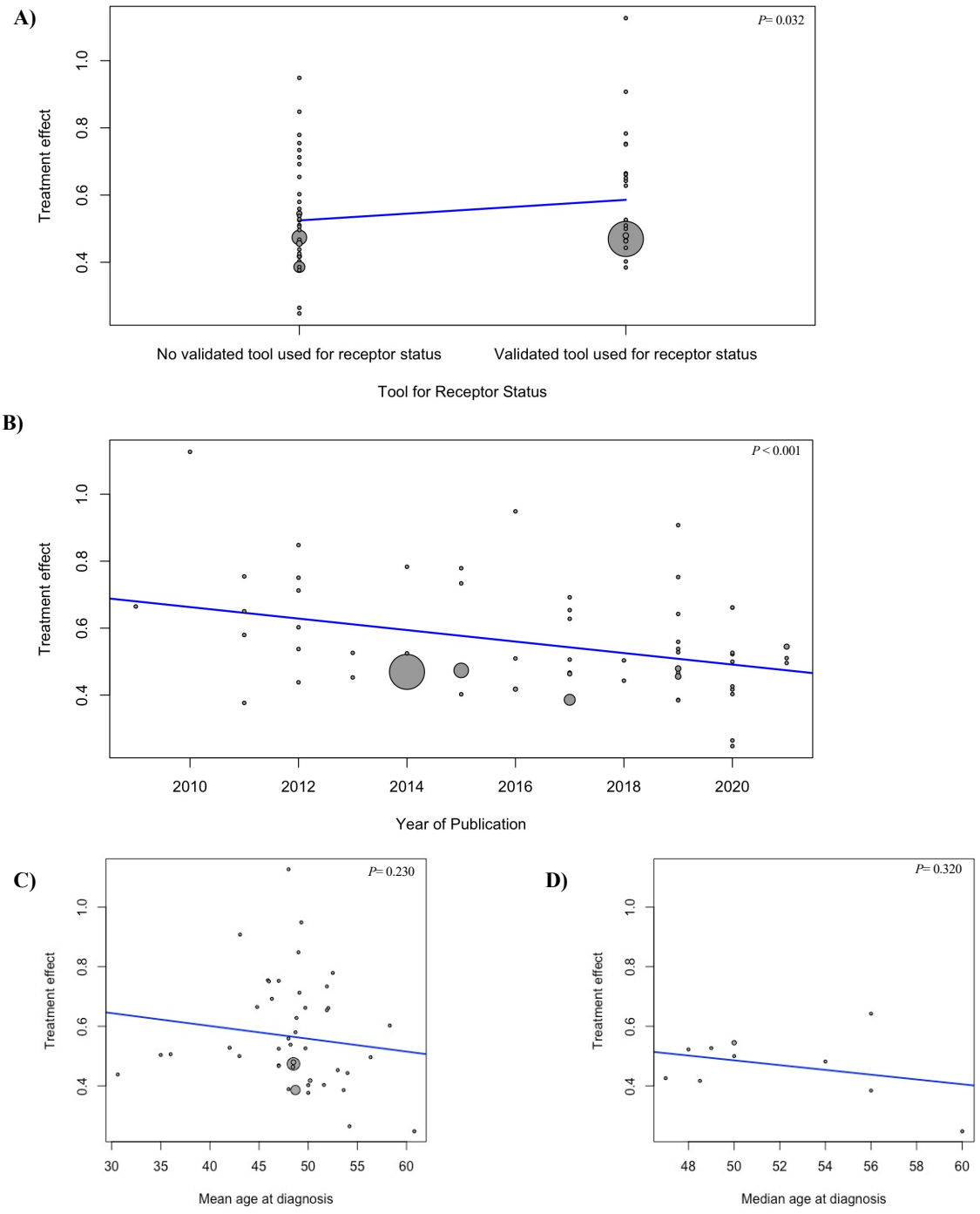
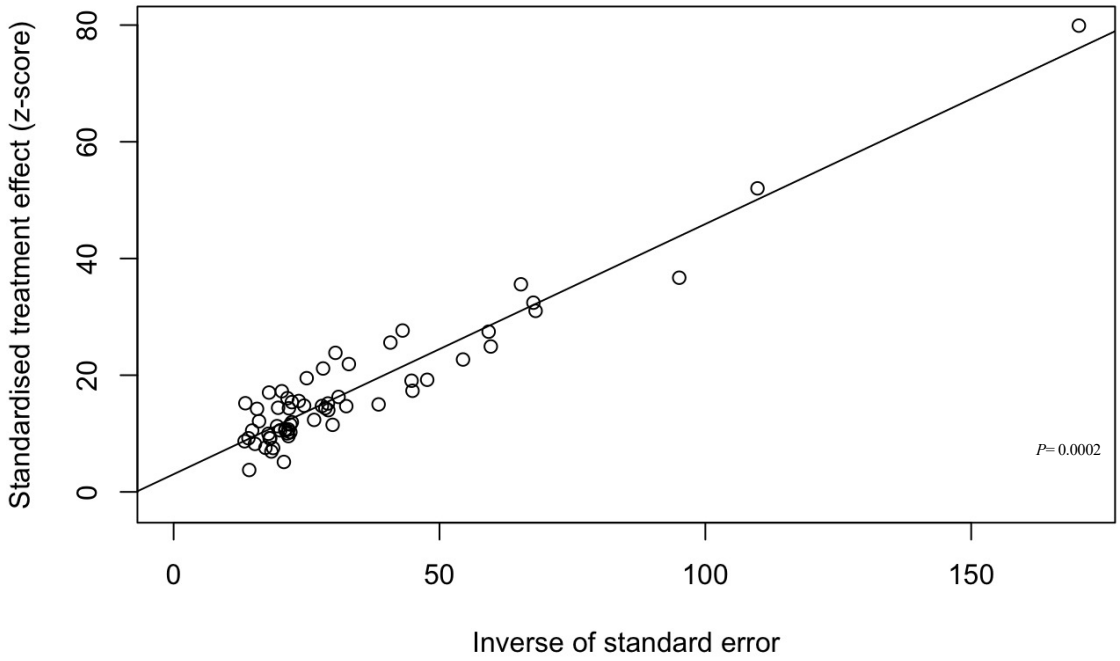


Figure S6: Egger's test shows high publication bias. Publication bias between studies for TNBC frequency investigated using Egger's test and a significant association was found, inferring significant publication bias. P <0.05 indicates significant association.



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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 2
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 3,4,6; Table S1
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 3,4,6; Figure 1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table S1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 3,4,6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 3,4,6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 3,4,6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 3,4,6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 3,4,6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 5
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	N/A
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 4,5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 4,5
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 4,5
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 5
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 4

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 5
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 5; Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	N/A
Study characteristics	17	Cite each included study and present its characteristics.	Table S2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table S2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table S2
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Table S2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	All forest plots
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	All forest plots
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Figure S6
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	All forest plots
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 6-9
	23b	Discuss any limitations of the evidence included in the review.	Page 6-9
	23c	Discuss any limitations of the review processes used.	Page 6-9
	23d	Discuss implications of the results for practice, policy, and future research.	Page 6-9
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 4
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	N/A
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 5
Competing interests	26	Declare any competing interests of review authors.	Page 5
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Table S2

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Triple Negative Breast Cancer Prevalence in Africa: a Systematic Review and Meta-analysis

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Triple Negative Breast Cancer Prevalence in Africa: a Systematic Review and Meta-analysis

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Abstract:

Objective: The aggressive triple negative breast cancer (TNBC) subtype disproportionately affects women of African ancestry across the diaspora, but its frequency across Africa has not been widely studied. This study seeks to estimate the frequency of TNBC among African populations.

Design: Systematic review and meta-analysis using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework.

Data sources: PubMed, EMBASE, African Journals Online and Web of Science were searched on April 25th 2021.

Eligibility criteria for selecting studies: We included studies that use breast cancer tissue samples from indigenous African women with sample size of eligible participants ≥ 40 and full receptor status for all three receptors (ER/PR/HER2) reported.

Data extraction and synthesis: Two independent reviewers extracted data and assessed risk of bias using the modified assessment tool by Hoy D. *et al.* (2012) for prevalence studies. A random-effects meta-analysis was performed, and data were pooled using the inverse-variance method and logit transformation. Pooled frequencies were reported with 95% confidence intervals calculated with the Clopper-Pearson method and heterogeneity quantified with I^2 statistic. GRADE assessed the certainty of the evidence.

Results: 1808 potentially eligible studies were identified of which 67 were included in the systematic review and 60 were included in the meta-analysis. Pooled TNBC frequency across African countries represented was estimated to be 27.0%; 95% CI: 24.0%- 30.2%, I^2= 94%. Pooled TNBC frequency was highest across West Africa, 45.7% [n= 15, 95% CI 38.8- 52.8, I^2= 91%] and lowest in Central Africa, 14.9% [n= 1, 95% CI 8.9- 24.1]. Estimates for TNBC were higher for studies that used Allred guidelines for ER/PR status compared to ASCO/CAP guidelines, and for studies that used older versions of ASCO/CAP guidelines for assessing HER2 status. Certainty of evidence was assessed to be very low using GRADE approach.

Conclusion: TNBC frequency was variable with the highest frequency reported in West Africa. Greater emphasis should be placed on establishing protocols for assessing receptor status due to the variability among studies.

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Strengths and limitations of this study:

- Previous studies have reported higher ER-negativity among West African countries compared to East African countries, but no review or meta-analysis has been solely focused on the frequency of TNBC across the African continent.
- Certainty of evidence was assessed using the GRADE approach
- Evidence of high frequency of TNBC and poor prognostic clinical factors highlight the need for early breast cancer detection across the continent.
- There was evidence of variable methods used to assess receptor status across the African continent.
- Only 20 out of 54 African countries were included in this study where there was a lack of cancer registry and population-based data, which suggests that our estimate for TNBC frequency across the continent is not fully accurate; however, given the available data, this is the best estimate from included studies.

Keywords

Triple negative breast cancer, TNBC, Africa, breast cancer, women of African ancestry

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Patient and Public Involvement

Patients were not involved in this study.

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Introduction

Breast cancer (BCa) mortality rates have markedly increased across Africa where estimated age-standardized rates in 2020 ranged from 20 deaths per 100,000 women across Northern Africa to 27 deaths per 100,000 women in Western Africa [1-3]. Breast cancer has thus been dubbed an emerging epidemic in Africa [4]. Indeed, a recent systematic review and meta-analysis found that female BCa incidence rates increased from 23.1 to 26.3 per 100,000 between 2000 and 2015 across the continent [5]. Women of African ancestry (WAA) across the globe and indigenous African women are more likely to receive a poorer BCa prognosis compared to women of other ancestries [6]. Multiple studies often posit poorer prognosis as a result of healthcare systems across Africa, where there is limited capacity and health infrastructure, *e.g.* inadequate screening and diagnostic services [7]. However, BCa prevalence by subtype, as defined by receptor status, is strikingly different for indigenous African women compared to the BCa prevalence profile of Western countries and is not simply explained by limited access to healthcare [6].

Immunohistochemistry (IHC) is routinely used to classify BCa into molecular subtypes according to the presence or absence of the estrogen (ER) and progesterone (PR) receptors and the human epidermal growth factor receptor 2 (HER2). Triple negative breast cancer (TNBC) is characterized by the lack of expression of all three biomarkers (ER, PR, HER2), which makes TNBC untreatable with targeted therapies such as Tamoxifen and Herceptin [8, 9]. This BCa subtype is often associated with earlier disease-onset, advanced-stage tumors and aggressive disease progression when compared with other BCa subtypes [10, 11]. Additionally, TNBC has been shown to disproportionately affect African women, and younger WAA and Hispanic women in North America [11], where the prevalence of TNBC in WAA has been estimated to be more than twice the prevalence in non-Hispanic White women [12]. There is also a higher mortality rate from TNBC and more advanced stage at diagnosis in WAA [13]. Thus, it is important to investigate what seems to be an ancestral predisposition to TNBC since the reasons for this disparity in TNBC prevalence and outcomes are not fully understood. Studies to date have not compiled adequate information on TNBC frequency or considered reported frequency and routine practices associated with diagnosis and treatment across the African continent.

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This systematic review and meta-analysis aims to increase understanding and knowledge regarding the frequency of TNBC across the African continent. The paucity of data and information on TNBC in Africa underscores the importance and urgency of such a review. A previous review reported higher ER-negativity among West African countries compared to East African countries [6] and a previous meta-analysis investigated ER-positivity across Africa [14] but no review or meta-analysis has been solely focused on the frequency of TNBC cases across the African continent. This review complements current biomedical research on TNBC and provides context for areas where TNBC research continues to expand. Improved understanding of TNBC frequency in continental Africa can further inform strategies for BCa detection and management for WAA globally. Due to shared ancestry between North American WAA and indigenous West African women [15], we hypothesize that there will be higher TNBC prevalence rates in countries across West Africa compared to other regions (North, East, Central, Southern) across Africa.

METHODS

Search Strategy and Selection Criteria

We used Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines as a framework for our systematic review and meta-analysis [16] as well as Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines [17]. On April 25th 2021, we searched PubMed, EMBASE, African Journals Online and Web of Science for relevant articles without date or language restrictions. Start date of the search was from inception of each database. A detailed version of our search strategy used in PubMed was modified for other databases. Search strategy and these modifications can be found in Table S1. Briefly, all search terms were Medical Subject Heading terms, including TNBC terms ("TNBC", "triple-negative*", "triple negative") and terms for African countries ("Africa", "African" and names of all 54 African countries) and outcome variables ("rate*", "prevalence", "epidemiology"). We included all studies that met the inclusion criteria. The inclusion criteria were as follows: studies that use breast cancer tissue samples from indigenous African women of any age, in any

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care setting and at any geographic location; sample size of eligible participants ≥ 40 (as slightly more stringent criteria since normal distribution could be assumed at n= 30 [18]); studies that demonstrate at least one of the following: report on receptor status of breast tumours including ER, PR, and HER2; any primary study including but not limited to observational studies, cross-sectional studies and case-control studies.

We excluded: editorials, single case reports, case series, and commentaries; studies that assessed diagnostic measures and treatment options for women with TNBC in the absence of assessment of its frequency; studies conducted in non-African nations without assessment of indigenous African TNBC rates or that of first-generation African immigrants. Study selection began by screening titles and abstracts of articles collected after employing the search strategy. The full text of these articles was then reviewed to assess inclusion. Two data abstractors independently reviewed articles at both the title/abstract and full text review stages. When there were discrepancies, a consensus was made in consultation with a third reviewer. The protocol for this review was not registered. Non-English studies were included after translation through Google Translate followed by verification of translation by a French speaker.

Quality Assessment

Studies which passed full text review (Table S2) were evaluated for risk of bias using a tool developed by Hoy and colleagues specifically intended for prevalence-studies [19]. Each study was assessed according to ten items assessing internal validity (Table S3) and assigned to have either low (score of 1) or high (score of 0) risk of bias for each question by two independent reviewers. A third reviewer mediated discrepancy and a final score per question was agreed upon. Studies were then classified based on the total score for all questions in the quality assessment tool as having a high (<=5), moderate (6-8), or low (>= 8) risk of bias.

Assessing the Certainty of Evidence

The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework [20] was used to assess the overall certainty of available evidence on the frequency of TNBC

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across the African continent. This framework considered factors such as study design, risk of bias, inconsistency, indirectness, imprecision, and publication bias.

Data Analysis

After reviewing full-text articles, a heat map was constructed with the number of studies, TNBC frequency and number of participants per country across African populations for unique studies using Google Sheets. All meta-analyses, meta-regressions and sensitivity analyses were completed using R (version 4.0.2) [21]. Using the *metaprop* package in R, we conducted a meta-analysis of TNBC frequency among indigenous African women with breast cancer, stratified by country, region, risk of bias assessment, year of publication and the use of a validated tool for assessing receptor status. Logit transformation was used to stabilize the variances and a random effects model with inverse-variance method for pooling frequencies and Clopper-Pearson method for calculating confidence intervals were used for our meta-analyses. Pooled TNBC frequency was estimated separately per country and per region as two studies included data from more than one region. When studies investigated African and non-African participants, only data from African participants were included in meta-analyses. Heterogeneity between studies was assessed with Cochrane's Q, I^2, and H statistics. Meta-regression was done to explore heterogeneity using *metareg* package. We used Egger's test to investigate publication bias and small study effects using the *metabias* package.

RESULTS

Of the 1808 records identified, 1032 remained after removing duplicates from the various databases. After screening titles and abstracts, 932 records were excluded due to irrelevance. The full text for the remaining 109 records were screened and an additional 42 studies were excluded because they did not meet eligibility criteria, leaving 67 relevant studies for inclusion (Figure 1). Our search strategy identified eligible studies from 20 countries across continental Africa. Nigeria and Tunisia had the highest number of eligible studies (8 studies each), followed by Morocco (7 studies), Algeria, Egypt, and South

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Africa (6 studies each), Ghana and Kenya (4 studies each) and South Africa (3 studies); there were only two studies from Uganda and only one study each from Botswana, Democratic Republic of Congo, Ethiopia, Guinea, Malawi, Mali, Mozambique, Rwanda, Senegal, and Sudan (Table S2). Five studies included data from multiple countries and regions across the continent (East and West Africa, Algeria, Egypt, Ethiopia, Ghana, Morocco, Namibia, Senegal, South Africa, and Tunisia). Five studies were translated from French to English and were all based in North Africa. Summary of clinical data can be found in Table 1 and Table S2.

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Table 1: Summary of study variables for included studies across Africa and by Region

Study variables	All studies (*n*= 67)	North African studies (*n*= 29)	East African studies (*n*= 10)	Southern African studies (*n*= 8)	Central African study (*n*= 1)	West African studies (*n*= 17)	West and East African studies (*n*= 2)
Mean age [SD]	49.21 [5.25] (*n*= 42)	48.47 [6.35] (*n*= 17)	48.65 [3.34] (*n*= 7)	56.09 [2.53] (*n*= 5)	50.00 [n/a] (*n*= 1)	47.67 [2.65] (*n*= 12)	not investigated
Median age [IQR]	48.75 [4.50] (*n*= 18)	47.00 [5.72] (*n*= 9)	48.75 [0.87] (*n*= 4)	56.00 [3.00] (*n*=3)	49.00 [n/a] (*n*=1)	not investigated	56.00 [n/a] (*n*= 1)
Grade 1 tumors, median [IQR]	9.56% [7.34] (*n*= 52)	8.44% [6.90] (*n*= 23)	10.66% [9.35] (*n*= 8)	11.40% [1.95] (*n*= 6)	1.15% [n/a] (*n*= 1)	6.06% [11.18] (*n*= 13)	W: 9.56% [n/a], E: 15.25% [n/a] (*n*= 1)
Grade 2 tumors, median [IQR]	51.11% [16.97] (*n*= 53)	57.54% [19.03] (*n*= 23)	44.44% [6.90] (*n*= 8)	46.76% [16.32] (*n*= 7)	51.72% [n/a] (*n*= 1)	51.22% [20.51] (*n*= 13)	W: 33.52% [n/a], E: 30.81% [n/a] (*n*= 1)
Grade 3 tumors, median [IQR]	38.68% [15.04] (*n*= 58)	33.76% [16.40] (*n*= 27)	41.11% [13.92] (*n*= 8)	39.53% [16.90] (*n*= 7)	47.13% [n/a] (*n*= 1)	38.83% [18.89] (*n*= 14)	W 56.92% [n/a], E 43.95% [n/a] (*n*= 1)
Positive lymph node status, median [IQR]	64.68% [20.99] (*n*= 42)	58.50% [15.40] 23)	70.53% [11.91] (*n*= 6)	64.66% [28.00] (*n*= 3)	not investigated	91.38% [25.07] (*n*= 10)	not investigated
Premenopausal, median [IQR]	58..86% [18.16] (*n*= 30)	58.40% [13.87] (*n*= 13)	57.00% [17.19] (*n*= 5)	37.10% [n/a] (n= 1)	60.92% [n/a] (*n*= 1)	68.48% [17.26] (*n*= 7)	not investigated
TNBC frequency, median [IQR]	25.00% [19.41] (*n*= 67)	23.00% [9.02] (*n*= 29)	25.00% [13.08] (*n*=10)	20.43% [7.32] (*n*= 8)	14.94% [n/a] (*n*= 1)	49.4-% [13.60] (*n*= 17)	n/a

* two studies investigated both West and East African populations in their respective manuscripts, W- West African participants, E- East African participants
n/a- not applicable

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All 67 studies reported TNBC frequency from specific hospital/health facility settings, 34 of which were conducted via academic and academic/university teaching hospitals. Fourteen studies were prospective whereas the others (n= 53) were retrospective studies. Additionally, 20 studies included some form of biased sampling (*e.g*., all metastatic cases, tissue microarrays, age cut-offs) whereas the remaining studies included were either random sampling or population-based. Of the included studies, eight (12%), thirty-seven (55%) and twenty-two (33%) were classified as low, moderate and high risk of bias, respectively, (Figure S1-S2, Table S3-S4) after using the risk of bias assessment tool for prevalence studies by Hoy *et al.* [19]. Most studies were scored as high risk of bias due to data acquisition (*e.g.,* study population). However, according to criteria set by the Hoy *et al.* risk of bias tool, data were interpreted appropriately for most studies (e.g., having a clear definition of TNBC, appropriate numerator/denominator for frequencies). The eight studies that were low risk of bias, were based in Algeria (*n*= 1), Botswana (*n*= 1), Malawi (*n*= 1), Rwanda (*n*= 1), South Africa (*n*= 2) or multiple countries (*n*= 2).

After identifying unique study populations per country (*n*= 60, Figure 1), TNBC estimates from the meta-analysis, number of studies and participants per country were highlighted (Figure 2). Overall TNBC frequency from included studies representing countries across the African continent was estimated to be 27.0%; 95% CI: 24.0%- 30.2%, I^2= 94% (Table S5). Pooled TNBC frequency estimates per country (Table S5) ranged from 14.0% in South Africa (95% CI: 9.6%-19.8%, I^2 = 75%) to 57.2% in Ghana (95% CI: 43.6%-69.8%, I^2 = 82%). When investigating estimates per region (Figure 3), TNBC frequency was lowest in Central Africa (n=1, 14.1%; 95% CI 8.9%-24.1%) and highest in West Africa (n=15, 45.7%; 95%CI: 38.8%-52.8%, I^2 = 91%). For these two analyses with pooled estimates per country and per region, heterogeneity (I^2) was estimated at 94%, indicative of high between study variability. When investigating the effect of risk of bias on study estimates, none was observed (Figure S3). Pooled TNBC estimates were also stratified by use of a validated tool for receptor status testing. Pooled TNBC frequency was higher (n= 25, 30.1%; 95%CI: 24.4%-34.7%, I^2 = 95%) in studies that reported the use of a validated tool for assessing receptor expression, when compared to studies that did not (n= 35, 24.7%; 95%CI: 21.6%-28.1%, I^2 = 93%)

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(Figure S4). Between study heterogeneity was high (I^2= 94%) and meta-regression showed that these estimates were not statistically significant (Figure S5A, P = 0.057, β coefficient= 0.267). When investigating TNBC estimates by the tool used for ER/PR and HER2 cutoffs, pooled TNBC frequency was higher in studies that used the Allred 1998 and Reiner's scale scoring for ER/PR cutoffs and in older versions of ASCO/CAP guidelines when compared to more recent versions for HER2 status (Figure 4). There was also an association between publication year and TNBC estimates where meta-regression showed a decrease in effect estimate with increasing publication year (Figure S5B, $P < 0.001$, β coefficient= -0.075). Influence analyses showed that two studies contributed largely to heterogeneity and influence on overall estimates (Figure S6). After conducting Egger's test (Figure S7, $P < 0.002$) and funnel plot analysis (Figure S8), a significant publication bias was identified.

The overall certainty of evidence for this systematic review and meta-analysis was judged as very low using the GRADE approach. Our judgement was downgraded due to risk of bias, indirectness, and publication bias.

Discussion

In this systematic review and meta-analysis of 67 studies on African women with BCa, we found that there was a high frequency of TNBC (27.0%) in cases reported across Africa although it varied depending on country and region. TNBC frequency was highest in West African populations (45.7%) compared to other regions across continental Africa (14.9%-22.7%). This is consistent with increased TNBC/ER-negative prevalence observed in populations with high West African ancestry [15, 22, 23] in the Caribbean [24, 25] and in North America [26]. In a USA population-based study (2010-2014), TNBC prevalence in non-Hispanic White women was estimated to be ~8% whereas it was ~15% in non-Hispanic Black women [23]. Additionally, TNBC prevalence was estimated to be ~8% and ~25% in White and Black women respectively in a UK cancer registry-based population in London [27]. These high frequencies of TNBC across Africa and the African diaspora are concerning as triple negative breast tumors have a greater

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propensity to metastasize to vital organs such as the brain [28] and are typically more aggressive due to lack of targeted therapies.

When investigating clinical factors, the reported mean and/or median age at diagnosis was under 50 in 35 out of the 47 studies reporting age. Young age at diagnosis (under age 40) has been previously reported to be associated with triple negative and HER2-positive cancers as well as more aggressive clinical outcome [29]. Indeed, this poor prognosis of TNBC patients was evident as most of the included studies reported a high percentage of grade 3 tumours, lymph node positivity and TNBC frequency. It must however be noted that a younger age at diagnosis is also routinely observed in lower- and middle-income countries as this is also reflective of the population structure [30]. Therefore, this observed lower age at diagnosis may be indicative of the population distribution rather than the intrinsic aggressive biology of the tumors. To consider this possibility, we investigated associations with mean and median age at diagnosis and effect estimates and found no association (Figure S5C p= 0.209, β coefficient= -0.022, Figure S5D p= 0.311, β coefficient= -0.039). More advanced stage tumors and lymph node involvement at presentation may also be attributable to poor infrastructure and lack of BCa awareness and screening. A recent study of BCa across sub-Saharan Africa found that the majority of cases diagnosed were late-stage, emphasizing the need for early diagnosis [31]. Two separate studies from Nigeria and Ghana both found that most of the information obtained about BCa was from mass media and there was a general poor knowledge of BCa-associated risk factors [32, 33]. The Ghanaian-based study also found that the rate of breast self-examination (BSE), and clinical breast examination (CBE) were higher than that of obtaining mammograms [33] which emphasizes the need to promote screening programs in a culturally-relevant setting. It should be noted however that mammography has been associated with a two times higher chance of detecting ER-positive BCa compared to ER-negative BCa [34] which might be contributing to the relatively higher TNBC

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frequency observed across West African countries when compared to Southern African countries where mammography is more accessible.

Many studies were excluded on the account of not assessing ER, PR and HER2 status. With respect to receptor status, 30 out of the 67 studies used validated guidelines (ASCO/CAP [35-40], Allred [41] or Reiner's Scale [42]) for receptor status cut-offs. Such variability in classifying receptor status (i.e., the use of other guidelines with different cut-offs) affects the resulting treatment for patients with a diagnosis and how TNBC frequencies are calculated in each study. After stratifying studies by use of ASCO-CAP guidelines which account for specimen fixation and cut-offs for ER/PR expression at 1%, there was a decrease in pooled TNBC frequency (24.4%) compared to those that used Allred or Reiner's Scale (cutoff at 10%) resulting in TNBC frequency of 42.9% and 38.0% respectively (Figure 4A). A similar trend was observed for ASCO/CAP guidelines with respect to assessing HER2 status where more recent guidelines correlated with lower TNBC frequency compared to older guidelines (Figure 4B). Thus, meta-regression with publication year was done and indeed there was an association with effect estimates ($P < 0.002$, β coefficient= -0.075, Figure S5B). A similar trend was recently reported for East Africa-based studies conducted before and after 2013; ER/PR positivity was lower before 2013 compared to after 2013 [43]. The variability in how receptor status is assessed highlights a need for increased capacity to conduct immunohistochemical receptor status testing to further enhance BCa diagnoses and classification. It must be noted however, that IHC might not be feasible for many of the hospitals/health centers across Africa and international collaborations should be encouraged to assist with building such capacity. One Nigerian study noted cost to be a barrier – IHC was performed on only 31% of the reported cases [44]. It also stated that ASCO/CAP guidelines could not be adhered to for HER2 due to the high cost of fluorescence *in situ* hybridization in the case of an equivocal HER2 score. In contrast, South Africa has an extensive healthcare system with a comprehensive standardized national public health system for routinely assessing breast

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cancer receptor status [45]. The disparity in access to diagnostic and therapeutic tools across the continent could also be contributing to the lack of receptor status data and higher BCa burden reported here.

To our knowledge, this is the first systematic review and meta-analysis with an in-depth analysis on TNBC frequency across continental Africa. However, there are some limitations to be considered. IHC and specimen collection and processing are not equally accessible, and neither are they uniformly done across the African continent. Specimen fixation, storage time of the samples and other pre-analytical IHC variables have been previously shown to impact accuracy of IHC results [46]. It was estimated that up to 20% of IHC results globally are inaccurate based off of these pre-analytical variables [47]. The true frequency of TNBC is therefore not ascertainable due to lack of population-based data. Our search strategy identified representation from 20 African countries to be included with low representation from Central Africa (n=1) which was similarly observed in a recent systematic review and meta-analysis that investigated BCa incidence across Africa with 22 African countries and low representation from Central Africa (n=2) [5]. Another caveat to consider is that there are no validated search strategies for observational studies thus, some studies may be missed. As expected, there was considerable heterogeneity in our meta-analyses. High heterogeneity could be due to the large number of studies, or the varying ethnicities across Africa. We graded our confidence in the evidence presented as "very low" using the GRADE assessment due to high heterogeneity, high risk of bias studies included and low representation across the continent (indirectness); however, given the available data, this is the best estimate of TNBC frequency across the African continent.

This study provides the closest estimate of TNBC frequency across the different regions of continental Africa. Considerations should be made at the country-level to address IHC protocols and adherence to ASCO-CAP guidelines wherever possible. There is a clear disparity across the continent (with respect to diagnostic and therapeutic tools) that needs to be effectively addressed to prevent BCa burden. Priority should also be given to implementing culturally relevant BCa awareness programs as these have been proven to increase cancer awareness knowledge and thus could decrease preventable deaths from BCa

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[48]. There is also a dearth of knowledge across the continent about BCa subtype prevalence in general. This should be addressed as soon as possible by the establishment of cancer registries before the burden of BCa and other chronic diseases drastically increase with the epidemiological transition that has already started to take place across Africa.

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Juliet M. Daniel, Ph.D.: conceptualization, study design, data interpretation, supervision, funding acquisition, review & editing.

Competing Interests:
None declared

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Ethical approval statement:
Not necessary for this manuscript.

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Figure legends:

Figure 1: PRISMA flowchart accounting for all articles included in the narrative synthesis and meta-analysis.

Figure 2: Pooled TNBC frequencies out of all BCa subtypes from studies done across African countries. Data represent pooled TNBC frequencies among all BCa subtypes reported in unique studies done in populations from stated countries. Pooled frequencies were calculated if the country had more than one study as stated within the meta-analysis. Estimates do not account for heterogeneity, IHC cut-offs and size of the respective populations.

Figure 3: Pooled TNBC frequency in Africa by region. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

Figure 4: Pooled TNBC frequency in Africa by tool used for (A) ER/PR status (B) HER2 status. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

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Figure 1: PRISMA flowchart accounting for all articles included in the narrative synthesis and meta-analysis

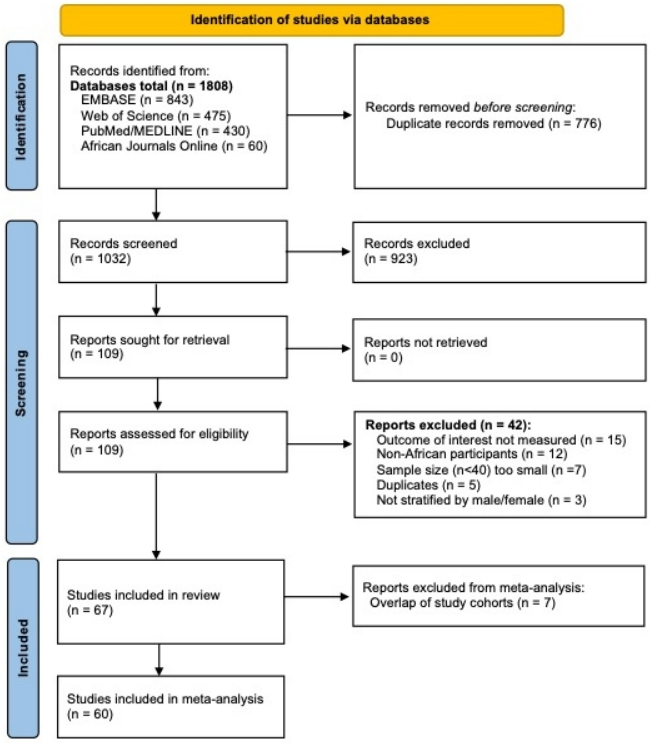


Figure 1: PRISMA flowchart accounting for all articles included in the narrative synthesis and meta-analysis.

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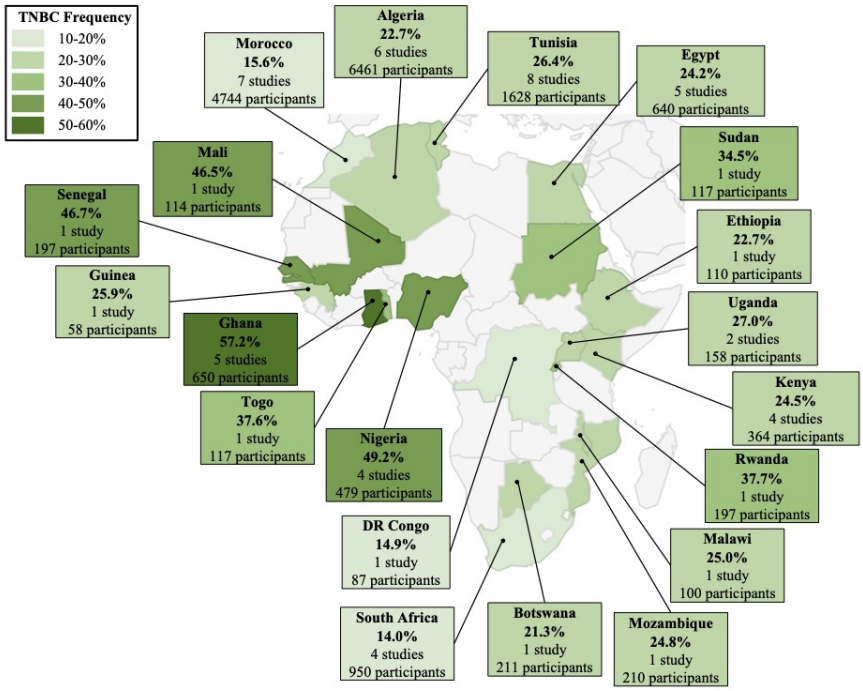


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Figure 3: Pooled TNBC frequency in Africa by region. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

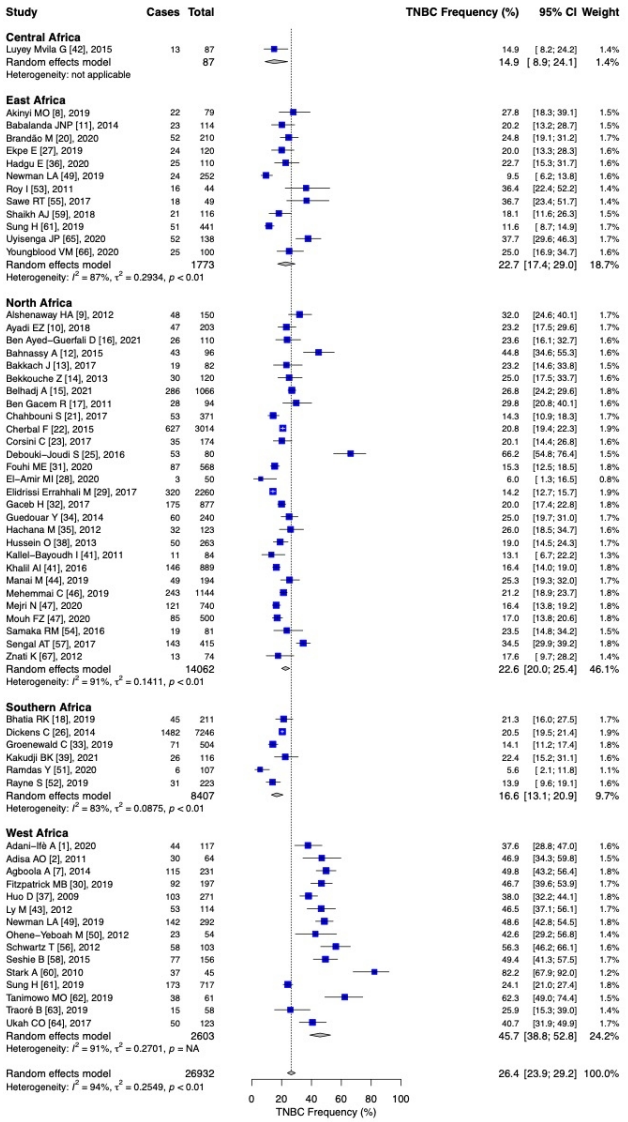


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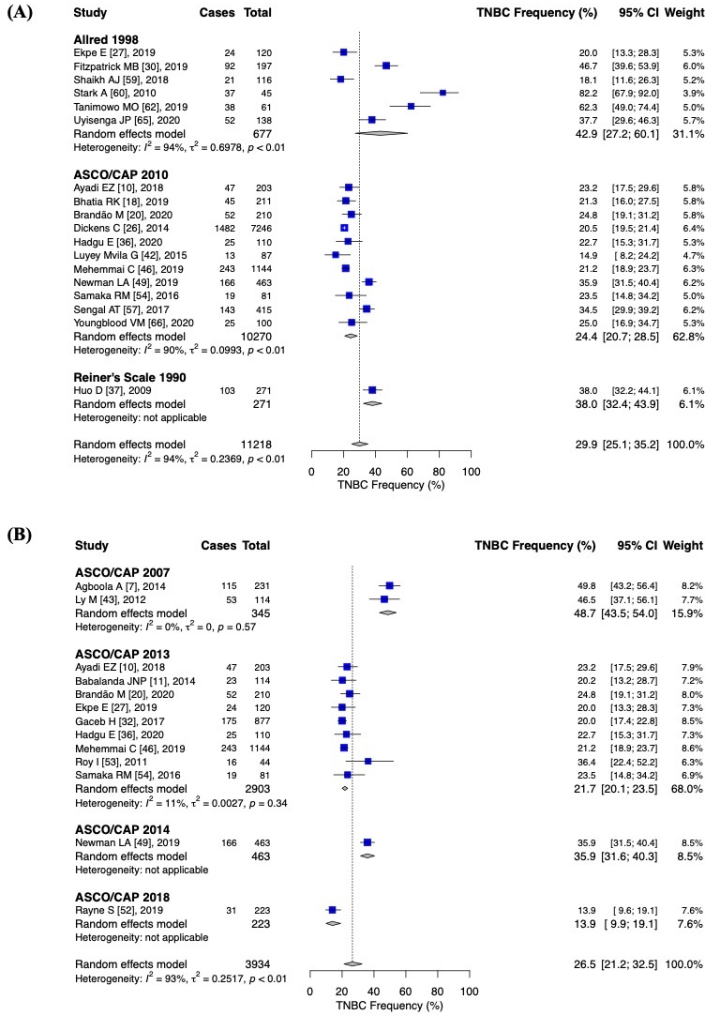


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SUPPORTING DOCUMENTS

Table S1: Search strategy used for Databases

Database	Search terms
Pubmed/MEDLINE	(Africa OR African OR "West* Africa" OR "East* Africa" OR "South* Africa" OR "North* Africa" OR "Central Africa" OR Algeria OR Angola OR Benin OR Botswana OR "Burkina Faso" OR Burundi OR Cabo Verde OR Cameroon OR Central African Republic OR "CAR" OR Chad OR Comoros OR "DRC" OR Congo OR "Republic of Congo" OR "Ivory Coast" OR Cote d'Ivoire OR Djibouti OR Egypt OR "Equatorial Guinea" OR Eritrea OR Eswatini OR Swaziland OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR "Guinea-Bissau" OR Kenya OR Lesotho OR Liberia OR Libya OR Madagascar [PL] OR Malawi OR Mali OR Mauritania OR Mauritanie OR Mauritius OR Morocco OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR "Sao Tome and Principe" OR Senegal OR Seychelles OR "Sierra Leone" OR Somalia OR "South Africa" OR "South Sudan" OR Sudan OR Tanzania OR Togo OR Tunisia OR Uganda OR Zambia OR Zimbabwe) AND (TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND (rate* OR prevalence OR incidence OR epidemiology)
Web of Science	TS = (Africa or African or "West* Africa" or "East* Africa" or "South* Africa" or "North* Africa" or "Central Africa" or Algeria or Angola OR Benin OR Botswana OR "Burkina Faso" OR Burundi OR Cabo Verde OR Cameroon OR Central African Republic OR "CAR" OR Chad OR Comoros OR "DRC" OR Congo OR "Republic of Congo" OR "Ivory Coast" OR Cote d'Ivoire OR Djibouti OR Egypt OR "Equatorial Guinea" OR Eritrea OR Eswatini OR Swaziland OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR "Guinea-Bissau" OR Kenya OR Lesotho OR Liberia OR Libya OR Madagascar [PL] OR Malawi OR Mali OR Mauritania OR Mauritanie OR Mauritius OR Morocco OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR "Sao Tome and Principe" OR Senegal OR Seychelles OR "Sierra Leone" OR Somalia OR "South Africa" OR "South Sudan" OR Sudan OR Tanzania OR Togo OR Tunisia OR Uganda OR Zambia OR Zimbabwe) AND TS= (TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND TS= (rate* OR prevalence OR incidence OR epidemiology)
Embase	(Africa OR African OR "West* Africa" OR "East* Africa" OR "South* Africa" OR "North* Africa" OR "Central Africa" OR Algeria OR Angola OR Benin OR Botswana OR "Burkina Faso" OR Burundi OR Cabo Verde OR Cameroon OR Central African Republic OR "CAR" OR Chad OR Comoros OR "DRC" OR Congo OR "Republic of Congo" OR "Ivory Coast" OR Cote d'Ivoire OR Djibouti OR Egypt OR "Equatorial Guinea" OR Eritrea OR Eswatini OR Swaziland OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR "Guinea-Bissau" OR Kenya OR Lesotho OR Liberia OR Libya OR Madagascar OR Malawi OR Mali OR Mauritania OR Mauritanie OR Mauritius OR Morocco OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR "Sao Tome and Principe" OR Senegal OR Seychelles OR "Sierra Leone" OR Somalia OR "South Africa" OR "South Sudan" OR Sudan OR Tanzania OR Togo OR Tunisia OR Uganda OR Zambia OR Zimbabwe) AND (TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND (rate* OR prevalence OR incidence OR epidemiology)
African Journals Online (AJOL)	(TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND (rate* OR prevalence OR incidence OR epidemiology)

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Table S2: Summary of data from included studies

Author	Population	Region	Mean age ± SD	Median age (min-max)	Grade 3 tumors	Positive Nodes	Number of TNBC cases	Participants studied with known receptor status (all participants)	Frequency of TNBC	Risk of Bias Assessment
Adani-Ifè A, 2020 [1]	Togo	West Africa	52.05 ±12.38	(30-85)	42.70%	41.90%	44	117 (312)	37.60%	Moderate
Adisa AO, 2011 [2]	Nigeria	West Africa	45.90	(20-81)	/	/	30	64 (385)	46.88%	High
Agboola A, 2014 [3]	Nigeria	West Africa	/	/	35.70%	99.20%	99	155 (199)	54.36%	High
Agboola A, 2014 [4]	Nigeria	West Africa	/	/	/	94.20%	55	100 (308)	55.00%	Moderate
Agboola A, 2013 [5]	Nigeria	West Africa	/	/	/	92.70%	81	149 (308)	63.87%	High
Agboola A, 2016 [6]	Nigeria	West Africa	/	/	34.12%	91.76%	114	187 (255)	60.96%	Moderate
Agboola A, 2014 [7]	Nigeria	West Africa	/	/	38.50%	92.20%	115	231 (231)	49.78%	High
Akinyi MO, 2019 [8]	Kenya	East Africa	48.00 ±14.5	/	/	71%	22	79 (79)	27.85%	High
Alshenawy HA, 2012 [9]	Egypt	North Africa	58.30 ±10.1	/	20.00%	/	48	150 (150)	26.67%	Moderate
Ayadi EZ, 2018 [10]	Tunisia	North Africa	35.00 ±4.2	(17-40)	46.85%	64.10%	47	203 (238)	23.15%	Moderate
Babalanda JNP, 2014 [11]	Uganda	East Africa	47	(13-87)	53.51%	77.19%	23	114 (114)	20.18%	Moderate
Bahnassy A, 2015 [12]	Egypt	North Africa	TNBC: 51.91± 12.34 non-TNBC: 52.77 ± 12.13 Controls: 35 ± 13.94	TNBC: (30-78) non-TNBC: (27-81) Controls: (22-64)	15.63%	72.91%	43	96 (96)	44.79%	Moderate
Bakkach J, 2017 [13]	Morocco	North Africa	/	36 (/)	40.20%	58.50%	19	82 (331)	23.00%	High
Bekkouche Z, 2013 [14]	Algeria	North Africa	49.72	(30-84)	73.33%	35.83%	30	120 (120)	25.00%	High
Belhadj A, 2021 [15]	Algeria	North Africa	/	50 (21-88)	38.86%	64.70%	286	1066 (1140)	27.00%	Moderate
Ben Ayed-Guerfali D, 2021 [16]	Tunisia	North Africa	/	42.78 (27-65)	26.78%	59.70%	26	110 (134)	23.60%	High
Ben Gacem R, 2011 [17]	Tunisia	North Africa	48.70	47 (31-75)	43.62%	55.26%	28	94 (94)	29.79%	Moderate
Bhatia RK, 2019 [18]	Botswana	Southern Africa	/	54 (27-96)	25.76%	/	45	211 (384)	21.30%	Low

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Bogan D, 2017 [19]	Egypt	North Africa	52.38 ±12.17	(27-81)	13.94%	66.67%	43	96 (96)	44.79%	Moderate
Brandão M, 2020 [20]	Mozambique	East Africa	/	48 (/)	31.25%	61.43%	52	210 (262)	25.00%	Moderate
Chahbouni S, 2017 [21]	Morocco	North Africa	48.00	(22-99)	/	/	53	371 (390)	13.59%	High
Cherbal F, 2015 [22]	Algeria	North Africa	/	48.5 (22-84)	31.30%	/	627	3014 (3014)	20.80%	Moderate
Corsini C, 2017 [23]	Multiple (France, Algeria, Egypt, Morocco, Tunisia)	North Africa	/	/	44.74%	42%	35	174 (456)	20.10%	Moderate
Cubasch H, 2018 [24]	South Africa	Southern Africa	54.40 ±14.2	/	39.53%	/	126	581 (602)	21.70%	Low
Debouki-Joudi S, 2016 [25]	Tunisia	North Africa	49.30	(24-80)	26.22%	55.20%	53	80 (91)	66.25%	High
Dickens C, 2014 [26]	Multiple (South Africa & Namibia)	Southern Africa	/	/	33.87%	/	1482	7246 (12361)	20.45%	Low
Ekpe E, 2019 [27]	Kenya	East Africa	47.00 ±12	(21-79)	50.00%	/	24	120 (125)	20.00%	Moderate
El-Amir MI, 2020 [28]	Egypt	North Africa	54.20 ±10.6	(32-77)	18.00%	72%	3	50 (50)	6.00%	High
Elidrissi Errahhali M, 2017 [29]	Morocco	North Africa	48.70	(19-105)	18.00%	64.80%	320	2260 (2406)	14.20%	Moderate
Fitzpatrick MB, 2019 [30]	Senegal	West Africa	47.00 ±12.7	/	76.17%	/	92	197 (522)	46.70%	Moderate
Fouhi ME, 2020 [31]	Morocco	North Africa	51.60	(23-89)	32.60%	35.70%	87	568 (668)	15.30%	High
Gaceb H, 2017 [32]	Algeria	North Africa	48.44	(22-84)	28.50%	/	175	877 (877)	19.95%	Low
Groenewald C, 2019 [33]	South Africa	Southern Africa	53.60 ±13.4	/	/	68.10%	71	504 (505)	14.10%	Moderate
Guedouar Y, 2014 [34]	Algeria	North Africa	47.00	(23-78)	24.58%	38.33%	60	240 (240)	25.00%	High
Hachana M, 2012 [35]	Tunisia	North Africa	/	/	40.18%	21.40%	32	123 (123)	26.02%	Moderate
Hadgu E, 2020 [36]	Ethiopia	East Africa	43.00 ±14	50 (22-75)	50.00%	/	25	110 (112)	23.00%	Moderate
Huo D, 2009 [37]	Multiple (Nigeria & Senegal)	West Africa	44.80	/	39.15%	72%	103	271 (378)	27.00%	Low
Hussein O, 2013 [38]	Egypt	North Africa	53.00 ±11.6	52	17.39%	/	50	263 (263)	19.00%	High

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Kakudji BK, 2021 [39]	South Africa	Southern Africa	56.35 ±14.09	/	62.07%	64.66%	26	116 (116)	22.50%	High
Kallel-Bayoudh I, 2011 [40]	Tunisia	North Africa	50.00 ±11	(27-75)	35.44%	65%	11	84 (84)	13.00%	High
Khalil AI, 2016 [41]	Morocco	North Africa	50.20 ±11.34	(17-93)	35.48%	49.40%	146	889 (1277)	16.40%	Moderate
Luyey Mvila G, 2015 [42]	Democratic Republic of Congo	Central Africa	50.00	49 (29-82)	47.13%	/	13	87 (87)	14.94%	Moderate
Ly M, 2012 [43]	Mali	West Africa	46.00	(25-82)	78.18%	91%	53	114 (114)	46.00%	Moderate
Manai M, 2019 [44]	Tunisia	North Africa	/	42 (24-62)	42.20%	95%	49	194 (210)	25.00%	Moderate
McCormack VA, 2013 [45]	South Africa	Southern Africa	55.30	/	42.34%	/	209	1027 (1247)	20.40%	Low
Mehemmai C, 2019 [46]	Algeria	North Africa	/	48.5 (21-84)	26.99%	/	243	1144 (1162)	21.24%	Moderate
Mejri N, 2020 [47]	Tunisia	North Africa	/	48.5 IBCa 49 non-IBCa (/)	46.20% IBCa 36.10% non-IBCa	50%	121	740 (740)	16.40%	High
Mouh FZ, 2020 [48]	Morocco	North Africa	/	47 (/)	34.92%	60.10%	85	500 (500)	17.00%	High
Newman LA, 2019 [49]	Multiple (Ghana, Ethiopia, United States)	West and East Africa	/	56 (14-95)	/	/	Ghanaian: 142 Ethiopian: 24	Ghana: 292 (618) Ethiopia: 144 (252)	Ghana: 49.00% Ethiopia: 17.00%	High
Ohene-Yeboah M, 2012 [50]	Ghana	West Africa	49.10 ±15.3	(20-96)	53.70%	/	23	54 (330)	42.70%	Moderate
Ramdas Y, 2020 [51]	South Africa	Southern Africa	60.80 ±9.3	60 (43-86)	9.57%	12.10%	6	107 (107)	5.60%	High
Rayne S, 2019 [52]	South Africa	Southern Africa	/	56	51.10%	/	31	223 (365)	14.00%	Moderate
Roy I, 2011 [53]	Uganda	East Africa	/	/	40.00%	/	16	44 (64)	36.00%	Moderate
Samaka RM, 2016 [54]	Egypt	North Africa	/	(30-81)	44.44%	77.80%	19	81 (81)	23.50%	Moderate
Sawe RT, 2017 [55]	Kenya	East Africa	51.90	48.5 (16-84)	42.22%	/	18	49 (49)	36.70%	Moderate
Schwartz T, 2012 [56]	Ghana	West Africa	49.00 ±13.4	/	52.52%	/	58	103 (104)	56.30%	Moderate
Sengal AT, 2017 [57]	Sudan	North Africa	48.80 ±13.5	(20-90)	41.79%	52.70%	143	415 (560)	34.50%	Moderate
Seshie B, 2015 [58]	Ghana	West Africa	52.50 ±12.1	(24-77)	31.58%	66.00%	77	156 (165)	49.40%	Moderate
Shaikh AJ, 2018 [59]	Kenya	East Africa	54.00	(27-83)	33.33%	49%	21	116 (123)	17.00%	Moderate
Stark A, 2010 [60]	Ghana	West Africa	48.00 ±13.7	/	76.00%	/	37	45 (75)	82.20%	Moderate

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Sung H, 2019 [61]	Multiple (USA, Caribbean, West Africa, East Africa)	West and East Africa	/	/	West Africa: 56.92% East Africa: 43.95%	/	West Africa: 173 East Africa: 51	West Africa: 717 (811) East Africa: 441 (484)	West Africa: 24.10% East Africa: 11.60%	Moderate
Tanimowo MO, 2019 [62]	Nigeria	West Africa	43.07 ±11.19	(23-70)	29.51%	/	38	61 (61)	62.30%	High
Traoré B, 2019 [63]	Guinea	West Africa	48.20 ±10.9	(31-80)	36.21%	62.10%	15	58 (58)	25.90%	Moderate
Ukah CO, 2017 [64]	Nigeria	West Africa	46.30 ±12.3	(24-85)	26.83%	/	50	123 (397)	40.70%	Moderate
Uyisenga JP, 2020 [65]	Rwanda	East Africa	49.70 ±12.98	(17-86)	/	70.06%	52	138 (138)	37.70%	Low
Youngblood VM, 2020 [66]	Malawi	East Africa	/	49 (21-80)	37.00%	77%	25	100 (100)	25.00%	Low
Znati K, 2012 [67]	Morocco	North Africa	30.62	(18-35)	45.07%	51%	13	74 (74)	18.00%	High

/ data not stated

IBCa- inflammatory breast cancer

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Table S3: Questions for Risk of Bias Assessment, modified from Hoy *et al*., 2012

1. Was the study's target population a close representation of the national population in relation to relevant variables, e.g. age, sex, occupation?
2. Was the sampling frame a true or close representation of the target population?
3. Was some form of random selection used to select the sample, or was a census undertaken?
4. Was the likelihood of non-response bias minimal?
5. Were data collected directly from the subjects (as opposed to a proxy)?
6. Was an acceptable case definition used in the study? Was TNBC defined as ER, PR, HER2 negative?
7. Was the study instrument that measured the parameter of interest (e.g. appropriate guidelines for cut-offs ASCO-CAP) shown to have reliability and validity?
8. Was the same mode of data collection used for all subjects? Was the information from each patient collected in the same way (e.g clinical referral appointment)?
9. Was there a shortest prevalence or incidence period identified?
10.Were the numerator(s) and denominator(s) for the parameter of interest appropriate?

Table S4: Raw Values for Risk of Bias Assessment for Included Studies

Author	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	TOTAL	Overall ROB
Adani-Ifè A, 2020 [1]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
Adisa AO, 2011 [2]	0	0	1	0	1	0	0	0	1	0	3	**High**
Agboola A, 2014 [3]	0	0	0	0	1	1	1	1	1	0	5	**High**
Agboola A, 2014 [4]	0	0	1	0	1	1	1	1	1	1	7	**Moderate**
Agboola A, 2013 [5]	0	0	0	0	1	1	1	1	0	1	5	**High**
Agboola A, 2016 [6]	0	0	0	1	1	0	1	1	1	1	6	**Moderate**
Agboola A, 2014 [7]	0	0	0	0	1	0	1	1	1	1	5	**High**
Akinyi MO, 2019 [8]	0	0	0	1	1	1	0	1	0	0	4	**High**
Alshenawy HA, 2012 [9]	0	0	1	1	1	1	0	1	1	1	7	**Moderate**
Ayadi EZ, 2018 [10]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Babalanda JNP, 2014 [11]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Bahnassy A, 2015 [12]	0	0	1	1	1	1	0	1	1	1	7	**Moderate**
Bakkach J, 2017 [13]	0	0	0	1	1	0	0	1	1	0	4	**High**
Bekkouche Z, 2013 [14]	0	0	1	0	1	1	0	1	1	0	5	**High**

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Belhadj A, 2021 [15]	1	0	1	1	1	1	0	1	1	1	8	**Moderate**
Ben Ayed-Guerfali D, 2021 [16]	0	0	0	0	1	0	0	1	1	0	3	**High**
Ben Gacem R, 2011 [17]	0	0	1	1	1	1	0	1	0	1	6	**Moderate**
Bhatia RK, 2019 [18]	1	1	1	1	1	1	1	1	1	1	10	**Low**
Bogan D, 2017 [19]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Brandão M, 2020 [20]	0	0	1	1	1	1	1	0	1	1	7	**Moderate**
Chahbouni S, 2017 [21]	0	0	0	0	1	1	0	1	1	0	4	**High**
Cherbal F, 2015 [22]	1	1	1	0	1	1	0	1	1	1	8	**Moderate**
Corsini C, 2017 [23]	0	0	1	1	1	0	0	1	1	1	6	**Moderate**
Cubasch H, 2018 [24]	1	1	1	1	1	1	0	1	1	1	9	**Low**
Debouki-Joudi S, 2016 [25]	0	0	0	0	1	1	0	1	0	1	4	**High**
Dickens C, 2014 [26]	1	1	1	1	1	1	1	1	1	1	10	**Low**
Ekpe E, 2019 [27]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
El-Amir MI, 2020 [28]	0	0	0	0	1	0	0	1	1	1	4	**High**
Elidrissi Errahhali M, 2017 [29]	0	1	1	0	1	1	0	1	1	1	7	**Moderate**
Fitzpatrick MB, 2019 [30]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
Fouhi ME, 2020 [31]	0	1	0	0	1	0	0	1	1	0	4	**High**
Gaceb H, 2017 [32]	1	1	1	0	1	1	1	1	1	1	9	**Low**
Groenewald C, 2019 [33]	0	0	0	1	1	1	0	1	1	1	6	**Moderate**
Guedouar Y, 2014 [34]	0	1	0	0	1	1	0	1	1	0	5	**High**
Hachana M, 2012 [35]	0	0	1	1	1	1	0	1	1	1	7	**Moderate**
Hadgu E, 2020 [36]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Huo D, 2009 [37]	0	1	1	1	1	1	1	1	1	1	9	**Low**
Hussein O, 2013 [38]	0	0	1	0	1	0	0	1	1	1	5	**High**
Kakudji BK, 2021 [39]	0	0	0	1	1	1	0	0	1	1	5	**High**
Kallel-Bayoudh I, 2011 [40]	0	0	0	0	1	1	0	1	1	1	5	**High**
Khalil AI, 2016 [41]	0	1	1	0	1	1	0	1	1	1	7	**Moderate**
Luyey Mvila G, 2015 [42]	0	0	1	0	1	1	1	1	1	1	7	**Moderate**
Ly M, 2012 [43]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Manai M, 2019 [44]	0	1	1	0	1	1	0	1	1	1	7	**Moderate**
McCormack VA, 2013 [45]	1	1	1	1	1	1	0	1	1	1	9	**Low**
Mehemmai C, 2019 [46]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Mejri N, 2020 [47]	0	0	0	0	1	0	0	1	1	0	3	**High**
Mouh FZ, 2020 [48]	0	0	0	0	1	1	0	1	1	1	5	**High**
Newman LA, 2019 [49]	0	0	0	0	1	0	1	1	1	1	5	**High**
Ohene-Yeboah M, 2012 [50]	0	0	1	0	1	1	0	1	1	1	6	**Moderate**

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Ramdas Y, 2020 [51]	0	0	0	1	1	0	0	1	1	1	5	**High**
Rayne S, 2019 [52]	0	0	1	0	1	1	1	0	1	1	6	**Moderate**
Roy I, 2011 [53]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
Samaka RM, 2016 [54]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Sawe RT, 2017 [55]	0	1	1	0	1	1	0	1	1	0	6	**Moderate**
Schwartz T, 2012 [56]	0	0	0	1	1	1	0	1	1	1	6	**Moderate**
Sengal AT, 2017 [57]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Seshie B, 2015 [58]	0	1	1	1	1	1	0	1	1	0	7	**Moderate**
Shaikh AJ, 2018 [59]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Stark A, 2010 [60]	0	0	1	0	1	1	1	1	1	0	6	**Moderate**
Sung H, 2019 [61]	0	1	1	1	1	1	0	1	1	1	8	**Moderate**
Tanimowo MO, 2019 [62]	0	0	0	1	1	0	1	0	1	1	5	**High**
Traoré B, 2019 [63]	0	1	1	0	1	1	0	1	1	0	6	**Moderate**
Ukah CO, 2017 [64]	0	1	0	1	1	1	0	1	1	1	7	**Moderate**
Uyisenga JP, 2020 [65]	0	1	1	1	1	1	1	1	1	1	9	**Low**
Youngblood VM, 2020 [66]	0	1	1	1	1	1	1	1	1	1	9	**Low**
Znati K, 2012 [67]	0	0	0	0	1	0	0	1	1	0	3	**High**

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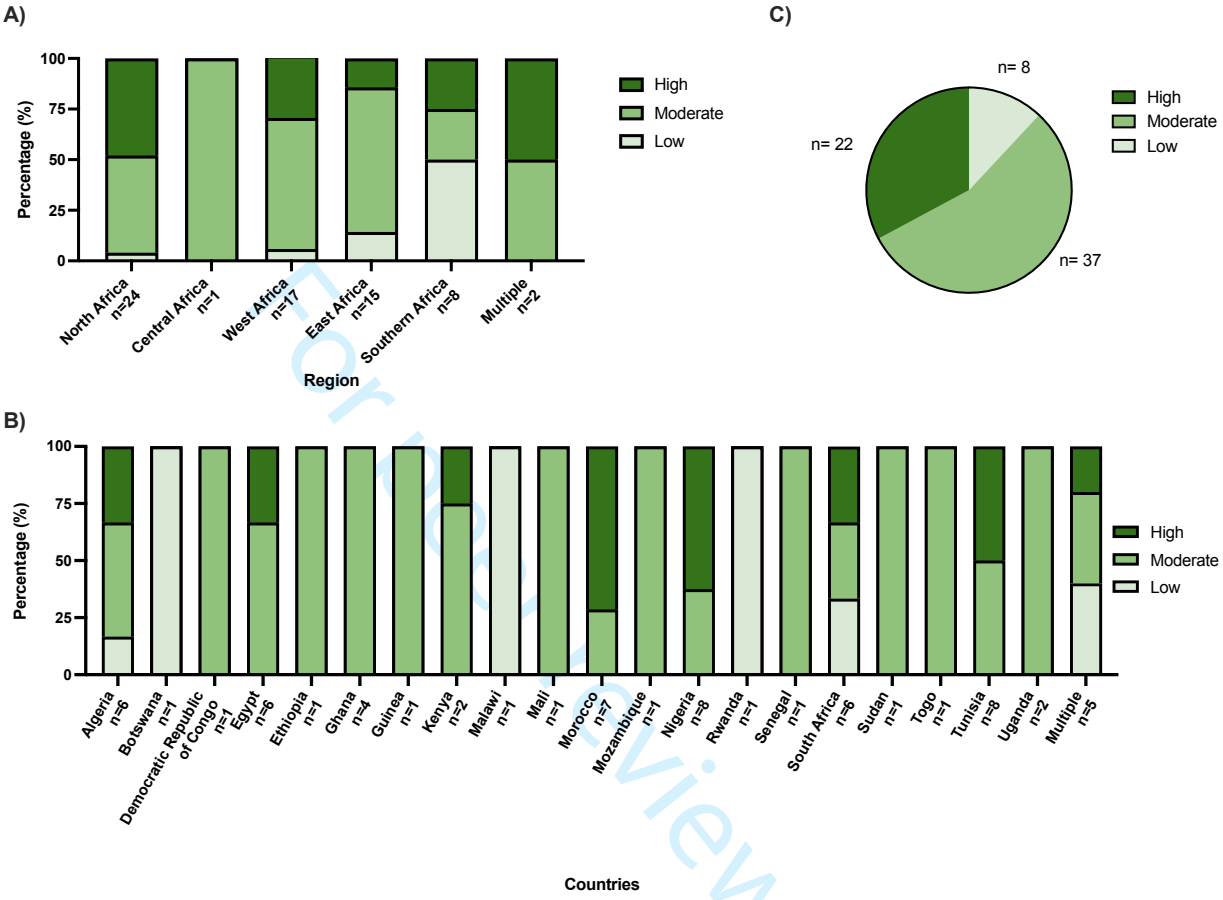
Table S5: Pooled TNBC Frequcny in Africa by Country

Country (no. of studies)	Total participants	Pooled TNBC Frequency, % [95% CI]	Heterogeneity (I^2), %	Weight, %
Algeria (*n*= 6)	6461	22.7 [20.3- 25.2]	77	12.0
Botswana (*n*= 1)	211	21.3 [16.3- 27.4]	not applicable	1.9
Democratic Republic of Congo (*n*= 1)	87	14.9 [8.9- 24.1]	not applicable	1.6
Egypt (*n*= 5)	640	24.5 [15.3- 36.7]	88	8.3
Ethiopia (*n*= 1)	110	22.7 [15.8- 31.5]	not applicable	1.8
Ghana (*n*= 4)	358	57.2 [43.6- 69.8]	82	6.8
Guinea (*n*= 1)	58	25.9 [16.2- 38.6]	not applicable	1.6
Kenya (*n*= 4)	364	24.5 [17.8- 32.8]	63	6.9
Malawi (*n*= 1)	100	25.0 [17.5- 34.4]	not applicable	1.8
Mali (*n*= 1)	114	46.5 [37.6- 55.7]	not applicable	1.9
Morocco (*n*= 7)	4744	15.6 [14.3- 17.1]	31	13.4
Mozambique (*n*= 1)	210	24.8 [19.4- 31.0]	not applicable	1.9
Nigeria (*n* = 4)	479	49.2 [41.4- 57.0]	61	7.3
Rwanda (*n*= 1)	138	37.7 [30.0- 46.0]	not applicable	1.9
Senegal (*n*= 1)	197	46.7 [39.8- 53.7]	not applicable	2.0
South Africa (*n*= 4)	950	14.0 [9.6- 19.8]	75	7.0
Sudan (*n*= 1)	415	34.5 [30.0- 39.2]	not applicable	2.0
Togo (*n*= 1)	117	37.6 [29.3- 46.7]	not applicable	1.9
Tunisia (*n*= 8)	1628	26.4 [18.5- 36.2]	92	14.6
Uganda (*n*= 2)	158	27.0 [14.3- 45.1]	77	3.4
Total	17539	27.0 [24.0- 30.2]	94	100

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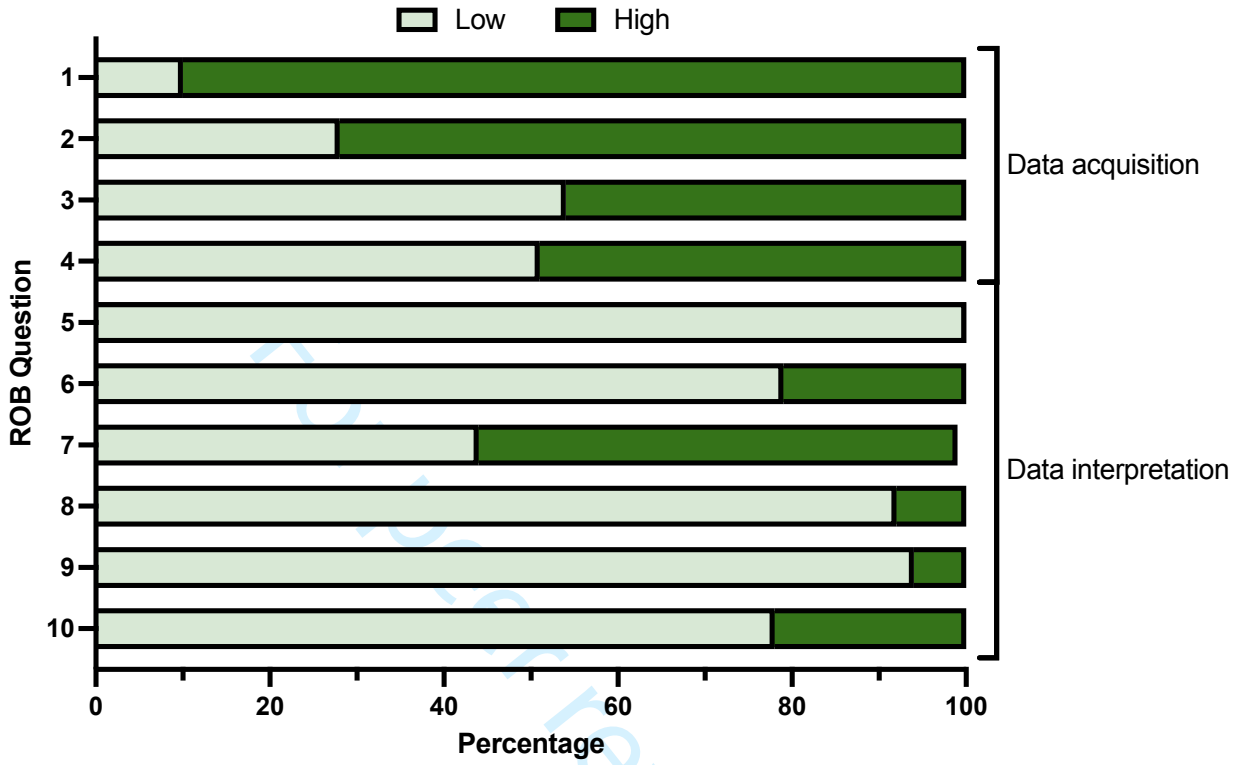
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Figure S1: Risk of bias assessment of all articles by (A) region, (B) country and (C) overall risk of bias assessment for all articles included. Southern Africa had the highest percentage of low risk of bias. Multiple denotes more than one country and/or region or was investigated within that study.



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Figure S2: Risk of bias (ROB) assessment per question for all articles included in systematic review. High risk of bias was noted specifically for questions on data acquisition



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Figure S3: Pooled TNBC frequency in Africa by risk of bias assessment. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

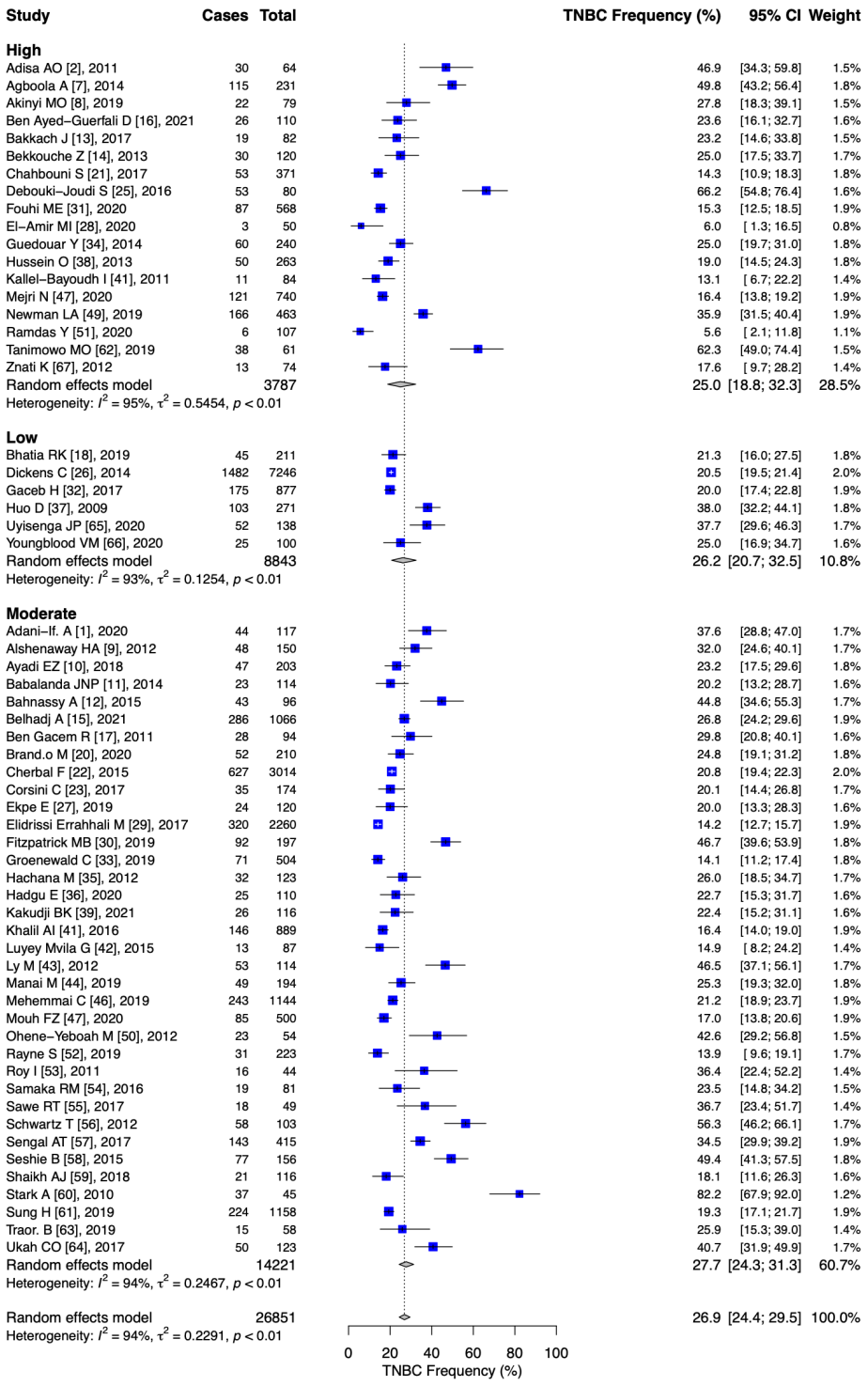
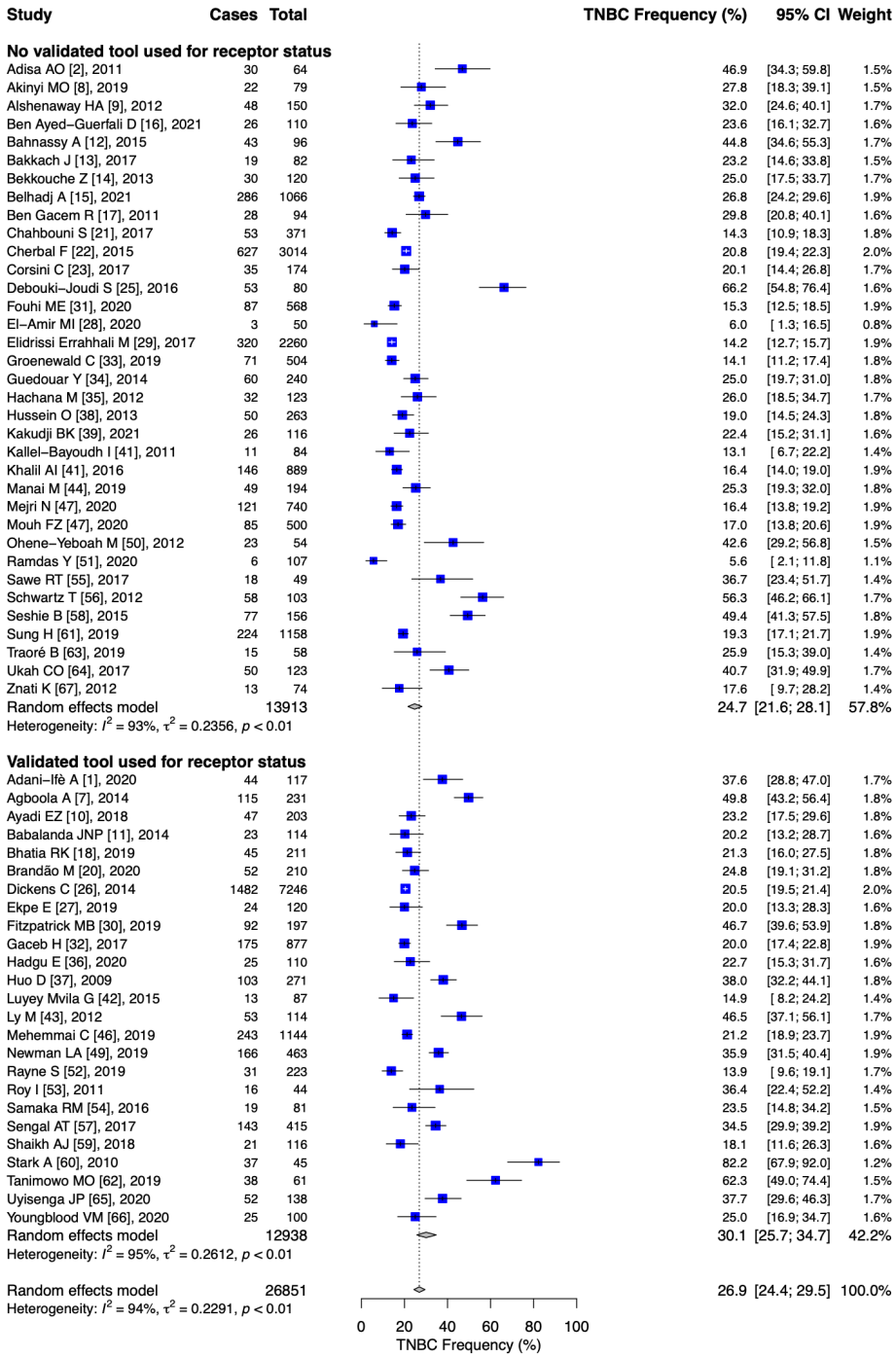


Figure S4: Pooled TNBC frequency in Africa by use of validated tool for receptor status. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.



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Figure S5: Meta-regression shows association with year of publication but not with age or use of validated tool for receptor status. The association of use of validated tool to assess receptor status (A), year of publication (B), mean age at diagnosis (C) and median age at diagnosis (D) on heterogeneity of treatment effects between studies was investigated by a meta-regression. P <0.05 indicates significant association.

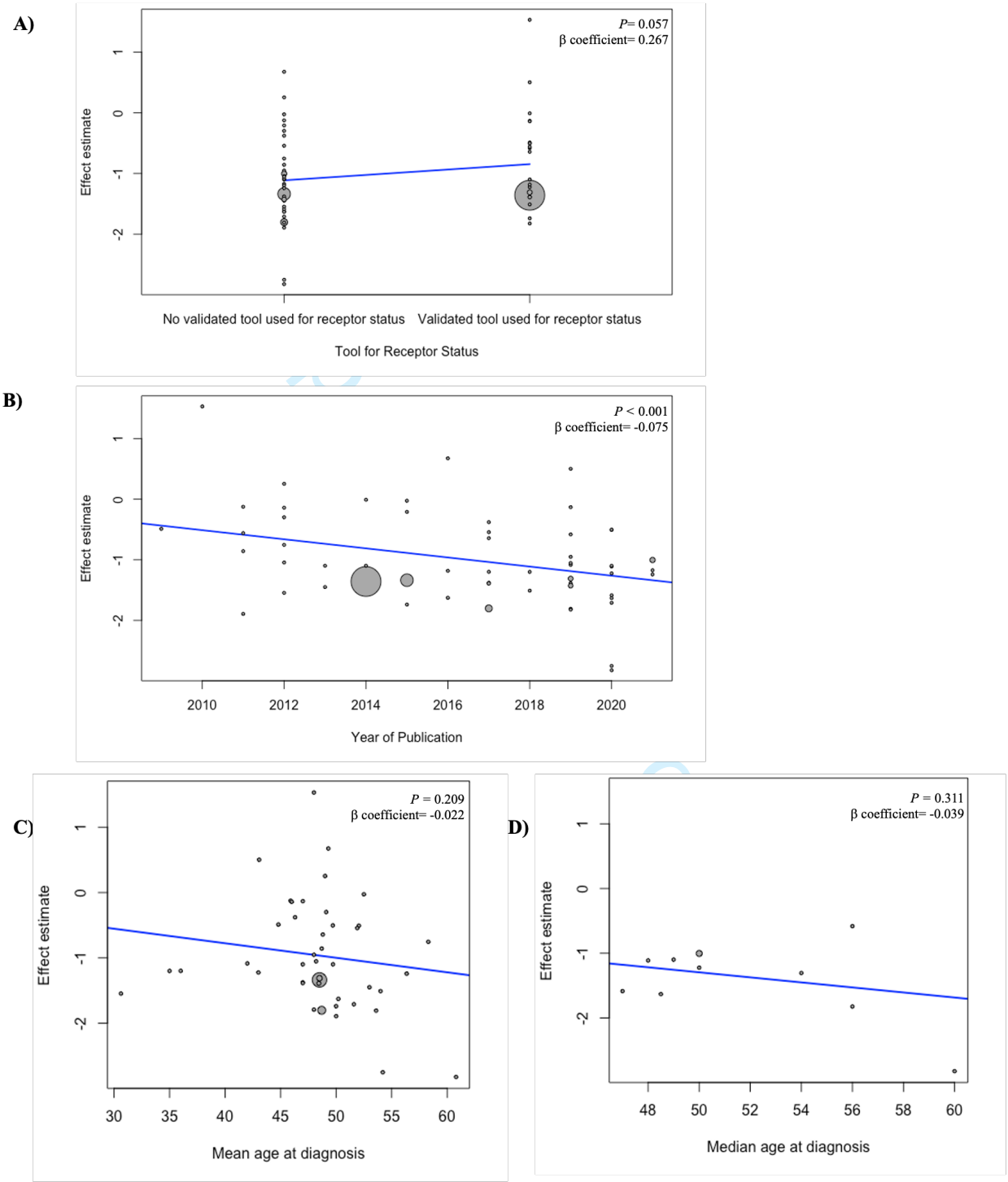
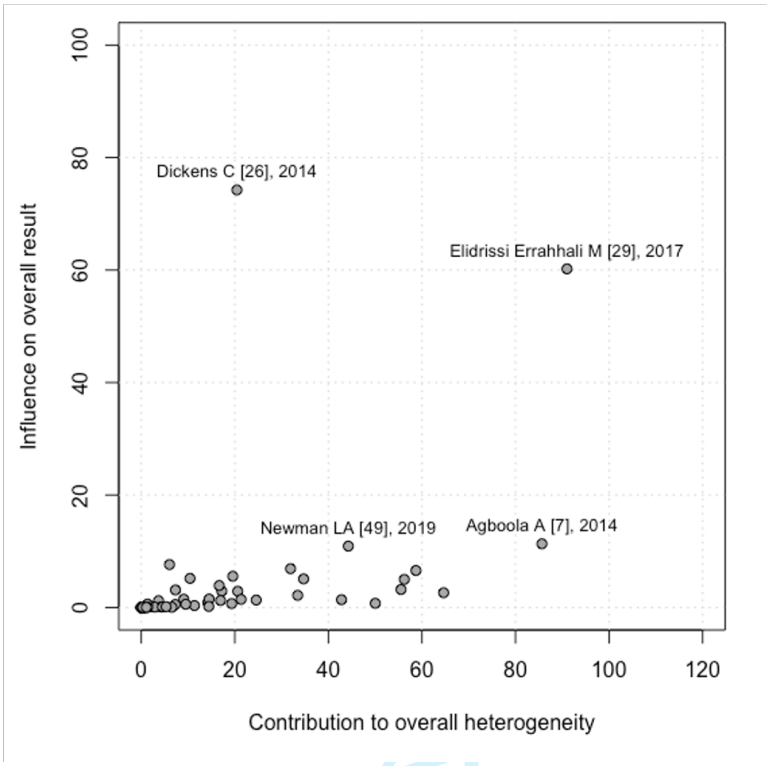


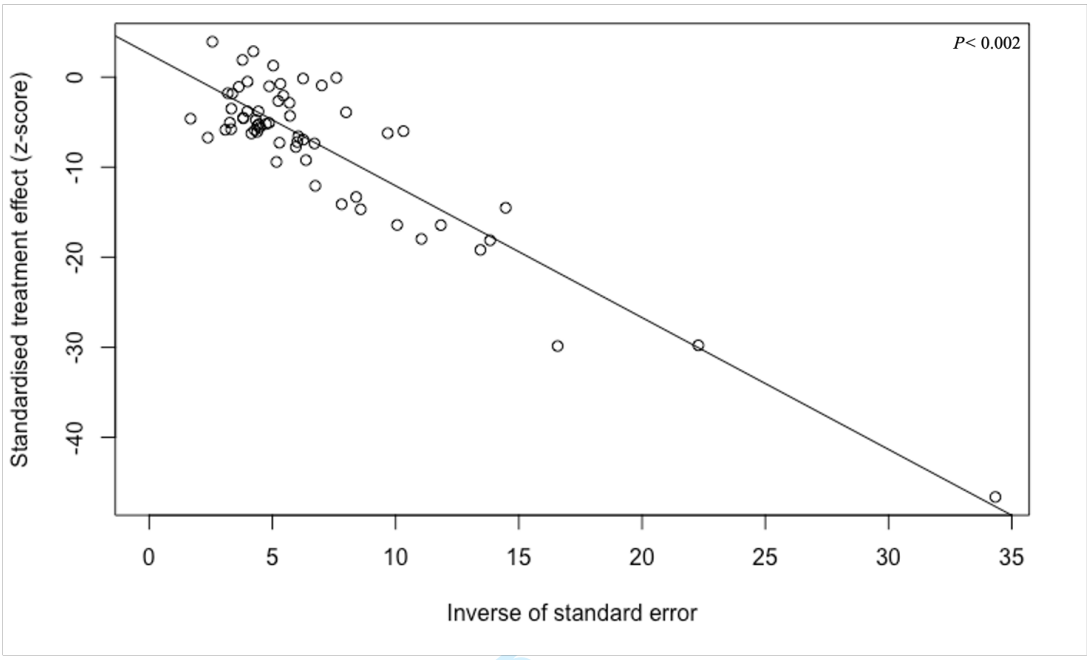
Figure S6: Influence analysis highlights studies that overly contribute to heterogeneity. Influence analysis via Baujat plot shows studies that two studies [7] and [29] contribute largely to overall heterogeneity.



view only

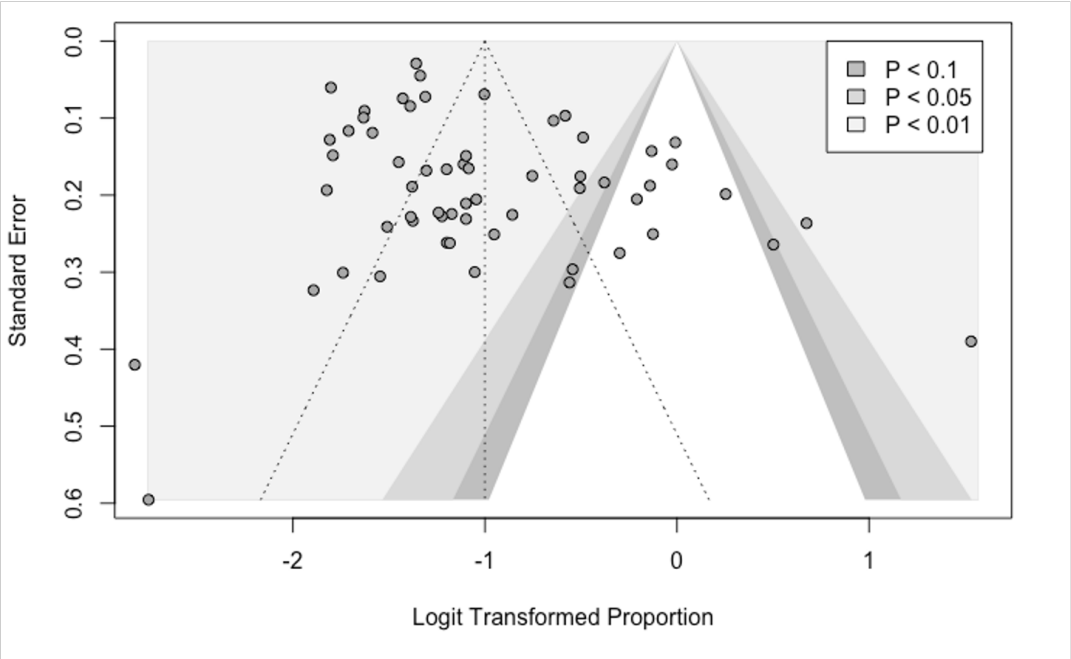
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Figure S7: Egger's test shows high publication bias. Publication bias between studies for TNBC frequency investigated using Egger's test and a significant association was found, inferring significant publication bias. P <0.05 indicates significant association.



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Figure S8: Funnel plot shows high publication bias. Publication bias between studies was investigated using funnel plot to visualize asymmetry and multiple studies do not follow the funnel shape inferring significant publication bias. P <0.05 indicates significant association.



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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 3~~1~~
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 6~~2~~
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 6, 7~~2~~
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 3~~,4,6~~7, 8; Table S1
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 3~~,4,6~~7, 8; Figure 1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table S1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 3~~,4,6~~7, 8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 3~~,4,6~~7, 8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 3~~,4,6~~7, 8,
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 3~~,4,6~~7, 8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 3~~,4,6~~8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 9~~5~~
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	N/A
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 4~~,5~~7,8,9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 4~~,5~~9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 9~~4,5~~
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 9~~5~~

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Section and Topic	Item #	Checklist item	Location where item is reported
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 84
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 85
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 9, 105; Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	N/A
Study characteristics	17	Cite each included study and present its characteristics.	Table S2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table S2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table S2
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Table S2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	All forest plots
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	All forest plots
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Figure S6-S8
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	All forest plotspage 13
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 136-179
	23b	Discuss any limitations of the evidence included in the review.	Page 6-913-17
	23c	Discuss any limitations of the review processes used.	Page 166-9
	23d	Discuss implications of the results for practice, policy, and future research.	Page 13-176-9
OTHER INFORMATION			
Registration and	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 4

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Section and Topic	Item #	Checklist item	Location where item is reported
protocol	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	N/A
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 185
Competing interests	26	Declare any competing interests of review authors.	Page 185
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Table S2

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Triple Negative Breast Cancer Prevalence in Africa: a Systematic Review and Meta-analysis

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Keywords:	Epidemiology < TROPICAL MEDICINE, ONCOLOGY, Breast tumours < ONCOLOGY

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Triple Negative Breast Cancer Prevalence in Africa: a Systematic Review and Meta-analysis

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Abstract:

Objective: The aggressive triple negative breast cancer (TNBC) subtype disproportionately affects women of African ancestry across the diaspora, but its frequency across Africa has not been widely studied. This study seeks to estimate the frequency of TNBC among African populations.

Design: Systematic review and meta-analysis using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework.

Data sources: PubMed, EMBASE, African Journals Online and Web of Science were searched on April 25th 2021.

Eligibility criteria for selecting studies: We included studies that use breast cancer tissue samples from indigenous African women with sample size of eligible participants ≥ 40 and full receptor status for all three receptors (ER/PR/HER2) reported.

Data extraction and synthesis: Two independent reviewers extracted data and assessed risk of bias using the modified assessment tool by Hoy D. *et al.* (2012) for prevalence studies. A random-effects meta-analysis was performed, and data were pooled using the inverse-variance method and logit transformation. Pooled frequencies were reported with 95% confidence intervals calculated with the Clopper-Pearson method and heterogeneity quantified with I^2 statistic. GRADE assessed the certainty of the evidence.

Results: 1808 potentially eligible studies were identified of which 67 were included in the systematic review and 60 were included in the meta-analysis. Pooled TNBC frequency across African countries represented was estimated to be 27.0%; 95% CI: 24.0%- 30.2%, I^2= 94%. Pooled TNBC frequency was highest across West Africa, 45.7% [n= 15, 95% CI 38.8- 52.8, I^2= 91%] and lowest in Central Africa, 14.9% [n= 1, 95% CI 8.9- 24.1]. Estimates for TNBC were higher for studies that used Allred guidelines for ER/PR status compared to ASCO/CAP guidelines, and for studies that used older versions of ASCO/CAP guidelines for assessing HER2 status. Certainty of evidence was assessed to be very low using GRADE approach.

Conclusion: TNBC frequency was variable with the highest frequency reported in West Africa. Greater emphasis should be placed on establishing protocols for assessing receptor status due to the variability among studies.

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Strengths and limitations of this study:

- Previous studies have reported higher ER-negativity among West African countries compared to East African countries, but no review or meta-analysis has been solely focused on the frequency of TNBC across the African continent.
- Certainty of evidence was assessed using the GRADE approach
- Evidence of high frequency of TNBC and poor prognostic clinical factors highlight the need for early breast cancer detection across the continent.
- There was evidence of variable methods used to assess receptor status across the African continent.
- Only 20 out of 54 African countries were included in this study where there was a lack of cancer registry and population-based data, which suggests that our estimate for TNBC frequency across the continent is not fully accurate; however, given the available data, this is the best estimate from included studies.

Keywords

Triple negative breast cancer, TNBC, Africa, breast cancer, women of African ancestry

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Patient and Public Involvement

Patients were not involved in this study.

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Introduction

Breast cancer (BCa) mortality rates have markedly increased across Africa where estimated age-standardized rates in 2020 ranged from 20 deaths per 100,000 women across Northern Africa to 27 deaths per 100,000 women in Western Africa [1-3]. Breast cancer has thus been dubbed an emerging epidemic in Africa [4]. Indeed, a recent systematic review and meta-analysis found that female BCa incidence rates increased from 23.1 to 26.3 per 100,000 between 2000 and 2015 across the continent [5]. Women of African ancestry (WAA) across the globe and indigenous African women are more likely to receive a poorer BCa prognosis compared to women of other ancestries [6]. Multiple studies often posit poorer prognosis as a result of healthcare systems across Africa, where there is limited capacity and health infrastructure, *e.g.* inadequate screening and diagnostic services [7]. However, BCa prevalence by subtype, as defined by receptor status, is strikingly different for indigenous African women compared to the BCa prevalence profile of Western countries and is not simply explained by limited access to healthcare [6].

Immunohistochemistry (IHC) is routinely used to classify BCa into molecular subtypes according to the presence or absence of the estrogen (ER) and progesterone (PR) receptors and the human epidermal growth factor receptor 2 (HER2). Triple negative breast cancer (TNBC) is characterized by the lack of expression of all three biomarkers (ER, PR, HER2), which makes TNBC untreatable with targeted therapies such as Tamoxifen and Herceptin [8, 9]. This BCa subtype is often associated with earlier disease-onset, advanced-stage tumors and aggressive disease progression when compared with other BCa subtypes [10, 11]. Additionally, TNBC has been shown to disproportionately affect African women, and younger WAA and Hispanic women in North America [11], where the prevalence of TNBC in WAA has been estimated to be more than twice the prevalence in non-Hispanic White women [12]. There is also a higher mortality rate from TNBC and more advanced stage at diagnosis in WAA [13]. Thus, it is important to investigate what seems to be an ancestral predisposition to TNBC since the reasons for this disparity in TNBC prevalence and outcomes are not fully understood. Studies to date have not compiled adequate information on TNBC frequency or considered reported frequency and routine practices associated with diagnosis and treatment across the African continent.

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This systematic review and meta-analysis aims to increase understanding and knowledge regarding the frequency of TNBC across the African continent. The paucity of data and information on TNBC in Africa underscores the importance and urgency of such a review. A previous review reported higher ER-negativity among West African countries compared to East African countries [6] and a previous meta-analysis investigated ER-positivity across Africa [14] but no review or meta-analysis has been solely focused on the frequency of TNBC cases across the African continent. This review complements current biomedical research on TNBC and provides context for areas where TNBC research continues to expand. Improved understanding of TNBC frequency in continental Africa can further inform strategies for BCa detection and management for WAA globally. Due to shared ancestry between North American WAA and indigenous West African women [15], we hypothesize that there will be higher TNBC prevalence rates in countries across West Africa compared to other regions (North, East, Central, Southern) across Africa.

METHODS

Search Strategy and Selection Criteria

We used Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines as a framework for our systematic review and meta-analysis [16] as well as Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines [17]. On April 25th 2021, we searched PubMed, EMBASE, African Journals Online and Web of Science for relevant articles without date or language restrictions. Start date of the search was from inception of each database. A detailed version of our search strategy used in PubMed was modified for other databases. Search strategy and these modifications can be found in Table S1. Briefly, all search terms were Medical Subject Heading terms, including TNBC terms ("TNBC", "triple-negative*", "triple negative") and terms for African countries ("Africa", "African" and names of all 54 African countries) and outcome variables ("rate*", "prevalence", "epidemiology"). We included all studies that met the inclusion criteria. The inclusion criteria were as follows: studies that use breast cancer tissue samples from indigenous African women of any age, in any

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care setting and at any geographic location; sample size of eligible participants ≥ 40 (as slightly more stringent criteria since normal distribution could be assumed at n= 30 [18]); studies that demonstrate at least one of the following: report on receptor status of breast tumours including ER, PR, and HER2; any primary study from which TNBC frequency could be estimated among BCa cases, including but not limited to observational studies, cross-sectional studies and case-control studies where controls were not included in TNBC frequency calculations.

We excluded: editorials, single case reports, case series, and commentaries; studies that assessed diagnostic measures and treatment options for women with TNBC in the absence of assessment of its frequency; studies conducted in non-African nations without assessment of indigenous African TNBC rates or that of first-generation African immigrants. Study selection began by screening titles and abstracts of articles collected after employing the search strategy. The full text of these articles was then reviewed to assess inclusion. Two data abstractors independently reviewed articles at both the title/abstract and full text review stages. When there were discrepancies, a consensus was made in consultation with a third reviewer. The protocol for this review was not registered. Non-English studies were included after translation through Google Translate followed by verification of translation by a French speaker.

Quality Assessment

Studies which passed full text review (Table S2) were evaluated for risk of bias using a tool developed by Hoy and colleagues specifically intended for prevalence-studies [19]. Each study was assessed according to ten items assessing internal validity (Table S3) and assigned to have either low (score of 1) or high (score of 0) risk of bias for each question by two independent reviewers. A third reviewer mediated discrepancy and a final score per question was agreed upon. Studies were then classified based on the total score for all questions in the quality assessment tool as having a high (<=5), moderate (6-8), or low (>= 8) risk of bias.

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Assessing the Certainty of Evidence

The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework [20] was used to assess the overall certainty of available evidence on the frequency of TNBC across the African continent. This framework considered factors such as study design, risk of bias, inconsistency, indirectness, imprecision, and publication bias.

Data Analysis

After reviewing full-text articles, a heat map was constructed with the number of studies, TNBC frequency and number of participants per country across African populations for unique studies using Google Sheets. All meta-analyses, meta-regressions and sensitivity analyses were completed using R (version 4.0.2) [21]. Using the *metaprop* package in R, we conducted a meta-analysis of TNBC frequency among indigenous African women with breast cancer, stratified by country, region, risk of bias assessment, year of publication and the use of a validated tool for assessing receptor status. Logit transformation was used to stabilize the variances and a random effects model with inverse-variance method for pooling frequencies and Clopper-Pearson method for calculating confidence intervals were used for our meta-analyses. Pooled TNBC frequency was estimated separately per country and per region as two studies included data from more than one region. When studies investigated African and non-African participants, only data from African participants were included in meta-analyses. Heterogeneity between studies was assessed with Cochrane's Q, I^2, and H statistics. Meta-regression was done to explore heterogeneity using *metareg* package. We used Egger's test to investigate publication bias and small study effects using the *metabias* package.

RESULTS

Of the 1808 records identified, 1032 remained after removing duplicates from the various databases. After screening titles and abstracts, 932 records were excluded due to irrelevance. The full text for the remaining 109 records were screened and an additional 42 studies were excluded because they did

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not meet eligibility criteria, leaving 67 relevant studies for inclusion (Figure 1). Our search strategy identified eligible studies from 20 countries across continental Africa. Nigeria and Tunisia had the highest number of eligible studies (8 studies each), followed by Morocco (7 studies), Algeria, Egypt, and South Africa (6 studies each), Ghana and Kenya (4 studies each) and South Africa (3 studies); there were only two studies from Uganda and only one study each from Botswana, Democratic Republic of Congo, Ethiopia, Guinea, Malawi, Mali, Mozambique, Rwanda, Senegal, and Sudan (Table S2). Five studies included data from multiple countries and regions across the continent (East and West Africa, Algeria, Egypt, Ethiopia, Ghana, Morocco, Namibia, Senegal, South Africa, and Tunisia). Five studies were translated from French to English and were all based in North Africa. Summary of clinical data can be found in Table 1 and Table S2.

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Table 1: Summary of study variables for included studies across Africa and by Region

Study variables	All studies (*n*= 67)	North African studies (*n*= 29)	East African studies (*n*= 10)	Southern African studies (*n*= 8)	Central African study (*n*= 1)	West African studies (*n*= 17)	West and East African studies (*n*= 2)
Mean age [SD]	49.21 [5.25] (*n*= 42)	48.47 [6.35] (*n*= 17)	48.65 [3.34] (*n*= 7)	56.09 [2.53] (*n*= 5)	50.00 [n/a] (*n*= 1)	47.67 [2.65] (*n*= 12)	not investigated
Median age [IQR]	48.75 [4.50] (*n*= 18)	47.00 [5.72] (*n*= 9)	48.75 [0.87] (*n*= 4)	56.00 [3.00] (*n*=3)	49.00 [n/a] (*n*=1)	not investigated	56.00 [n/a] (*n*= 1)
Grade 1 tumors, median [IQR]	9.56% [7.34] (*n*= 52)	8.44% [6.90] (*n*= 23)	10.66% [9.35] (*n*= 8)	11.40% [1.95] (*n*= 6)	1.15% [n/a] (*n*= 1)	6.06% [11.18] (*n*= 13)	W: 9.56% [n/a], E: 15.25% [n/a] (*n*= 1)
Grade 2 tumors, median [IQR]	51.11% [16.97] (*n*= 53)	57.54% [19.03] (*n*= 23)	44.44% [6.90] (*n*= 8)	46.76% [16.32] (*n*= 7)	51.72% [n/a] (*n*= 1)	51.22% [20.51] (*n*= 13)	W: 33.52% [n/a], E: 30.81% [n/a] (*n*= 1)
Grade 3 tumors, median [IQR]	38.68% [15.04] (*n*= 58)	33.76% [16.40] (*n*= 27)	41.11% [13.92] (*n*= 8)	39.53% [16.90] (*n*= 7)	47.13% [n/a] (*n*= 1)	38.83% [18.89] (*n*= 14)	W 56.92% [n/a], E 43.95% [n/a] (*n*= 1)
Positive lymph node status, median [IQR]	64.68% [20.99] (*n*= 42)	58.50% [15.40] 23)	70.53% [11.91] (*n*= 6)	64.66% [28.00] (*n*= 3)	not investigated	91.38% [25.07] (*n*= 10)	not investigated
Premenopausal, median [IQR]	58..86% [18.16] (*n*= 30)	58.40% [13.87] (*n*= 13)	57.00% [17.19] (*n*= 5)	37.10% [n/a] (n= 1)	60.92% [n/a] (*n*= 1)	68.48% [17.26] (*n*= 7)	not investigated
TNBC frequency, median [IQR]	25.00% [19.41] (*n*= 67)	23.00% [9.02] (*n*= 29)	25.00% [13.08] (*n*=10)	20.43% [7.32] (*n*= 8)	14.94% [n/a] (*n*= 1)	49.4-% [13.60] (*n*= 17)	n/a

* two studies investigated both West and East African populations in their respective manuscripts, W- West African participants, E- East African participants
n/a- not applicable

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All 67 studies reported TNBC frequency from specific hospital/health facility settings, 34 of which were conducted via academic and academic/university teaching hospitals. Fourteen studies were prospective whereas the others (n= 53) were retrospective studies. Additionally, 20 studies included some form of biased sampling (*e.g*., all metastatic cases, tissue microarrays, age cut-offs) whereas the remaining studies included were either random sampling or population-based. Of the included studies, eight (12%), thirty-seven (55%) and twenty-two (33%) were classified as low, moderate and high risk of bias, respectively, (Figure S1-S2, Table S3-S4) after using the risk of bias assessment tool for prevalence studies by Hoy *et al.* [19]. Most studies were scored as high risk of bias due to data acquisition (*e.g.,* study population). However, according to criteria set by the Hoy *et al.* risk of bias tool, data were interpreted appropriately for most studies (e.g., having a clear definition of TNBC, appropriate numerator/denominator for frequencies). The eight studies that were low risk of bias, were based in Algeria (*n*= 1), Botswana (*n*= 1), Malawi (*n*= 1), Rwanda (*n*= 1), South Africa (*n*= 2) or multiple countries (*n*= 2).

After identifying unique study populations per country (*n*= 60, Figure 1), TNBC estimates from the meta-analysis, number of studies and participants per country were highlighted (Figure 2). Overall TNBC frequency from included studies representing countries across the African continent was estimated to be 27.0%; 95% CI: 24.0%- 30.2%, I^2= 94% (Table S5). Pooled TNBC frequency estimates per country (Table S5) ranged from 14.0% in South Africa (95% CI: 9.6%-19.8%, I^2 = 75%) to 57.2% in Ghana (95% CI: 43.6%-69.8%, I^2 = 82%). When investigating estimates per region (Figure 3), TNBC frequency was lowest in Central Africa (n=1, 14.1%; 95% CI 8.9%-24.1%) and highest in West Africa (n=15, 45.7%; 95%CI: 38.8%-52.8%, I^2 = 91%). For these two analyses with pooled estimates per country and per region, heterogeneity (I^2) was estimated at 94%, indicative of high between study variability. When investigating the effect of risk of bias on study estimates, none was observed (Figure S3). Pooled TNBC estimates were also stratified by use of a validated tool for receptor status testing. Pooled TNBC frequency was higher (n= 25, 30.1%; 95%CI: 24.4%-34.7%, I^2 = 95%) in studies that reported the use of a validated tool for assessing receptor expression, when compared to studies that did not (n= 35, 24.7%; 95%CI: 21.6%-28.1%, I^2 = 93%)

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(Figure S4). Between study heterogeneity was high (I^2= 94%) and meta-regression showed that these estimates were not statistically significant (Figure S5A, P = 0.057, β coefficient= 0.267). When investigating TNBC estimates by the tool used for ER/PR and HER2 cutoffs, pooled TNBC frequency was higher in studies that used the Allred 1998 and Reiner's scale scoring for ER/PR cutoffs and in older versions of ASCO/CAP guidelines when compared to more recent versions for HER2 status (Figure 4). There was also an association between publication year and TNBC estimates where meta-regression showed a decrease in effect estimate with increasing publication year (Figure S5B, $P < 0.001$, β coefficient= -0.075). Influence analyses showed that two studies contributed largely to heterogeneity and influence on overall estimates (Figure S6). After conducting Egger's test (Figure S7, $P < 0.002$) and funnel plot analysis (Figure S8), a significant publication bias was identified.

The overall certainty of evidence for this systematic review and meta-analysis was judged as very low using the GRADE approach. Our judgement was downgraded due to risk of bias, indirectness, and publication bias.

Discussion

In this systematic review and meta-analysis of 67 studies on African women with BCa, we found that there was a high frequency of TNBC (27.0%) in cases reported across Africa although it varied depending on country and region. TNBC frequency was highest in West African populations (45.7%) compared to other regions across continental Africa (14.9%-22.7%). This is consistent with increased TNBC/ER-negative prevalence observed in populations with high West African ancestry [15, 22, 23] in the Caribbean [24, 25] and in North America [26]. In a USA population-based study (2010-2014), TNBC prevalence in non-Hispanic White women was estimated to be ~8% whereas it was ~15% in non-Hispanic Black women [23]. Additionally, TNBC prevalence was estimated to be ~8% and ~25% in White and Black women respectively in a UK cancer registry-based population in London [27]. These high frequencies of TNBC across Africa and the African diaspora are concerning as triple negative breast tumors have a greater

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propensity to metastasize to vital organs such as the brain [28] and are typically more aggressive due to lack of targeted therapies.

When investigating clinical factors, the reported mean and/or median age at diagnosis was under 50 in 35 out of the 47 studies reporting age. Young age at diagnosis (under age 40) has been previously reported to be associated with triple negative and HER2-positive cancers as well as more aggressive clinical outcome [29]. Indeed, this poor prognosis of TNBC patients was evident as most of the included studies reported a high percentage of grade 3 tumours, lymph node positivity and TNBC frequency. It must however be noted that a younger age at diagnosis is also routinely observed in lower- and middle-income countries as this is also reflective of the population structure [30]. Therefore, this observed lower age at diagnosis may be indicative of the population distribution rather than the intrinsic aggressive biology of the tumors. To consider this possibility, we investigated associations with mean and median age at diagnosis and effect estimates and found no association (Figure S5C p= 0.209, β coefficient= -0.022, Figure S5D p= 0.311, β coefficient= -0.039). More advanced stage tumors and lymph node involvement at presentation may also be attributable to poor infrastructure and lack of BCa awareness and screening. A recent study of BCa across sub-Saharan Africa found that the majority of cases diagnosed were late-stage, emphasizing the need for early diagnosis [31]. Two separate studies from Nigeria and Ghana both found that most of the information obtained about BCa was from mass media and there was a general poor knowledge of BCa-associated risk factors [32, 33]. The Ghanaian-based study also found that the rate of breast self-examination (BSE), and clinical breast examination (CBE) were higher than that of obtaining mammograms [33] which emphasizes the need to promote screening programs in a culturally-relevant setting. It should be noted however that mammography has been associated with a two times higher chance of detecting ER-positive BCa compared to ER-negative BCa [34] which might be contributing to the relatively higher TNBC

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frequency observed across West African countries when compared to Southern African countries where mammography is more accessible.

Many studies were excluded on the account of not assessing ER, PR and HER2 status. With respect to receptor status, 30 out of the 67 studies used validated guidelines (ASCO/CAP [35-40], Allred [41] or Reiner's Scale [42]) for receptor status cut-offs. Such variability in classifying receptor status (i.e., the use of other guidelines with different cut-offs) affects the resulting treatment for patients with a diagnosis and how TNBC frequencies are calculated in each study. After stratifying studies by use of ASCO-CAP guidelines which account for specimen fixation and cut-offs for ER/PR expression at 1%, there was a decrease in pooled TNBC frequency (24.4%) compared to those that used Allred or Reiner's Scale (cutoff at 10%) resulting in TNBC frequency of 42.9% and 38.0% respectively (Figure 4A). A similar trend was observed for ASCO/CAP guidelines with respect to assessing HER2 status where more recent guidelines correlated with lower TNBC frequency compared to older guidelines (Figure 4B). Thus, meta-regression with publication year was done and indeed there was an association with effect estimates ($P < 0.002$, β coefficient= -0.075, Figure S5B). A similar trend was recently reported for East Africa-based studies conducted before and after 2013; ER/PR positivity was lower before 2013 compared to after 2013 [43]. The variability in how receptor status is assessed highlights a need for increased capacity to conduct immunohistochemical receptor status testing to further enhance BCa diagnoses and classification. It must be noted however, that IHC might not be feasible for many of the hospitals/health centers across Africa and international collaborations should be encouraged to assist with building such capacity. One Nigerian study noted cost to be a barrier – IHC was performed on only 31% of the reported cases [44]. It also stated that ASCO/CAP guidelines could not be adhered to for HER2 due to the high cost of fluorescence *in situ* hybridization in the case of an equivocal HER2 score. In contrast, South Africa has an extensive healthcare system with a comprehensive standardized national public health system for routinely assessing breast

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cancer receptor status [45]. The disparity in access to diagnostic and therapeutic tools across the continent could also be contributing to the lack of receptor status data and higher BCa burden reported here.

To our knowledge, this is the first systematic review and meta-analysis with an in-depth analysis on TNBC frequency across continental Africa. However, there are some limitations to be considered. IHC and specimen collection and processing are not equally accessible, and neither are they uniformly done across the African continent. Specimen fixation, storage time of the samples and other pre-analytical IHC variables have been previously shown to impact accuracy of IHC results [46]. It was estimated that up to 20% of IHC results globally are inaccurate based off of these pre-analytical variables [47]. The true frequency of TNBC is therefore not ascertainable due to lack of population-based data. Our search strategy identified representation from 20 African countries to be included with low representation from Central Africa (n=1) which was similarly observed in a recent systematic review and meta-analysis that investigated BCa incidence across Africa with 22 African countries and low representation from Central Africa (n=2) [5]. Another caveat to consider is that there are no validated search strategies for observational studies thus, some studies may be missed. As expected, there was considerable heterogeneity in our meta-analyses. High heterogeneity could be due to the large number of studies, or the varying ethnicities across Africa. We graded our confidence in the evidence presented as "very low" using the GRADE assessment due to high heterogeneity, high risk of bias studies included and low representation across the continent (indirectness); however, given the available data, this is the best estimate of TNBC frequency across the African continent.

This study provides the closest estimate of TNBC frequency across the different regions of continental Africa. Considerations should be made at the country-level to address IHC protocols and adherence to ASCO-CAP guidelines wherever possible. There is a clear disparity across the continent (with respect to diagnostic and therapeutic tools) that needs to be effectively addressed to prevent BCa burden. Priority should also be given to implementing culturally relevant BCa awareness programs as these have been proven to increase cancer awareness knowledge and thus could decrease preventable deaths from BCa

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[48]. There is also a dearth of knowledge across the continent about BCa subtype prevalence in general. This should be addressed as soon as possible by the establishment of cancer registries before the burden of BCa and other chronic diseases drastically increase with the epidemiological transition that has already started to take place across Africa.

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Data sharing statement:
The data underlying this article are available in the article and in its online supplementary material.

Ethical approval statement:
Not necessary for this manuscript.

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Figure legends:

Figure 1: PRISMA flowchart accounting for all articles included in the narrative synthesis and meta-analysis.

Figure 2: Pooled TNBC frequencies out of all BCa subtypes from studies done across African countries. Data represent pooled TNBC frequencies among all BCa subtypes reported in unique studies done in populations from stated countries. Pooled frequencies were calculated if the country had more than one study as stated within the meta-analysis. Estimates do not account for heterogeneity, IHC cut-offs and size of the respective populations.

Figure 3: Pooled TNBC frequency in Africa by region. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

Figure 4: Pooled TNBC frequency in Africa by tool used for (A) ER/PR status (B) HER2 status. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

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Figure 1: PRISMA flowchart accounting for all articles included in the narrative synthesis and meta-analysis

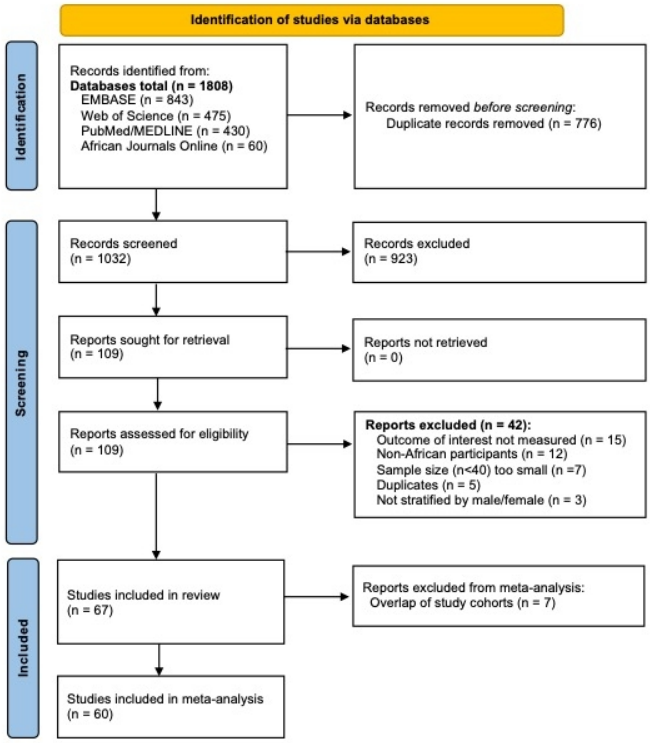


Figure 1: PRISMA flowchart accounting for all articles included in the narrative synthesis and meta-analysis.

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Figure 2: Pooled TNBC frequencies out of all BCa subtypes from studies done across African countries. Data represent pooled TNBC frequencies among all BCa subtypes reported in unique studies done in populations from stated countries. Pooled frequencies were calculated if the country had more than one study as stated within the meta-analysis. Estimates do not account for heterogeneity, IHC cut-offs and size of the respective populations

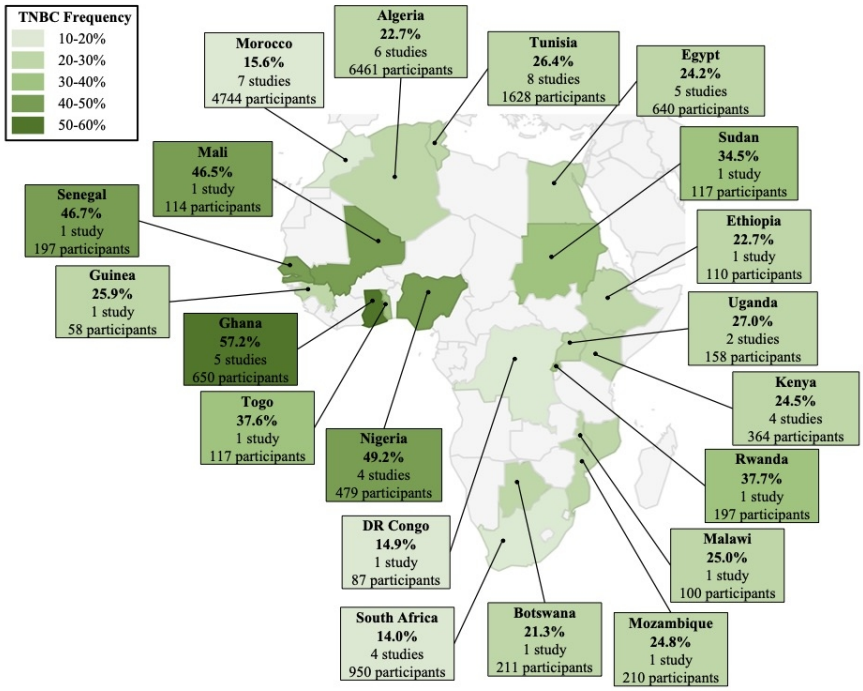


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Figure 3: Pooled TNBC frequency in Africa by region. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

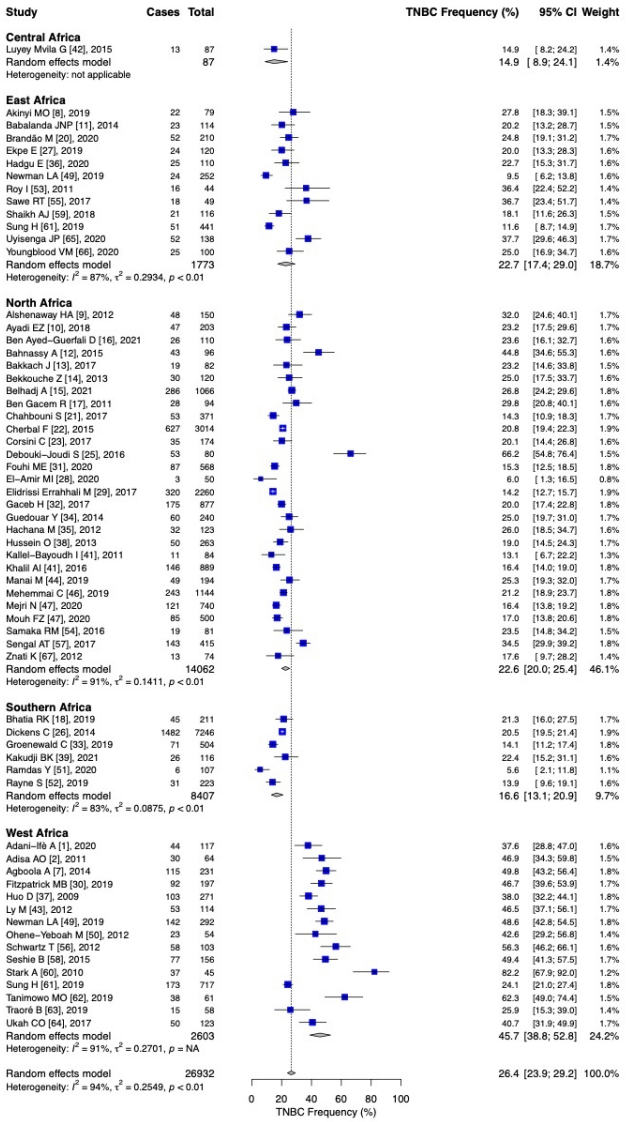


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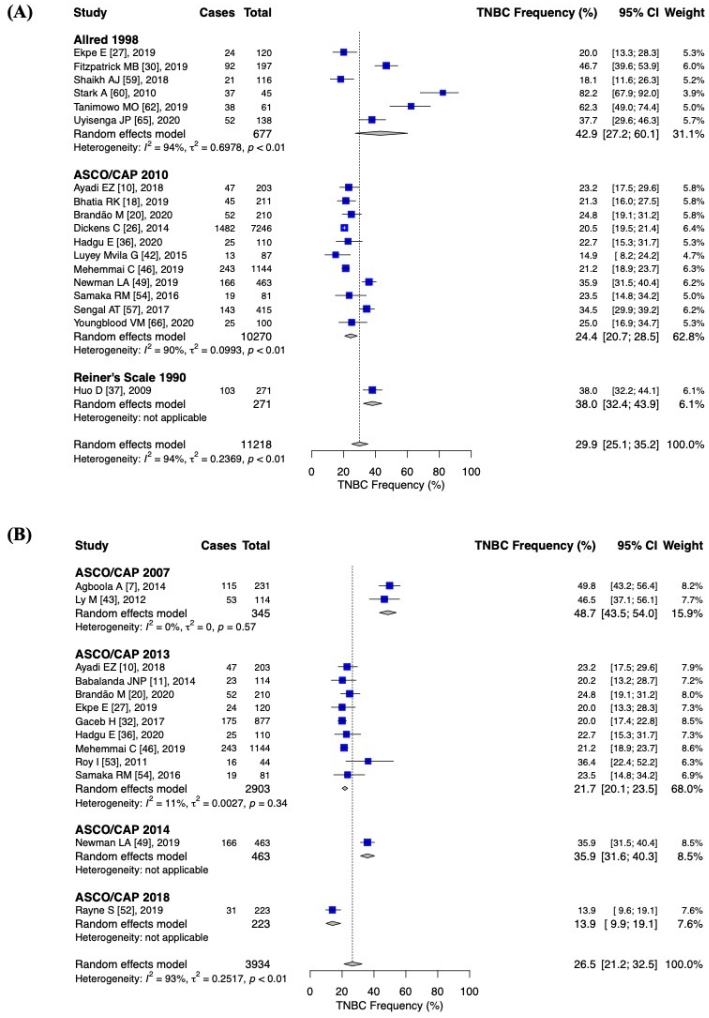


Figure 4: Pooled TNBC frequency in Africa by tool used for (A) ER/PR status (B) HER2 status. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

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SUPPORTING DOCUMENTS

Table S1: Search strategy used for Databases

Database	Search terms
Pubmed/MEDLINE	(Africa OR African OR "West* Africa" OR "East* Africa" OR "South* Africa" OR "North* Africa" OR "Central Africa" OR Algeria OR Angola OR Benin OR Botswana OR "Burkina Faso" OR Burundi OR Cabo Verde OR Cameroon OR Central African Republic OR "CAR" OR Chad OR Comoros OR "DRC" OR Congo OR "Republic of Congo" OR "Ivory Coast" OR Cote d'Ivoire OR Djibouti OR Egypt OR "Equatorial Guinea" OR Eritrea OR Eswatini OR Swaziland OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR "Guinea-Bissau" OR Kenya OR Lesotho OR Liberia OR Libya OR Madagascar [PL] OR Malawi OR Mali OR Mauritania OR Mauritanie OR Mauritius OR Morocco OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR "Sao Tome and Principe" OR Senegal OR Seychelles OR "Sierra Leone" OR Somalia OR "South Africa" OR "South Sudan" OR Sudan OR Tanzania OR Togo OR Tunisia OR Uganda OR Zambia OR Zimbabwe) AND (TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND (rate* OR prevalence OR incidence OR epidemiology)
Web of Science	TS = (Africa or African or "West* Africa" or "East* Africa" or "South* Africa" or "North* Africa" or "Central Africa" or Algeria or Angola OR Benin OR Botswana OR "Burkina Faso" OR Burundi OR Cabo Verde OR Cameroon OR Central African Republic OR "CAR" OR Chad OR Comoros OR "DRC" OR Congo OR "Republic of Congo" OR "Ivory Coast" OR Cote d'Ivoire OR Djibouti OR Egypt OR "Equatorial Guinea" OR Eritrea OR Eswatini OR Swaziland OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR "Guinea-Bissau" OR Kenya OR Lesotho OR Liberia OR Libya OR Madagascar [PL] OR Malawi OR Mali OR Mauritania OR Mauritanie OR Mauritius OR Morocco OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR "Sao Tome and Principe" OR Senegal OR Seychelles OR "Sierra Leone" OR Somalia OR "South Africa" OR "South Sudan" OR Sudan OR Tanzania OR Togo OR Tunisia OR Uganda OR Zambia OR Zimbabwe) AND TS= (TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND TS= (rate* OR prevalence OR incidence OR epidemiology)
Embase	(Africa OR African OR "West* Africa" OR "East* Africa" OR "South* Africa" OR "North* Africa" OR "Central Africa" OR Algeria OR Angola OR Benin OR Botswana OR "Burkina Faso" OR Burundi OR Cabo Verde OR Cameroon OR Central African Republic OR "CAR" OR Chad OR Comoros OR "DRC" OR Congo OR "Republic of Congo" OR "Ivory Coast" OR Cote d'Ivoire OR Djibouti OR Egypt OR "Equatorial Guinea" OR Eritrea OR Eswatini OR Swaziland OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR "Guinea-Bissau" OR Kenya OR Lesotho OR Liberia OR Libya OR Madagascar OR Malawi OR Mali OR Mauritania OR Mauritanie OR Mauritius OR Morocco OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR "Sao Tome and Principe" OR Senegal OR Seychelles OR "Sierra Leone" OR Somalia OR "South Africa" OR "South Sudan" OR Sudan OR Tanzania OR Togo OR Tunisia OR Uganda OR Zambia OR Zimbabwe) AND (TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND (rate* OR prevalence OR incidence OR epidemiology)
African Journals Online (AJOL)	(TNBC OR "triple-negative*" OR "triple negative*" OR "triple negative") AND (rate* OR prevalence OR incidence OR epidemiology)

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Table S2: Summary of data from included studies

Author	Population	Region	Mean age ± SD	Median age (min-max)	Grade 3 tumors	Positive Nodes	Number of TNBC cases	Participants studied with known receptor status (all participants)	Frequency of TNBC	Risk of Bias Assessment
Adani-Ifè A, 2020 [1]	Togo	West Africa	52.05 ±12.38	(30-85)	42.70%	41.90%	44	117 (312)	37.60%	Moderate
Adisa AO, 2011 [2]	Nigeria	West Africa	45.90	(20-81)	/	/	30	64 (385)	46.88%	High
Agboola A, 2014 [3]	Nigeria	West Africa	/	/	35.70%	99.20%	99	155 (199)	54.36%	High
Agboola A, 2014 [4]	Nigeria	West Africa	/	/	/	94.20%	55	100 (308)	55.00%	Moderate
Agboola A, 2013 [5]	Nigeria	West Africa	/	/	/	92.70%	81	149 (308)	63.87%	High
Agboola A, 2016 [6]	Nigeria	West Africa	/	/	34.12%	91.76%	114	187 (255)	60.96%	Moderate
Agboola A, 2014 [7]	Nigeria	West Africa	/	/	38.50%	92.20%	115	231 (231)	49.78%	High
Akinyi MO, 2019 [8]	Kenya	East Africa	48.00 ±14.5	/	/	71%	22	79 (79)	27.85%	High
Alshenawy HA, 2012 [9]	Egypt	North Africa	58.30 ±10.1	/	20.00%	/	48	150 (150)	26.67%	Moderate
Ayadi EZ, 2018 [10]	Tunisia	North Africa	35.00 ±4.2	(17-40)	46.85%	64.10%	47	203 (238)	23.15%	Moderate
Babalanda JNP, 2014 [11]	Uganda	East Africa	47	(13-87)	53.51%	77.19%	23	114 (114)	20.18%	Moderate
Bahnassy A, 2015 [12]	Egypt	North Africa	TNBC: 51.91± 12.34 non-TNBC: 52.77 ± 12.13 Controls: 35 ± 13.94	TNBC: (30-78) non-TNBC: (27-81) Controls: (22-64)	15.63%	72.91%	43	96 (96)	44.79%	Moderate
Bakkach J, 2017 [13]	Morocco	North Africa	/	36 (/)	40.20%	58.50%	19	82 (331)	23.00%	High
Bekkouche Z, 2013 [14]	Algeria	North Africa	49.72	(30-84)	73.33%	35.83%	30	120 (120)	25.00%	High
Belhadj A, 2021 [15]	Algeria	North Africa	/	50 (21-88)	38.86%	64.70%	286	1066 (1140)	27.00%	Moderate
Ben Ayed-Guerfali D, 2021 [16]	Tunisia	North Africa	/	42.78 (27-65)	26.78%	59.70%	26	110 (134)	23.60%	High
Ben Gacem R, 2011 [17]	Tunisia	North Africa	48.70	47 (31-75)	43.62%	55.26%	28	94 (94)	29.79%	Moderate
Bhatia RK, 2019 [18]	Botswana	Southern Africa	/	54 (27-96)	25.76%	/	45	211 (384)	21.30%	Low

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Bogan D, 2017 [19]	Egypt	North Africa	52.38 ±12.17	(27-81)	13.94%	66.67%	43	96 (96)	44.79%	Moderate
Brandão M, 2020 [20]	Mozambique	East Africa	/	48 (/)	31.25%	61.43%	52	210 (262)	25.00%	Moderate
Chahbouni S, 2017 [21]	Morocco	North Africa	48.00	(22-99)	/	/	53	371 (390)	13.59%	High
Cherbal F, 2015 [22]	Algeria	North Africa	/	48.5 (22-84)	31.30%	/	627	3014 (3014)	20.80%	Moderate
Corsini C, 2017 [23]	Multiple (France, Algeria, Egypt, Morocco, Tunisia)	North Africa	/	/	44.74%	42%	35	174 (456)	20.10%	Moderate
Cubasch H, 2018 [24]	South Africa	Southern Africa	54.40 ±14.2	/	39.53%	/	126	581 (602)	21.70%	Low
Debouki-Joudi S, 2016 [25]	Tunisia	North Africa	49.30	(24-80)	26.22%	55.20%	53	80 (91)	66.25%	High
Dickens C, 2014 [26]	Multiple (South Africa & Namibia)	Southern Africa	/	/	33.87%	/	1482	7246 (12361)	20.45%	Low
Ekpe E, 2019 [27]	Kenya	East Africa	47.00 ±12	(21-79)	50.00%	/	24	120 (125)	20.00%	Moderate
El-Amir MI, 2020 [28]	Egypt	North Africa	54.20 ±10.6	(32-77)	18.00%	72%	3	50 (50)	6.00%	High
Elidrissi Errahhali M, 2017 [29]	Morocco	North Africa	48.70	(19-105)	18.00%	64.80%	320	2260 (2406)	14.20%	Moderate
Fitzpatrick MB, 2019 [30]	Senegal	West Africa	47.00 ±12.7	/	76.17%	/	92	197 (522)	46.70%	Moderate
Fouhi ME, 2020 [31]	Morocco	North Africa	51.60	(23-89)	32.60%	35.70%	87	568 (668)	15.30%	High
Gaceb H, 2017 [32]	Algeria	North Africa	48.44	(22-84)	28.50%	/	175	877 (877)	19.95%	Low
Groenewald C, 2019 [33]	South Africa	Southern Africa	53.60 ±13.4	/	/	68.10%	71	504 (505)	14.10%	Moderate
Guedouar Y, 2014 [34]	Algeria	North Africa	47.00	(23-78)	24.58%	38.33%	60	240 (240)	25.00%	High
Hachana M, 2012 [35]	Tunisia	North Africa	/	/	40.18%	21.40%	32	123 (123)	26.02%	Moderate
Hadgu E, 2020 [36]	Ethiopia	East Africa	43.00 ±14	50 (22-75)	50.00%	/	25	110 (112)	23.00%	Moderate
Huo D, 2009 [37]	Multiple (Nigeria & Senegal)	West Africa	44.80	/	39.15%	72%	103	271 (378)	27.00%	Low
Hussein O, 2013 [38]	Egypt	North Africa	53.00 ±11.6	52	17.39%	/	50	263 (263)	19.00%	High

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Kakudji BK, 2021 [39]	South Africa	Southern Africa	56.35 ±14.09	/	62.07%	64.66%	26	116 (116)	22.50%	High
Kallel-Bayoudh I, 2011 [40]	Tunisia	North Africa	50.00 ±11	(27-75)	35.44%	65%	11	84 (84)	13.00%	High
Khalil AI, 2016 [41]	Morocco	North Africa	50.20 ±11.34	(17-93)	35.48%	49.40%	146	889 (1277)	16.40%	Moderate
Luyey Mvila G, 2015 [42]	Democratic Republic of Congo	Central Africa	50.00	49 (29-82)	47.13%	/	13	87 (87)	14.94%	Moderate
Ly M, 2012 [43]	Mali	West Africa	46.00	(25-82)	78.18%	91%	53	114 (114)	46.00%	Moderate
Manai M, 2019 [44]	Tunisia	North Africa	/	42 (24-62)	42.20%	95%	49	194 (210)	25.00%	Moderate
McCormack VA, 2013 [45]	South Africa	Southern Africa	55.30	/	42.34%	/	209	1027 (1247)	20.40%	Low
Mehemmai C, 2019 [46]	Algeria	North Africa	/	48.5 (21-84)	26.99%	/	243	1144 (1162)	21.24%	Moderate
Mejri N, 2020 [47]	Tunisia	North Africa	/	48.5 IBCa 49 non-IBCa (/)	46.20% IBCa 36.10% non-IBCa	50%	121	740 (740)	16.40%	High
Mouh FZ, 2020 [48]	Morocco	North Africa	/	47 (/)	34.92%	60.10%	85	500 (500)	17.00%	High
Newman LA, 2019 [49]	Multiple (Ghana, Ethiopia, United States)	West and East Africa	/	56 (14-95)	/	/	Ghanaian: 142 Ethiopian: 24	Ghana: 292 (618) Ethiopia: 144 (252)	Ghana: 49.00% Ethiopia: 17.00%	High
Ohene-Yeboah M, 2012 [50]	Ghana	West Africa	49.10 ±15.3	(20-96)	53.70%	/	23	54 (330)	42.70%	Moderate
Ramdas Y, 2020 [51]	South Africa	Southern Africa	60.80 ±9.3	60 (43-86)	9.57%	12.10%	6	107 (107)	5.60%	High
Rayne S, 2019 [52]	South Africa	Southern Africa	/	56	51.10%	/	31	223 (365)	14.00%	Moderate
Roy I, 2011 [53]	Uganda	East Africa	/	/	40.00%	/	16	44 (64)	36.00%	Moderate
Samaka RM, 2016 [54]	Egypt	North Africa	/	(30-81)	44.44%	77.80%	19	81 (81)	23.50%	Moderate
Sawe RT, 2017 [55]	Kenya	East Africa	51.90	48.5 (16-84)	42.22%	/	18	49 (49)	36.70%	Moderate
Schwartz T, 2012 [56]	Ghana	West Africa	49.00 ±13.4	/	52.52%	/	58	103 (104)	56.30%	Moderate
Sengal AT, 2017 [57]	Sudan	North Africa	48.80 ±13.5	(20-90)	41.79%	52.70%	143	415 (560)	34.50%	Moderate
Seshie B, 2015 [58]	Ghana	West Africa	52.50 ±12.1	(24-77)	31.58%	66.00%	77	156 (165)	49.40%	Moderate
Shaikh AJ, 2018 [59]	Kenya	East Africa	54.00	(27-83)	33.33%	49%	21	116 (123)	17.00%	Moderate
Stark A, 2010 [60]	Ghana	West Africa	48.00 ±13.7	/	76.00%	/	37	45 (75)	82.20%	Moderate

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Sung H, 2019 [61]	Multiple (USA, Caribbean, West Africa, East Africa)	West and East Africa	/	/	West Africa: 56.92% East Africa: 43.95%	/	West Africa: 173 East Africa: 51	West Africa: 717 (811) East Africa: 441 (484)	West Africa: 24.10% East Africa: 11.60%	Moderate
Tanimowo MO, 2019 [62]	Nigeria	West Africa	43.07 ±11.19	(23-70)	29.51%	/	38	61 (61)	62.30%	High
Traoré B, 2019 [63]	Guinea	West Africa	48.20 ±10.9	(31-80)	36.21%	62.10%	15	58 (58)	25.90%	Moderate
Ukah CO, 2017 [64]	Nigeria	West Africa	46.30 ±12.3	(24-85)	26.83%	/	50	123 (397)	40.70%	Moderate
Uyisenga JP, 2020 [65]	Rwanda	East Africa	49.70 ±12.98	(17-86)	/	70.06%	52	138 (138)	37.70%	Low
Youngblood VM, 2020 [66]	Malawi	East Africa	/	49 (21-80)	37.00%	77%	25	100 (100)	25.00%	Low
Znati K, 2012 [67]	Morocco	North Africa	30.62	(18-35)	45.07%	51%	13	74 (74)	18.00%	High

/ data not stated

IBCa- inflammatory breast cancer

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Table S3: Questions for Risk of Bias Assessment, modified from Hoy *et al*., 2012

1. Was the study's target population a close representation of the national population in relation to relevant variables, e.g. age, sex, occupation?
2. Was the sampling frame a true or close representation of the target population?
3. Was some form of random selection used to select the sample, or was a census undertaken?
4. Was the likelihood of non-response bias minimal?
5. Were data collected directly from the subjects (as opposed to a proxy)?
6. Was an acceptable case definition used in the study? Was TNBC defined as ER, PR, HER2 negative?
7. Was the study instrument that measured the parameter of interest (e.g. appropriate guidelines for cut-offs ASCO-CAP) shown to have reliability and validity?
8. Was the same mode of data collection used for all subjects? Was the information from each patient collected in the same way (e.g clinical referral appointment)?
9. Was there a shortest prevalence or incidence period identified?
10.Were the numerator(s) and denominator(s) for the parameter of interest appropriate?

Table S4: Raw Values for Risk of Bias Assessment for Included Studies

Author	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	TOTAL	Overall ROB
Adani-Ifè A, 2020 [1]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
Adisa AO, 2011 [2]	0	0	1	0	1	0	0	0	1	0	3	**High**
Agboola A, 2014 [3]	0	0	0	0	1	1	1	1	1	0	5	**High**
Agboola A, 2014 [4]	0	0	1	0	1	1	1	1	1	1	7	**Moderate**
Agboola A, 2013 [5]	0	0	0	0	1	1	1	1	0	1	5	**High**
Agboola A, 2016 [6]	0	0	0	1	1	0	1	1	1	1	6	**Moderate**
Agboola A, 2014 [7]	0	0	0	0	1	0	1	1	1	1	5	**High**
Akinyi MO, 2019 [8]	0	0	0	1	1	1	0	1	0	0	4	**High**
Alshenawy HA, 2012 [9]	0	0	1	1	1	1	0	1	1	1	7	**Moderate**
Ayadi EZ, 2018 [10]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Babalanda JNP, 2014 [11]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Bahnassy A, 2015 [12]	0	0	1	1	1	1	0	1	1	1	7	**Moderate**
Bakkach J, 2017 [13]	0	0	0	1	1	0	0	1	1	0	4	**High**
Bekkouche Z, 2013 [14]	0	0	1	0	1	1	0	1	1	0	5	**High**

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Belhadj A, 2021 [15]	1	0	1	1	1	1	0	1	1	1	8	**Moderate**
Ben Ayed-Guerfali D, 2021 [16]	0	0	0	0	1	0	0	1	1	0	3	**High**
Ben Gacem R, 2011 [17]	0	0	1	1	1	1	0	1	0	1	6	**Moderate**
Bhatia RK, 2019 [18]	1	1	1	1	1	1	1	1	1	1	10	**Low**
Bogan D, 2017 [19]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Brandão M, 2020 [20]	0	0	1	1	1	1	1	0	1	1	7	**Moderate**
Chahbouni S, 2017 [21]	0	0	0	0	1	1	0	1	1	0	4	**High**
Cherbal F, 2015 [22]	1	1	1	0	1	1	0	1	1	1	8	**Moderate**
Corsini C, 2017 [23]	0	0	1	1	1	0	0	1	1	1	6	**Moderate**
Cubasch H, 2018 [24]	1	1	1	1	1	1	0	1	1	1	9	**Low**
Debouki-Joudi S, 2016 [25]	0	0	0	0	1	1	0	1	0	1	4	**High**
Dickens C, 2014 [26]	1	1	1	1	1	1	1	1	1	1	10	**Low**
Ekpe E, 2019 [27]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
El-Amir MI, 2020 [28]	0	0	0	0	1	0	0	1	1	1	4	**High**
Elidrissi Errahhali M, 2017 [29]	0	1	1	0	1	1	0	1	1	1	7	**Moderate**
Fitzpatrick MB, 2019 [30]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
Fouhi ME, 2020 [31]	0	1	0	0	1	0	0	1	1	0	4	**High**
Gaceb H, 2017 [32]	1	1	1	0	1	1	1	1	1	1	9	**Low**
Groenewald C, 2019 [33]	0	0	0	1	1	1	0	1	1	1	6	**Moderate**
Guedouar Y, 2014 [34]	0	1	0	0	1	1	0	1	1	0	5	**High**
Hachana M, 2012 [35]	0	0	1	1	1	1	0	1	1	1	7	**Moderate**
Hadgu E, 2020 [36]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Huo D, 2009 [37]	0	1	1	1	1	1	1	1	1	1	9	**Low**
Hussein O, 2013 [38]	0	0	1	0	1	0	0	1	1	1	5	**High**
Kakudji BK, 2021 [39]	0	0	0	1	1	1	0	0	1	1	5	**High**
Kallel-Bayoudh I, 2011 [40]	0	0	0	0	1	1	0	1	1	1	5	**High**
Khalil AI, 2016 [41]	0	1	1	0	1	1	0	1	1	1	7	**Moderate**
Luyey Mvila G, 2015 [42]	0	0	1	0	1	1	1	1	1	1	7	**Moderate**
Ly M, 2012 [43]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Manai M, 2019 [44]	0	1	1	0	1	1	0	1	1	1	7	**Moderate**
McCormack VA, 2013 [45]	1	1	1	1	1	1	0	1	1	1	9	**Low**
Mehemmai C, 2019 [46]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Mejri N, 2020 [47]	0	0	0	0	1	0	0	1	1	0	3	**High**
Mouh FZ, 2020 [48]	0	0	0	0	1	1	0	1	1	1	5	**High**
Newman LA, 2019 [49]	0	0	0	0	1	0	1	1	1	1	5	**High**
Ohene-Yeboah M, 2012 [50]	0	0	1	0	1	1	0	1	1	1	6	**Moderate**

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Ramdas Y, 2020 [51]	0	0	0	1	1	0	0	1	1	1	5	**High**
Rayne S, 2019 [52]	0	0	1	0	1	1	1	0	1	1	6	**Moderate**
Roy I, 2011 [53]	0	0	0	0	1	1	1	1	1	1	6	**Moderate**
Samaka RM, 2016 [54]	0	0	0	1	1	1	1	1	1	1	7	**Moderate**
Sawe RT, 2017 [55]	0	1	1	0	1	1	0	1	1	0	6	**Moderate**
Schwartz T, 2012 [56]	0	0	0	1	1	1	0	1	1	1	6	**Moderate**
Sengal AT, 2017 [57]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Seshie B, 2015 [58]	0	1	1	1	1	1	0	1	1	0	7	**Moderate**
Shaikh AJ, 2018 [59]	0	0	1	1	1	1	1	1	1	1	8	**Moderate**
Stark A, 2010 [60]	0	0	1	0	1	1	1	1	1	0	6	**Moderate**
Sung H, 2019 [61]	0	1	1	1	1	1	0	1	1	1	8	**Moderate**
Tanimowo MO, 2019 [62]	0	0	0	1	1	0	1	0	1	1	5	**High**
Traoré B, 2019 [63]	0	1	1	0	1	1	0	1	1	0	6	**Moderate**
Ukah CO, 2017 [64]	0	1	0	1	1	1	0	1	1	1	7	**Moderate**
Uyisenga JP, 2020 [65]	0	1	1	1	1	1	1	1	1	1	9	**Low**
Youngblood VM, 2020 [66]	0	1	1	1	1	1	1	1	1	1	9	**Low**
Znati K, 2012 [67]	0	0	0	0	1	0	0	1	1	0	3	**High**

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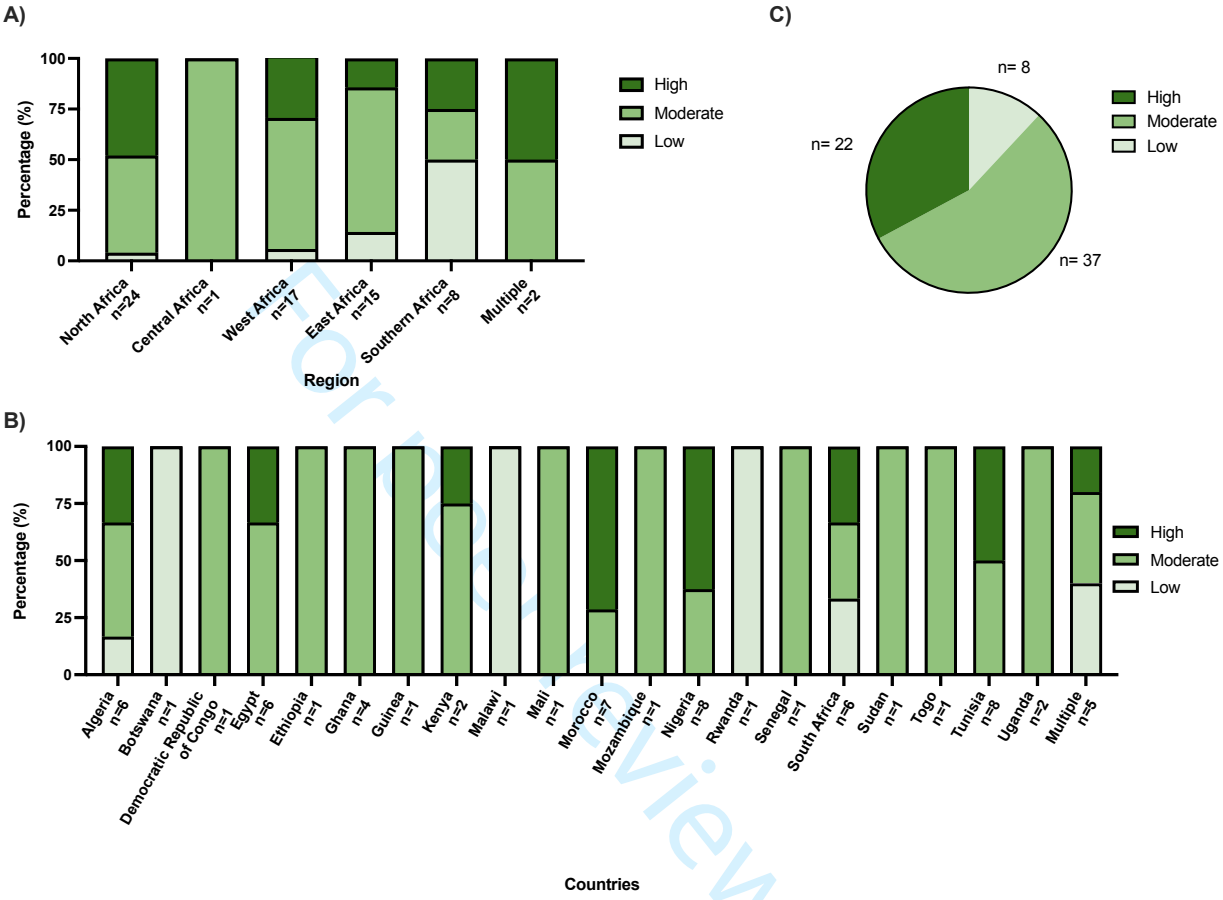
Table S5: Pooled TNBC Frequcny in Africa by Country

Country (no. of studies)	Total participants	Pooled TNBC Frequency, % [95% CI]	Heterogeneity (I^2), %	Weight, %
Algeria (*n*= 6)	6461	22.7 [20.3- 25.2]	77	12.0
Botswana (*n*= 1)	211	21.3 [16.3- 27.4]	not applicable	1.9
Democratic Republic of Congo (*n*= 1)	87	14.9 [8.9- 24.1]	not applicable	1.6
Egypt (*n*= 5)	640	24.5 [15.3- 36.7]	88	8.3
Ethiopia (*n*= 1)	110	22.7 [15.8- 31.5]	not applicable	1.8
Ghana (*n*= 4)	358	57.2 [43.6- 69.8]	82	6.8
Guinea (*n*= 1)	58	25.9 [16.2- 38.6]	not applicable	1.6
Kenya (*n*= 4)	364	24.5 [17.8- 32.8]	63	6.9
Malawi (*n*= 1)	100	25.0 [17.5- 34.4]	not applicable	1.8
Mali (*n*= 1)	114	46.5 [37.6- 55.7]	not applicable	1.9
Morocco (*n*= 7)	4744	15.6 [14.3- 17.1]	31	13.4
Mozambique (*n*= 1)	210	24.8 [19.4- 31.0]	not applicable	1.9
Nigeria (*n* = 4)	479	49.2 [41.4- 57.0]	61	7.3
Rwanda (*n*= 1)	138	37.7 [30.0- 46.0]	not applicable	1.9
Senegal (*n*= 1)	197	46.7 [39.8- 53.7]	not applicable	2.0
South Africa (*n*= 4)	950	14.0 [9.6- 19.8]	75	7.0
Sudan (*n*= 1)	415	34.5 [30.0- 39.2]	not applicable	2.0
Togo (*n*= 1)	117	37.6 [29.3- 46.7]	not applicable	1.9
Tunisia (*n*= 8)	1628	26.4 [18.5- 36.2]	92	14.6
Uganda (*n*= 2)	158	27.0 [14.3- 45.1]	77	3.4
Total	17539	27.0 [24.0- 30.2]	94	100

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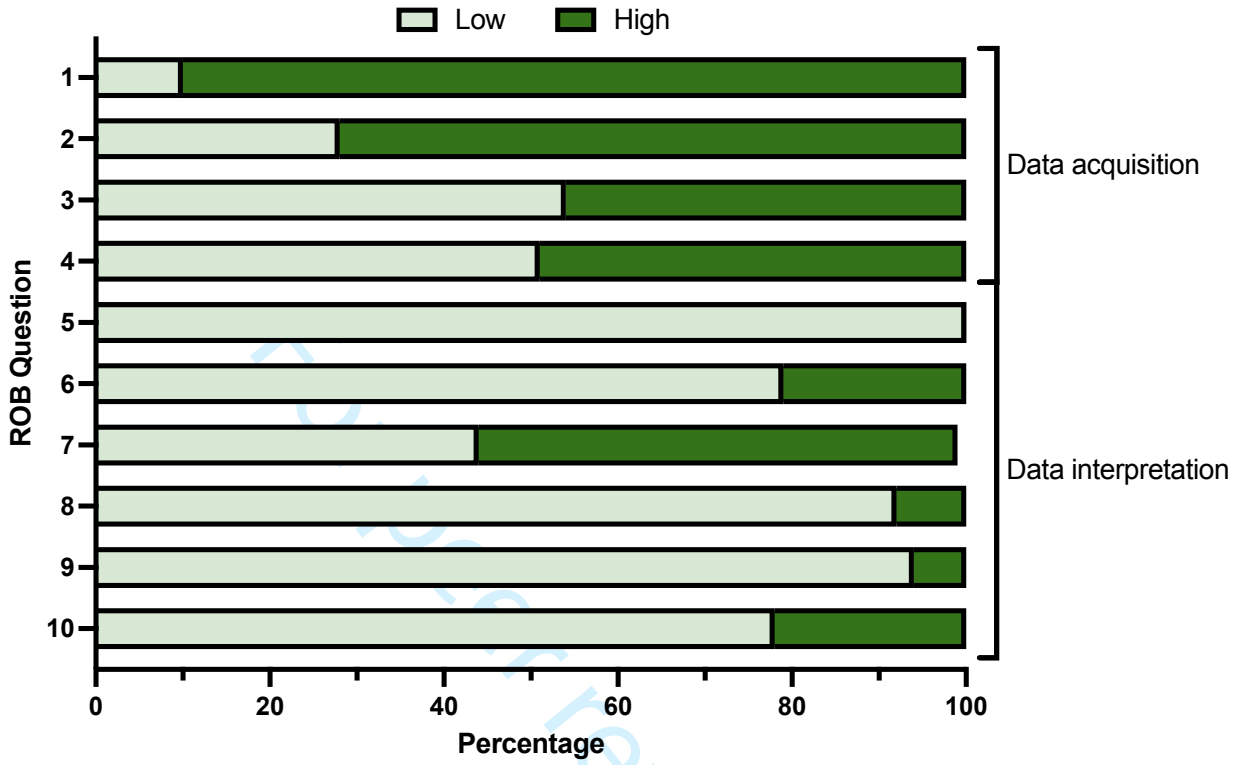
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Figure S1: Risk of bias assessment of all articles by (A) region, (B) country and (C) overall risk of bias assessment for all articles included. Southern Africa had the highest percentage of low risk of bias. Multiple denotes more than one country and/or region or was investigated within that study.



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Figure S2: Risk of bias (ROB) assessment per question for all articles included in systematic review. High risk of bias was noted specifically for questions on data acquisition



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Figure S3: Pooled TNBC frequency in Africa by risk of bias assessment. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.

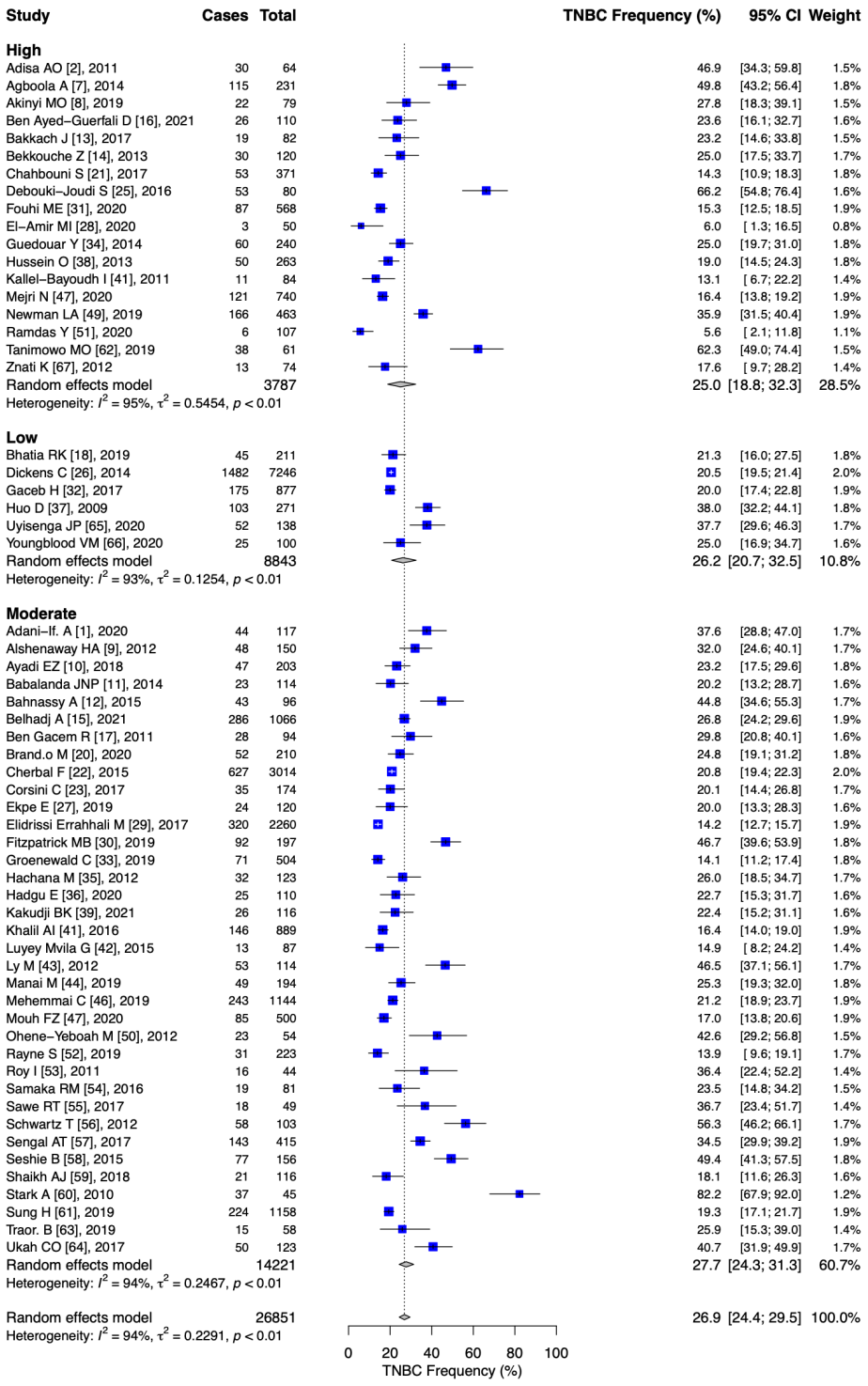
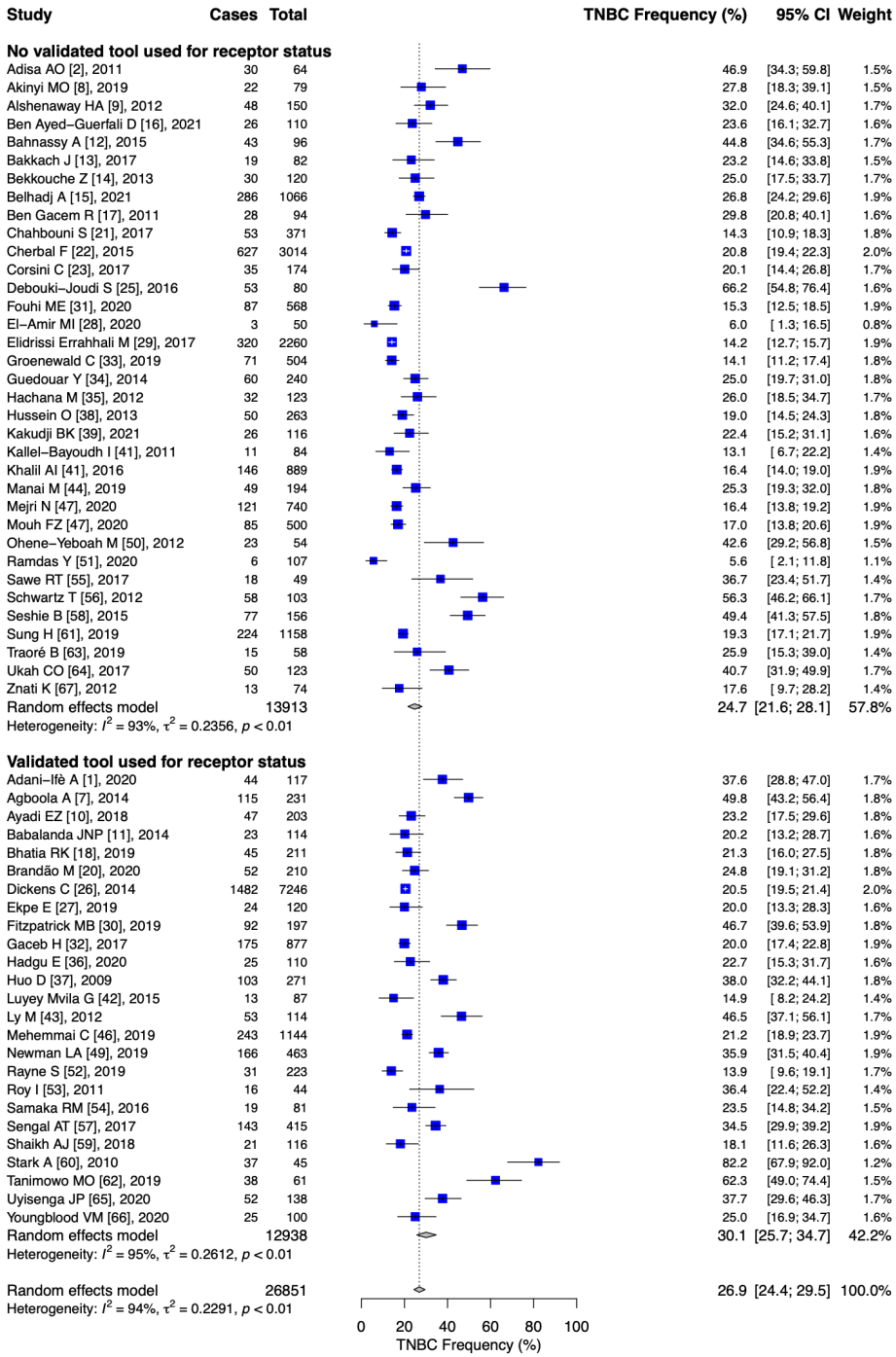


Figure S4: Pooled TNBC frequency in Africa by use of validated tool for receptor status. Cases are defined as participants in a study that were identified as triple negative, and total is the number of breast cancer participants with known receptor status in the study.



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Figure S5: Meta-regression shows association with year of publication but not with age or use of validated tool for receptor status. The association of use of validated tool to assess receptor status (A), year of publication (B), mean age at diagnosis (C) and median age at diagnosis (D) on heterogeneity of treatment effects between studies was investigated by a meta-regression. P <0.05 indicates significant association.

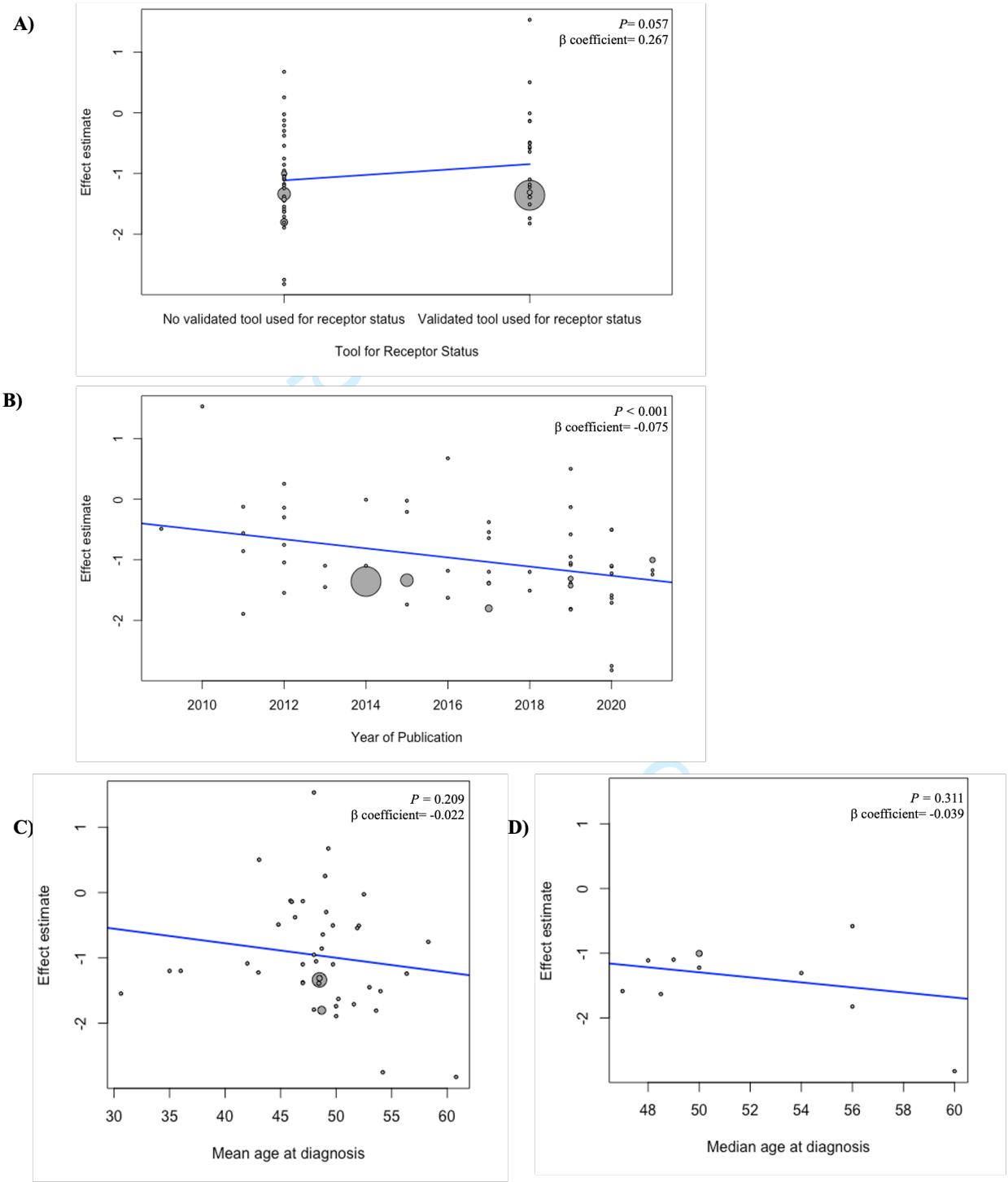
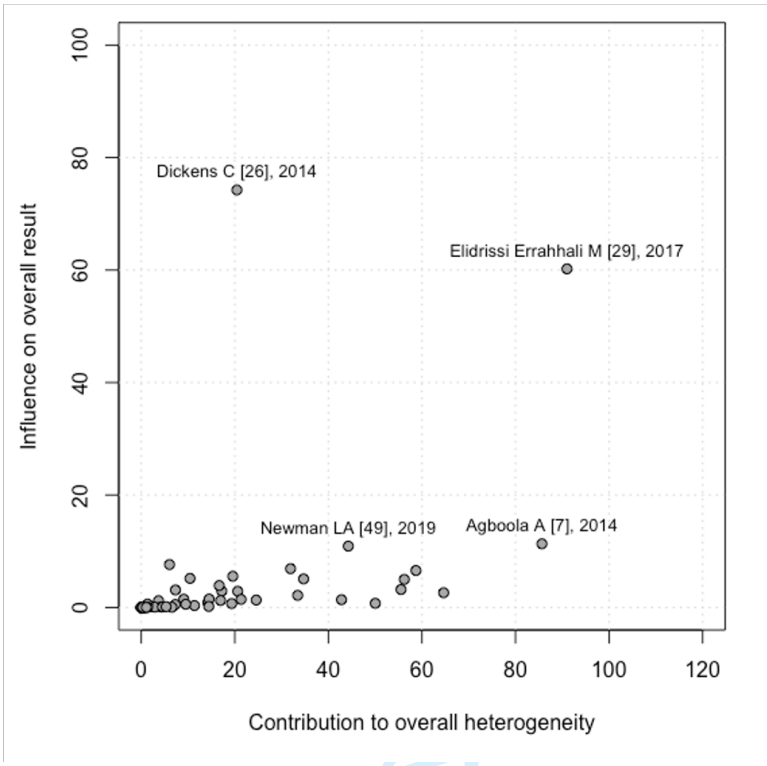


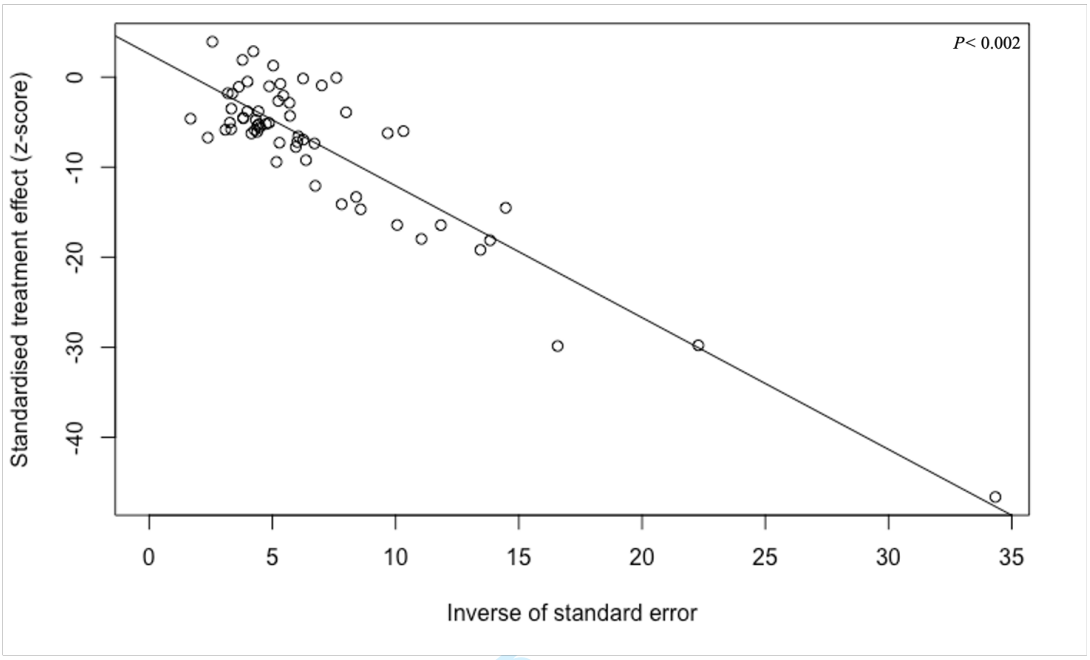
Figure S6: Influence analysis highlights studies that overly contribute to heterogeneity. Influence analysis via Baujat plot shows studies that two studies [7] and [29] contribute largely to overall heterogeneity.



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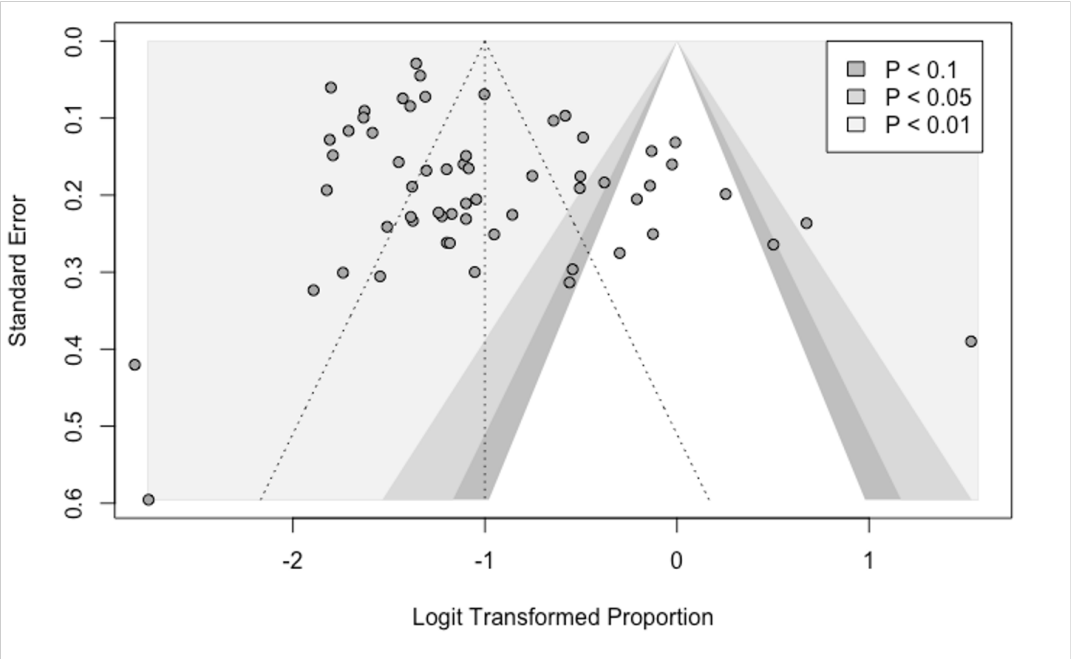
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Figure S7: Egger's test shows high publication bias. Publication bias between studies for TNBC frequency investigated using Egger's test and a significant association was found, inferring significant publication bias. P <0.05 indicates significant association.



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Figure S8: Funnel plot shows high publication bias. Publication bias between studies was investigated using funnel plot to visualize asymmetry and multiple studies do not follow the funnel shape inferring significant publication bias. P <0.05 indicates significant association.



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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 6
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 6, 7
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 7, 8; Table S1
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 7, 8; Figure 1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table S1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7, 8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 7, 8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 7, 8,
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 7, 8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 9
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	N/A
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 7,8,9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 9
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 8

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 8
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 9, 10; Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	N/A
Study characteristics	17	Cite each included study and present its characteristics.	Table S2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table S2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table S2
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Table S2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	All forest plots
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	All forest plots
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Figure S6-S8
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	page 13
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 13-17
	23b	Discuss any limitations of the evidence included in the review.	Page 13-17
	23c	Discuss any limitations of the review processes used.	Page 16
	23d	Discuss implications of the results for practice, policy, and future research.	Page 13-17
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 4
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	N/A
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 18
Competing interests	26	Declare any competing interests of review authors.	Page 18

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Table S2

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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